

BIOLOGICAL PHYSICS

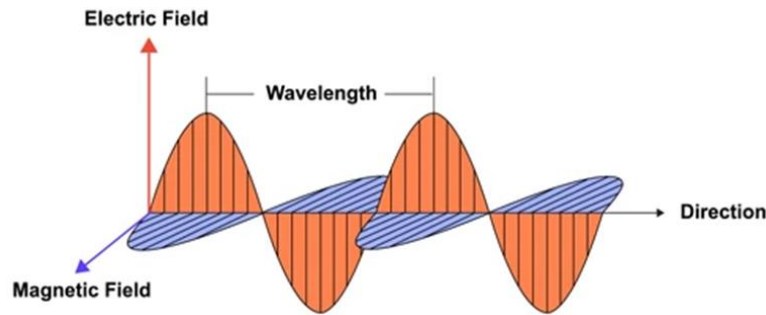
CHAPTER 5 –INTRODUCTION TO LIGHT AND SPECTROSCOPY

Poopalasingam Sivakumar
Department of Physics, SIUC

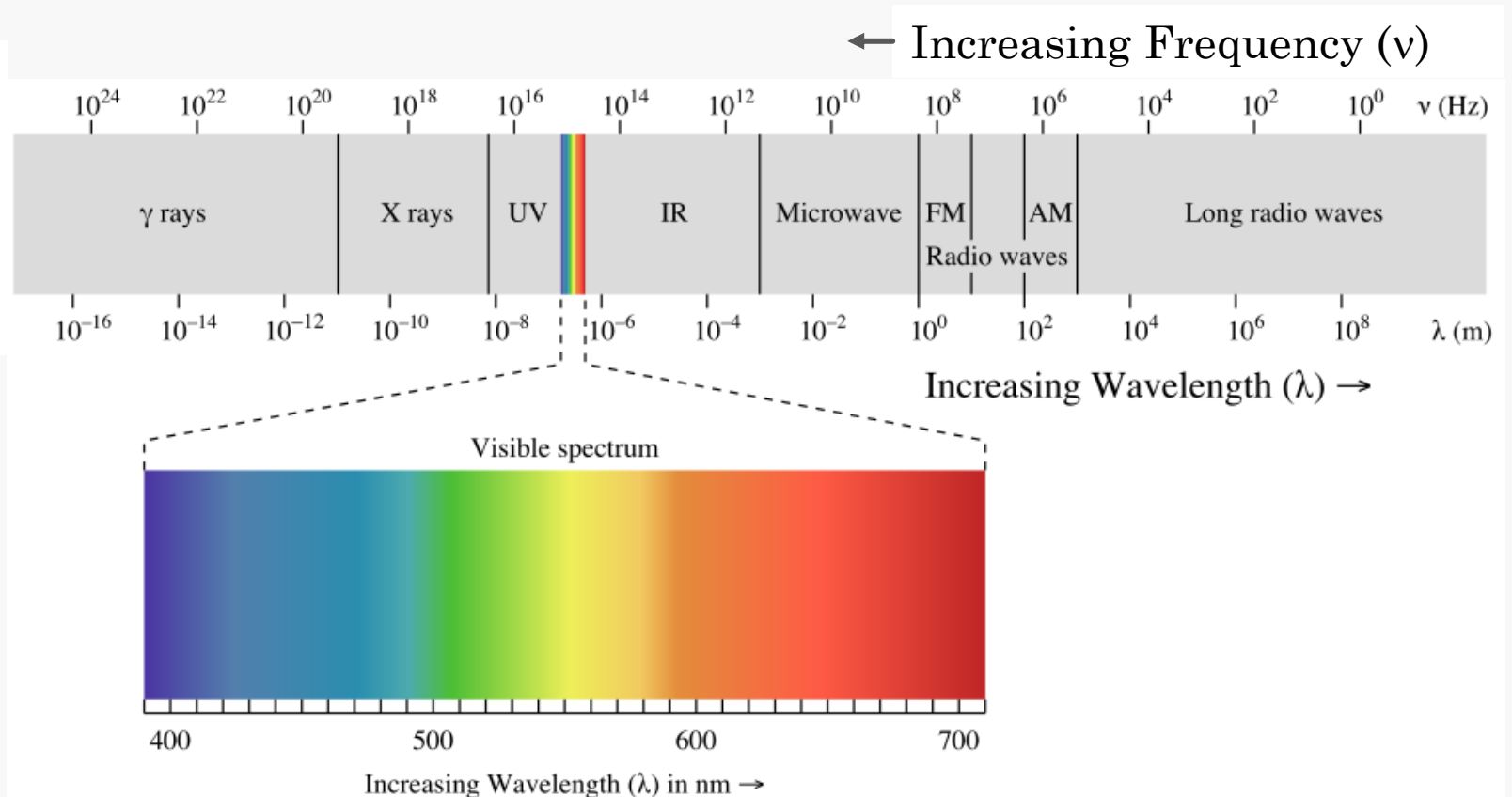


Electromagnetic Spectrum

- Light is part of EM spectrum.
- Photon is a massless particle and travel like wave with speed $c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \approx 3 \times 10^8 \text{ m/s}.$

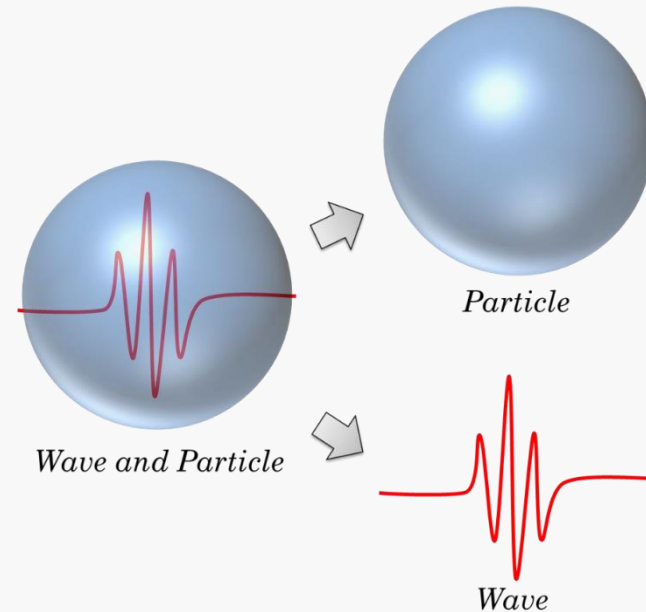
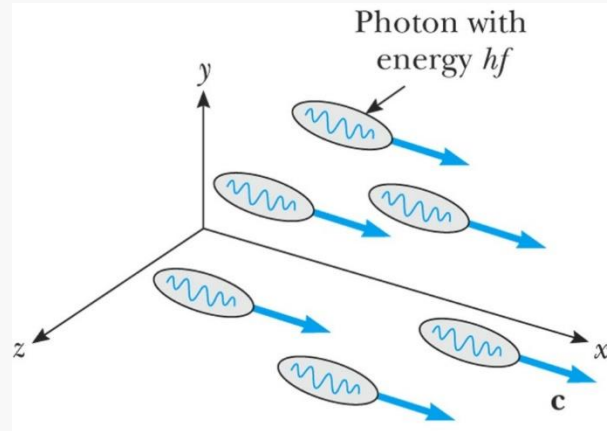
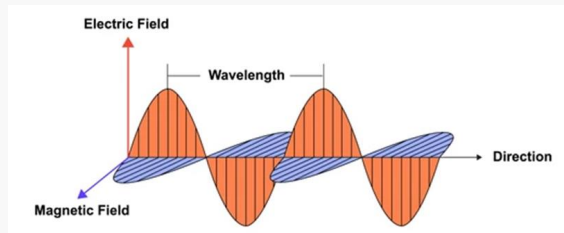


- γ -ray: nuclear
- X-ray: core-level electron
- UV: valence electrons
- Vis: valence electrons
- IR: molecular vibrations



Photons: The Dual Nature of Light

- **Photon** is the smallest quantity (quantum), also known as quanta.
- **Wave-Particle Duality:** Photons exhibit a unique ability to behave like both waves and particles depending on the experimental setup.
- **Energy of a Photon:** $E = hf = \frac{hc}{\lambda}$ (*Planck Hypothesis*).



Albert Einstein

Particle Nature of Photon

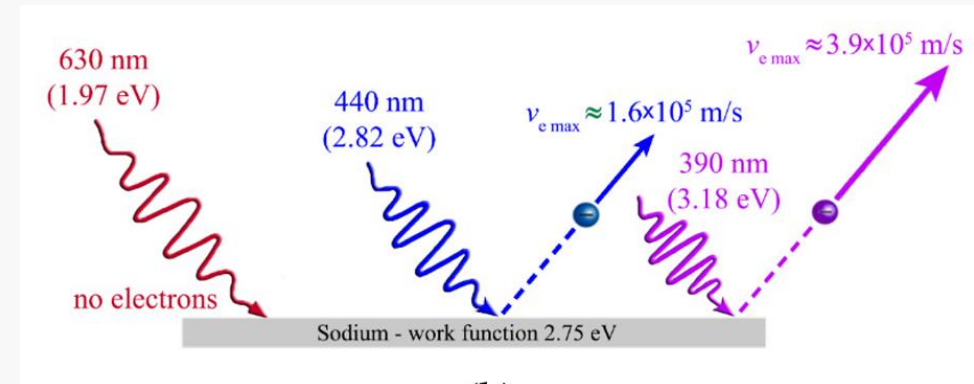
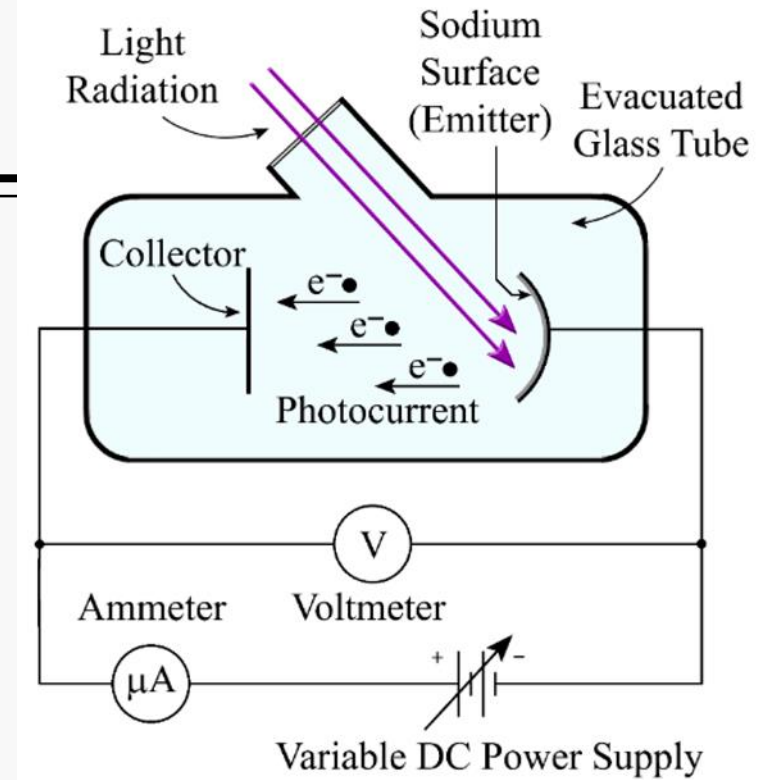
Photoelectric effect experiment is an evidence of particle nature of light in which electrons are emitted from metal surface when UV light shines on metal surface.

- i. Electron energy is independent of intensity radiation (larger amplitude waves).
- ii. Energy of electron depend on the wavelength of light but not on amplitude of the wave.

Albert Einstein proposed that energy of light is quantized. And the kinetic energy of electron can be expressed as:

$$K_e = hf - \phi$$

Where K_e is the kinetic energy of an escaping electron, and ϕ is work function of material.



Particles

In special relativity predict that for particles:

➤ **Energy** $E = \gamma m_o c^2 = K.E + m_o c^2 = \sqrt{p^2 c^2 + m_o^2 c^4}$

➤ **Momentum** $p = \frac{\sqrt{E^2 - m_o^2 c^4}}{c}$

➤ **Wavelength** $\lambda = \frac{h}{p} = \frac{hc}{\sqrt{E^2 - m_o^2 c^4}}$ (In 1924, Louis de Broglie published his speculation)

➤ **Velocity**

$$v = \frac{pc^2}{E} = c \sqrt{1 - \frac{m_o^2 c^4}{E^2}}$$

Photons

In case of photon, rest mass is zero :

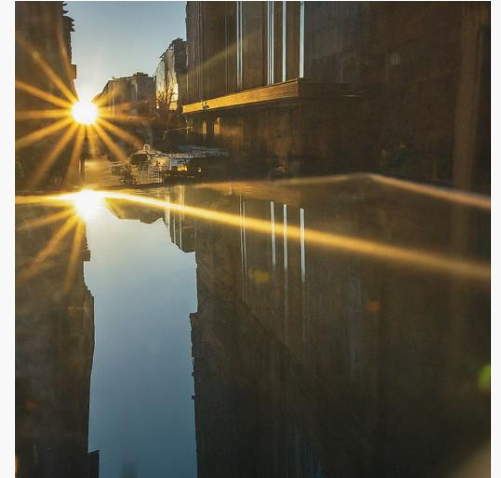
- *Momentum* $p = \frac{E}{c}$
- *Wave Vector* $k = \frac{2\pi}{\lambda} \Rightarrow p = \hbar k$ where $\hbar = \frac{h}{2\pi}$
- *Wavelength* $\lambda = \frac{h}{p} = \frac{hc}{E}$
- *Velocity* $v = \frac{pc^2}{E} = c$

Duality of Photon

Particle like model: Can describe phenomena like reflection. Light is conceptualized as a stream of tiny particles called photons. Each photon carries a specific amount of energy.

Wave Model:

- Light propagates as an electromagnetic wave.
- Waves carry energy without transporting matter.
- Essential for explaining interference and diffraction.
- When light moves between mediums, wavelength changes, frequency remains constant.
- Wave speed in a medium is independent of intensity.



Wavelength and wavenumber: inverse of the wavelength ($1/\lambda$) is called wavenumber (ν), which is expressed in cm^{-1} (number of cycles per unit length)

$$\nu \text{ (or } \bar{\nu}) = \frac{10^7}{\lambda \text{ (in nm)}}$$

Geometrical Optics

cycle time or period (τ): The time to complete one cycle.

$$\tau = \frac{\lambda}{v}$$

frequency: Number of cycles pass a point in a second.

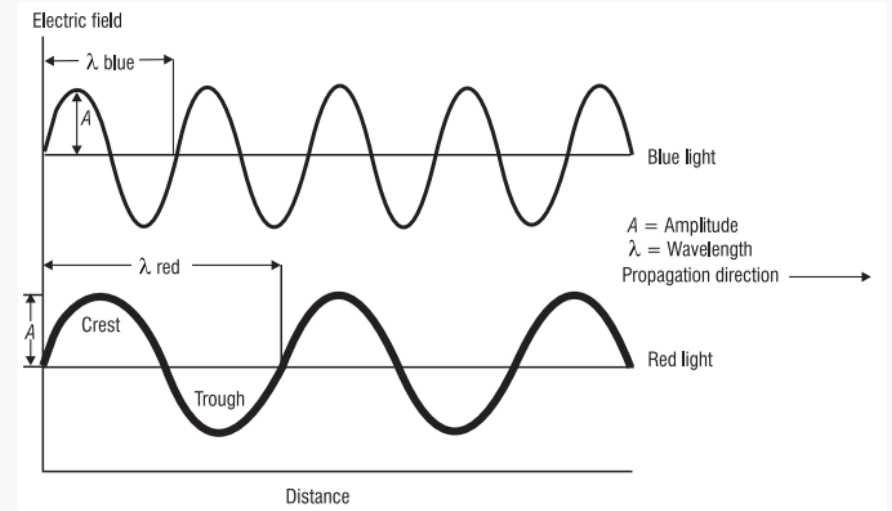
$$f = \frac{1}{\tau} = \frac{v}{\lambda}$$

In geometric optics, the wavelength (λ) of light is negligible compared to the size of the components in the optical system (lenses, mirrors, etc.).

$$\lim_{\lambda \rightarrow 0} \text{Physical Optics} = (\text{geometrical optics})$$

EM waves share six properties with all forms of wave motions:

Polarization, Superposition, Reflection, Refraction, Diffraction, and Interference.

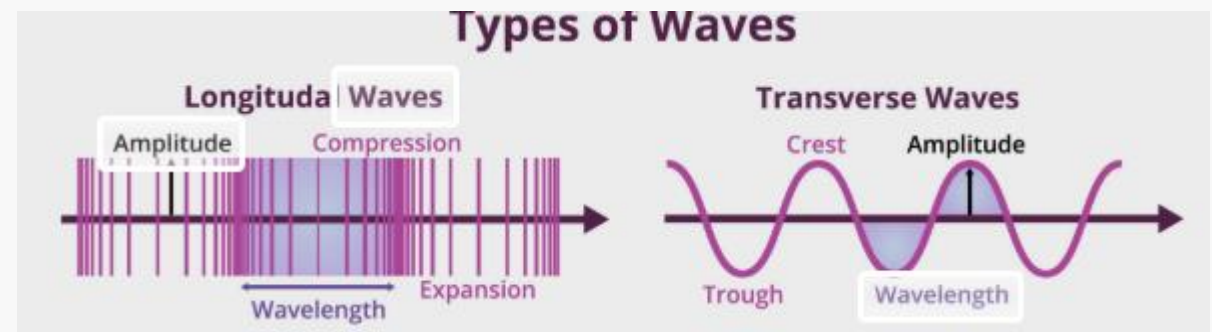
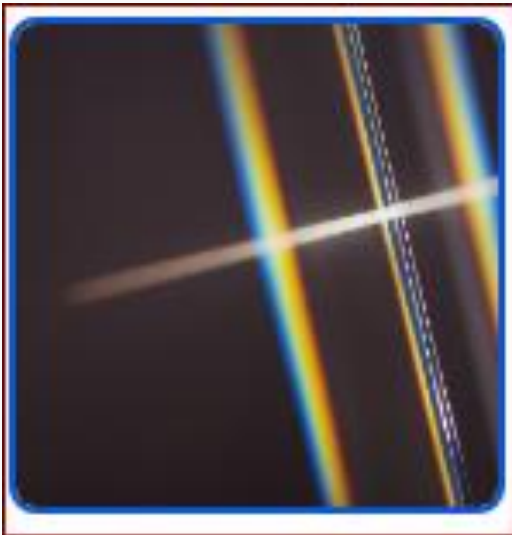


Example 5-1

Determine the frequency, energy, and wavenumber of a blue light, wavelength of 500 nm.

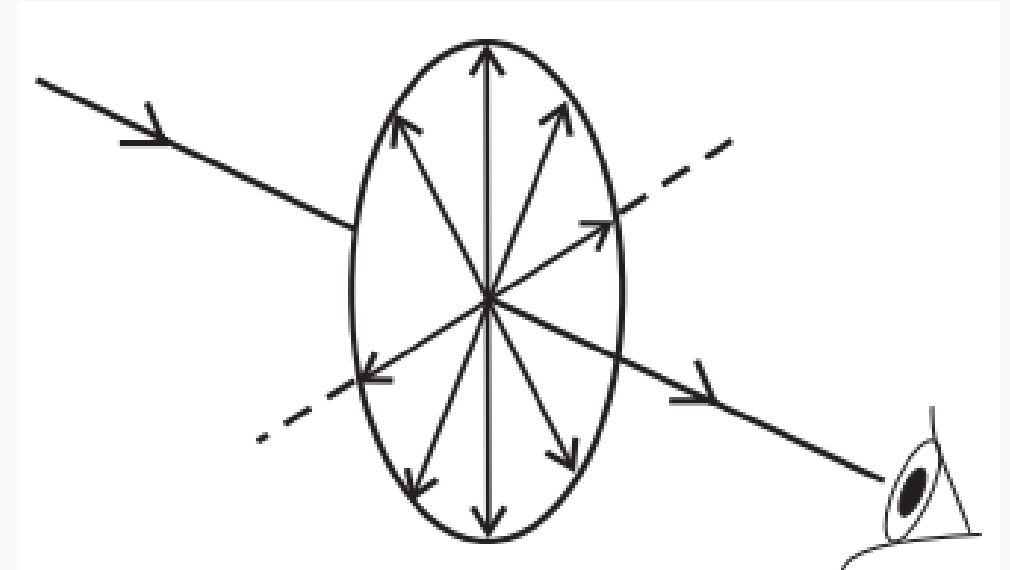
Light and Polarization

- Polarization arises from the direction of the E-field vector with respect to the direction of light's propagation.
- Because light is transverse wave (electric field vibrating perpendicular to its propagation motion), it is polarizable.
- In the other hand, sound wave (longitudinal), vibrate along its propagation direction and thus is not polarizable.



Unpolarized Light

- An ordinary light source consists of a very large number of randomly oriented atomic emitters. Each excited atom radiates a polarized wave-train for roughly 10^{-8} s.
- Because of rapid change, the resultant polarization is indiscernible, this wave is referred as **natural light**. This is known as *unpolarized*, no preferred orientation.
- **Randomly polarization** is probably a better way of representation than unpolarized light. For example, of unpolarized light source are incandescent bulbs, arc lamps, and sun.



Linear Polarization

Let's consider two orthogonal optical disturbances

$$\vec{E}_x(z, t) = \hat{i} E_{0x} \cos(kz - \omega t)$$

$$\vec{E}_y(z, t) = \hat{j} E_{0y} \cos(kz - \omega t + \varepsilon)$$

Where ε is the relative phase difference between the waves, both are traveling in the z-direction.

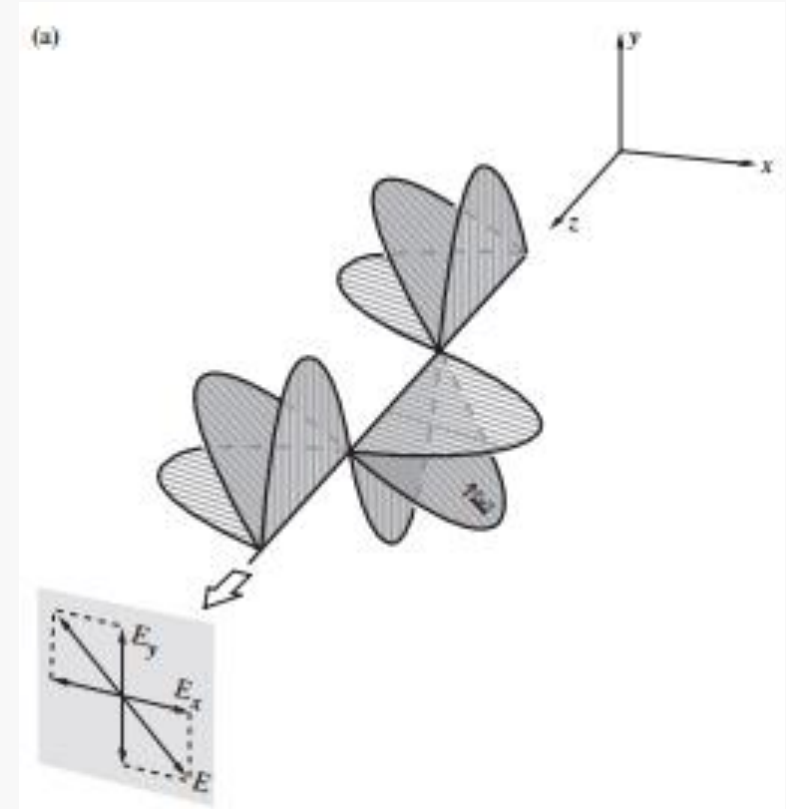
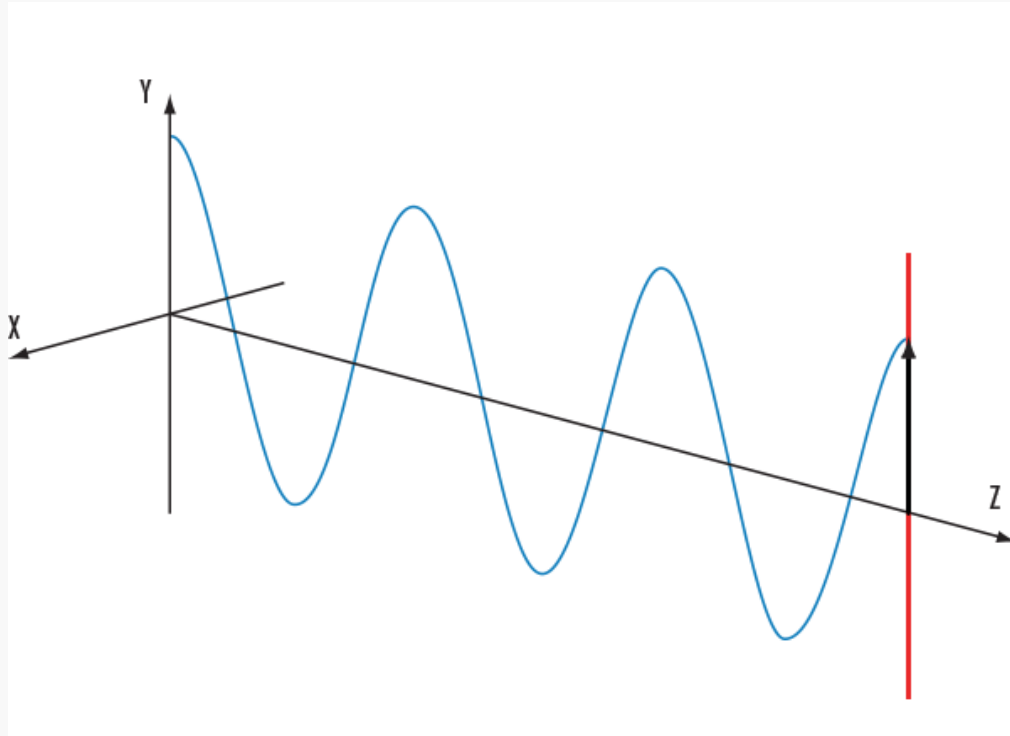
The resultant of optical disturbance is.

$$\vec{E}(z, t) = \vec{E}_x(z, t) + \vec{E}_y(z, t)$$

If ε is zero or an integral multiple of $\pm 2\pi$, the waves are in-phase, and becomes

$$\vec{E} = \left(\hat{i} \vec{E}_x(z, t) + \hat{j} \vec{E}_y(z, t) \right) \cos(kz - \omega t)$$

Linear Polarization



Circular Polarization

When both constituent waves have equal amplitudes (i.e., $E_{0x} = E_{0y} = E_0$), and their relative phase difference $\varepsilon = -\frac{\pi}{2} + 2m\pi$, where $m = 0, \pm 1, \pm 2, \dots$, it is said circular polarization. In other words, for any value of m (increasing or decreasing)

$$\begin{aligned}\vec{E}_x(z, t) &= \hat{i} E_{0x} \cos(kz - \omega t) \\ \vec{E}_y(z, t) &= \hat{j} E_{0y} \cos\left(kz - \omega t - \frac{\pi}{2}\right) = \hat{j} E_{0y} \left[\cos(kz - \omega t) \cos \frac{\pi}{2} + \sin(kz - \omega t) \sin \frac{\pi}{2} \right] \\ \vec{E}_y(z, t) &= \hat{j} E_{0y} \sin(kz - \omega t)\end{aligned}$$

Wave becomes:

$$\vec{E} = E_0(\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t))$$

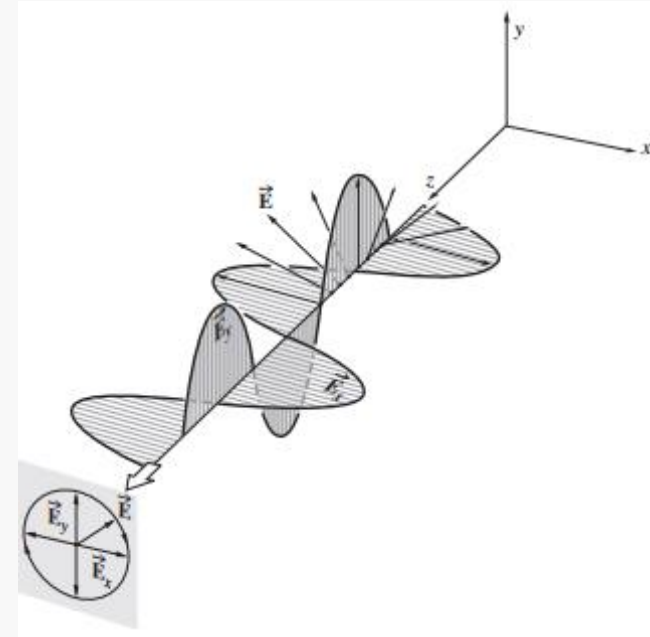
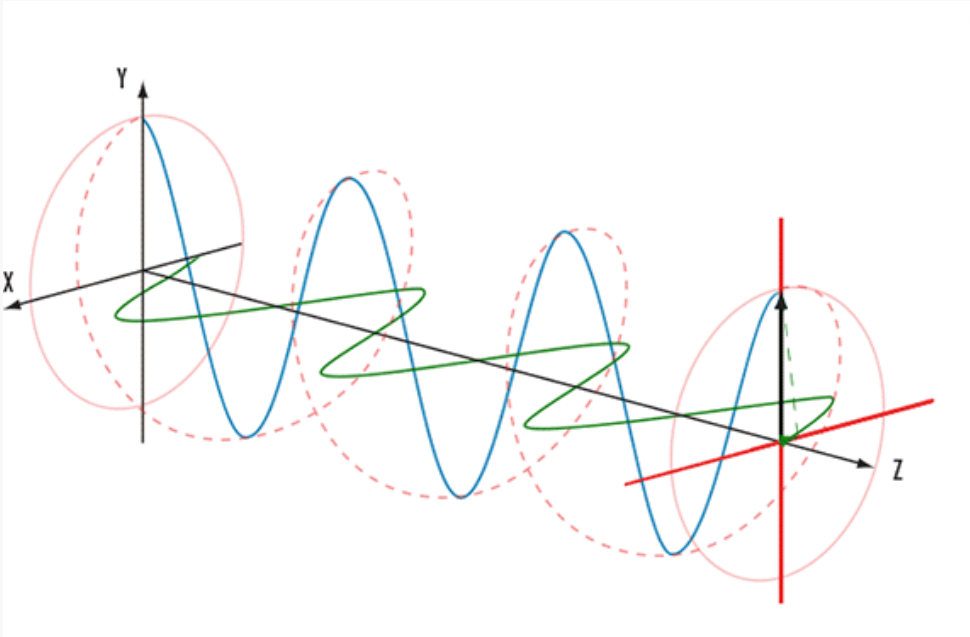
Notice that now the scalar amplitude of $\vec{E}$ is constant, and

$$\sqrt{\vec{E} \cdot \vec{E}} = E_0$$

$\vec{E}$ is rotating **clockwise** at angular frequency of ω . Such a wave is **right-circularly polarized** and generally it is referred as **right-circular light**.

Circular Polarization

- In other word, if $\varepsilon = \frac{\pi}{2} + 2m\pi$, where $m = 0, \pm 1, \pm 2, \dots$, then $\vec{E}$ rotates **counterclockwise** at angular frequency of ω . Such a wave is **left-circularly polarized** and generally it is referred as **left-circular light**.



Elliptical Polarization

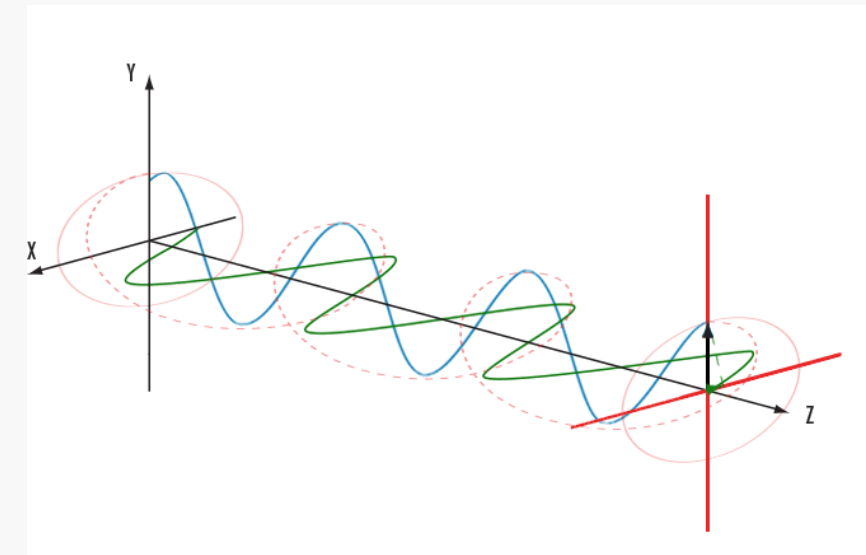
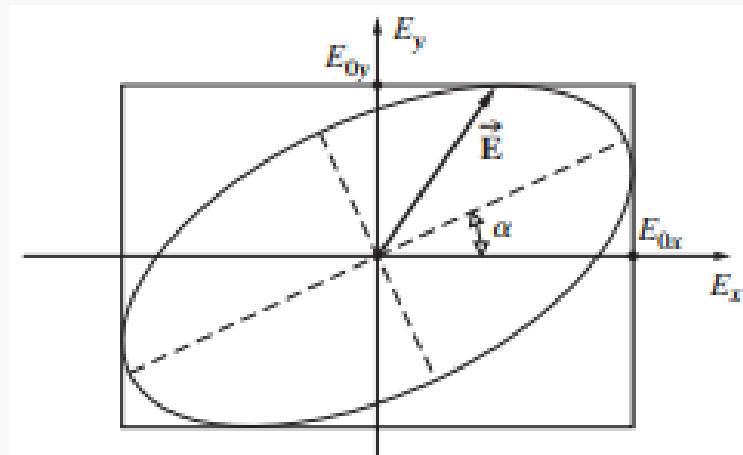
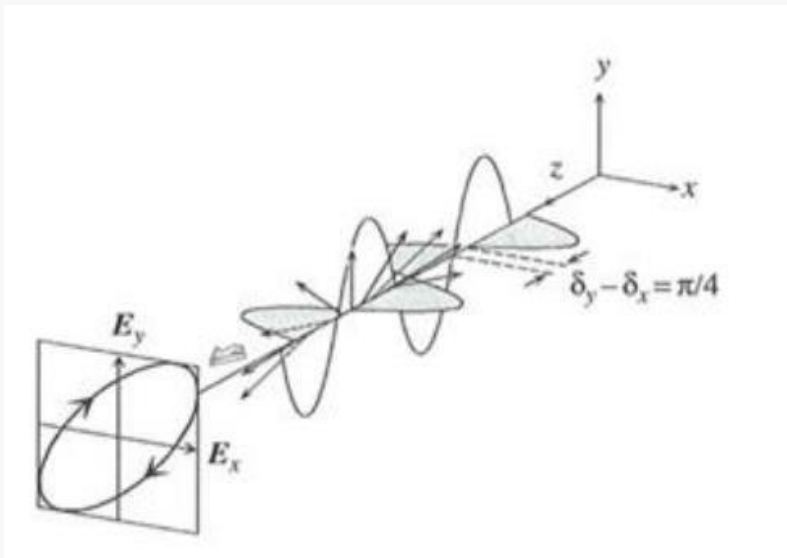
As far as the mathematical description is concerned, both linear and circular light may be considered to be special cases of **elliptically polarized** light or, more simply, **elliptical light**.

➤ We have learned that Two components

$$\vec{E}_x = E_0 \sin(kz - \omega t) \text{ and } \vec{E}_y = E_0 \sin(kz - \omega t + \phi)$$

➤ $\phi = 0, \pm\pi$ corresponds to linear polarization and $\phi = \pm\frac{\pi}{2}$ corresponds to circular polarization.

➤ What about the other values of ϕ , It is called **Elliptical Polarization** where the two components of $\vec{E}$ field have the different phase ϕ where $\phi \neq 0, \pm\frac{\pi}{2}, \pm\pi, \dots$

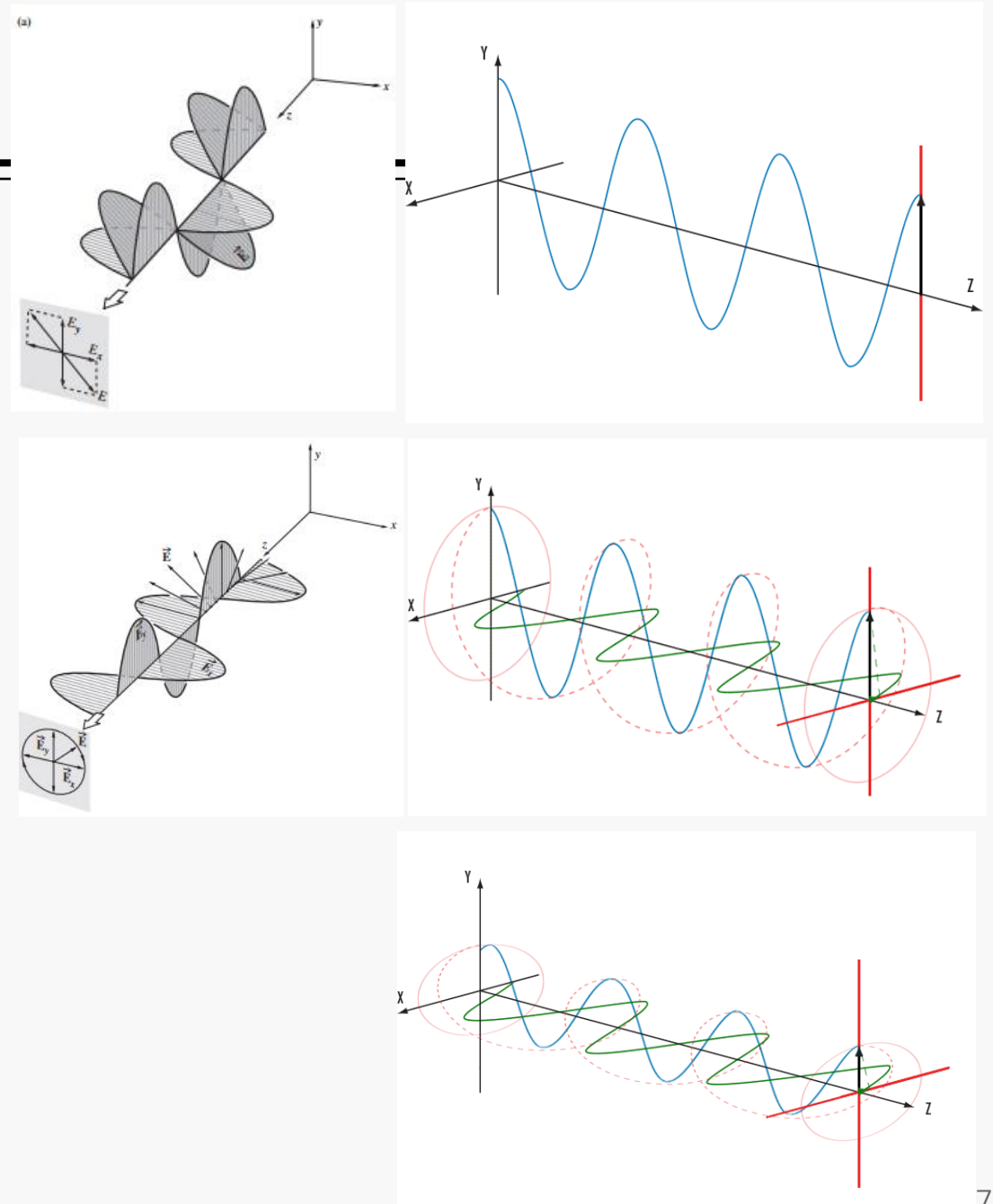


Recapitulate of Last Class

Both linear and circular light can be considered to be special cases of **elliptically polarized** light. We have learned that Two components

$$\vec{E}_x = E_0 \sin(kz - \omega t) \text{ and } \vec{E}_y = E_0 \sin(kz - \omega t + \phi)$$

- $\phi = 0, \pm\pi$ corresponds to linear polarization
- $\phi = \pm\frac{\pi}{2}$ corresponds to circular polarization.
- What about the other values of ϕ , It is called **Elliptical Polarization** where the two components of $\vec{E}$ field have the different phase ϕ where $\phi \neq 0, \pm\frac{\pi}{2}, \pm\pi, \dots$



Polarizers

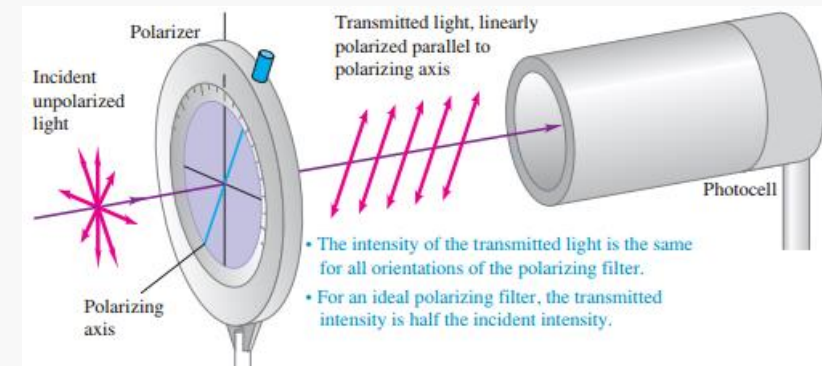
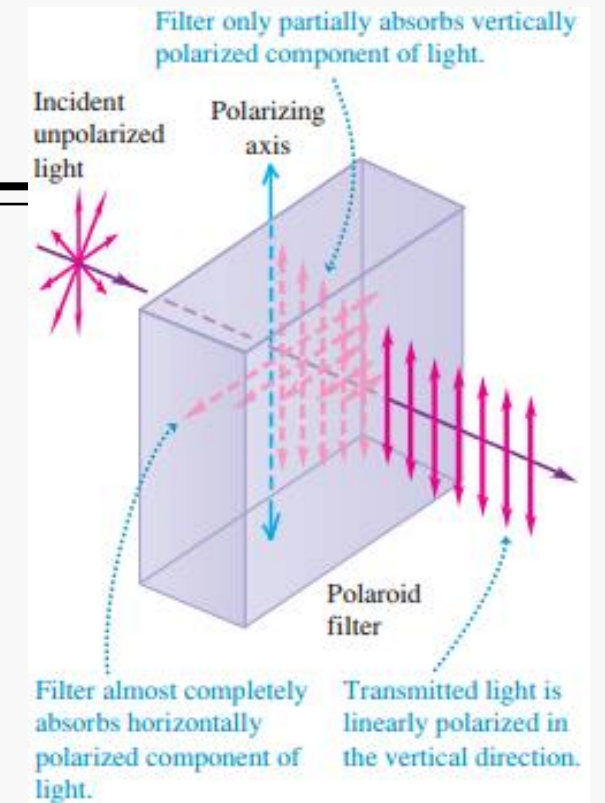
An optical instrument that separates light waves based on their polarization direction. All polarizers rely on an underlying physical mechanism to achieve polarization.

➤ The four main mechanisms are:

- Dichroism (Selective Absorption): Certain materials absorb light vibrating in one direction more strongly than another. (e.g., Polaroid filters)
 - Reflection: Light reflected at a specific angle (Brewster's angle) becomes partially or fully polarized. (e.g., glare reduction)
 - Scattering: Small particles can scatter light, and the scattered light is often polarized (e.g., blue sky).
 - Birefringence (Double Refraction): Some crystals split incoming light into two beams with different polarization directions and refractive indices. (e.g., calcite crystals)
- A crucial underlying property of all polarization methods is the presence of some form of asymmetry in the interaction with light. This asymmetry allows for the selective manipulation of different polarization states.

Understanding Polarized Light

- **Regular light:** Imagine light waves vibrating in all directions.
- **Polarized light:** This is like a rope shaking in only one direction, achieved using a polarizer. It filters light, allowing only specific vibrations to pass through.
- **Testing a Polarizer:**
 - Shine unpolarized light (e.g., from the sun) through the suspected polarizer.
 - Rotate the polarizer around a fixed axis (z-axis).
 - Monitor the light intensity with a detector (e.g., photocell).
- Key point: If the detector reading stays constant, the device is likely a linear polarizer. This is because unpolarized light has no preferential direction, so rotating the filter shouldn't affect the overall intensity.



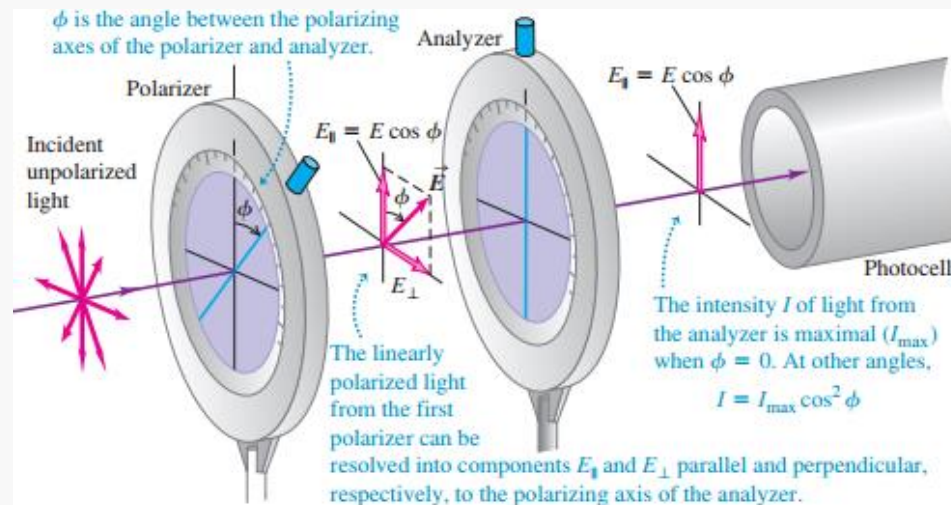
Malus's Law: Intensity of Transmitted Polarized Light

The Phenomenon: When plane-polarized light passes through a second polarizer (the analyzer), the intensity of the transmitted light depends on the relative orientation of their polarization axes.

Malus's Law (Published 1809 by Etienne Malus), Mathematically expressed as:

$$I(\theta) = \frac{c\varepsilon_0}{2} E_x^2 = \frac{c\varepsilon_0}{2} E_{01}^2 \cos^2 \theta \Rightarrow I(\theta) = I(0) \cos^2 \theta$$

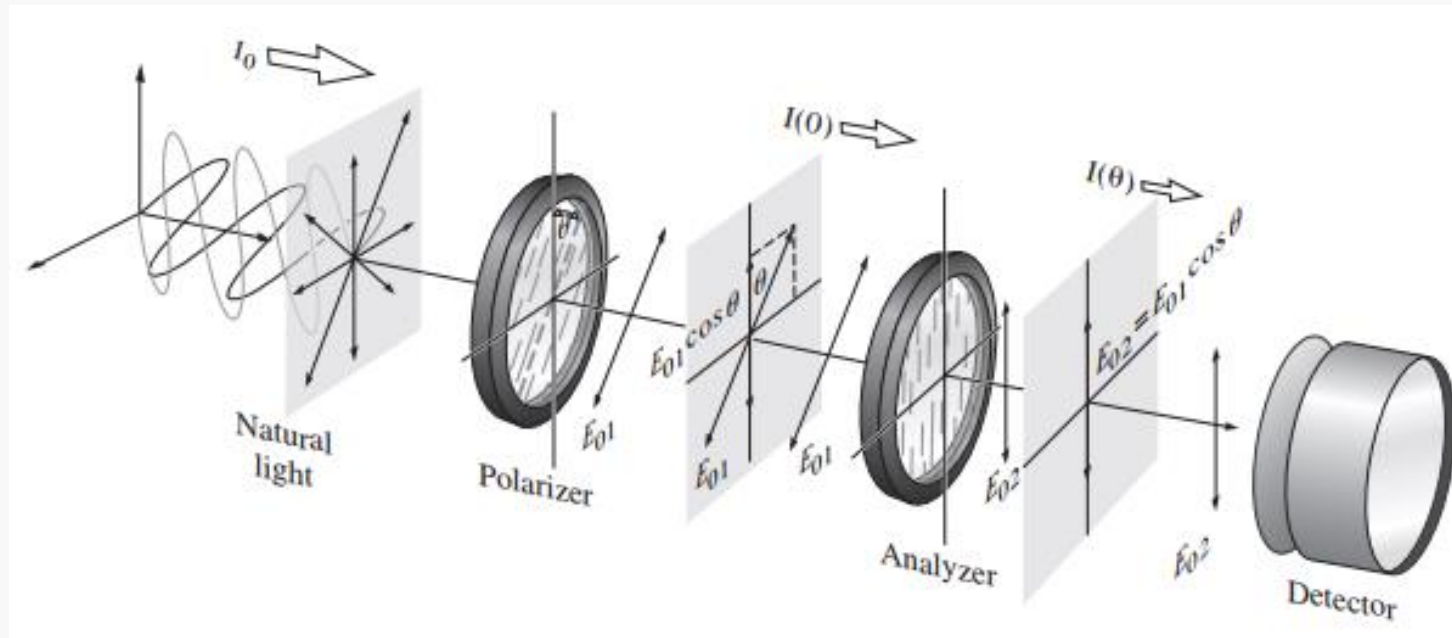
Where $I(0) = \frac{c\varepsilon_0}{2} E_{01}^2$, Maximum Intensity of the plane-polarized light incidence on the analyzer, $I(\theta) =$ Intensity of the light transmitted through the analyzer, and $\theta =$ Angle between the polarization axis of the incident light (from the polarizer) and the transmission axis of the analyzer.



Malus's Law: Intensity of Transmitted Polarized Light

Key points:

- If the axes are parallel ($\theta = 0^\circ$), the maximum intensity ($I = I_0$) is transmitted.
- When the axes are perpendicular ($\theta = 90^\circ$), the minimum intensity ($I = 0$) is transmitted, resulting in no light passing through.

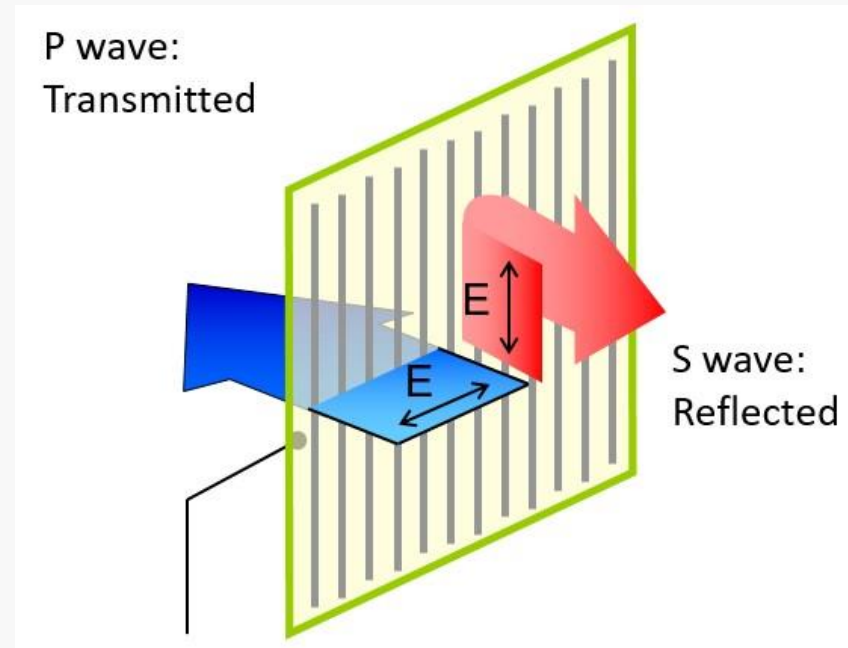


Example 3-4

Linearly polarized light (with an irradiance of 300 W m^2) aligned with its electric-field vector at $+55^\circ$ from the vertical impinges perpendicularly on an ideal sheet polarizer whose transmission axis is at $+10^\circ$ from the vertical. What fraction of the incoming light emerges?

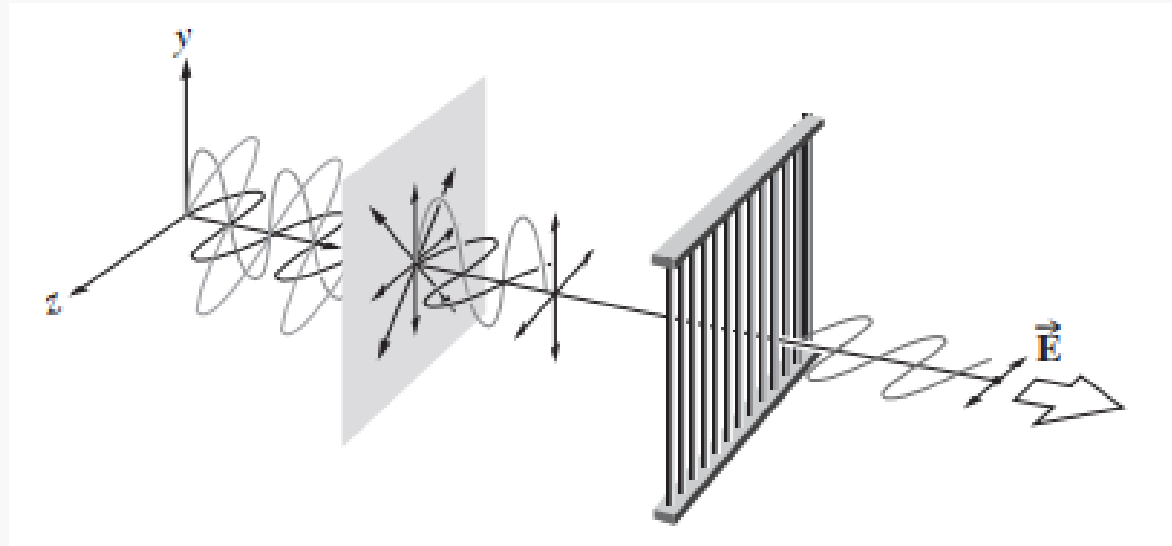
Wire-Grid Polarizer

Wire-Grid Polarizer works by allowing only light waves (electric field) with a specific orientation to pass through, while blocking those with a different orientation. When light passes through the grid, the electric field of the light wave interacts with the electrons in the metal wire.



Wire-Grid Polarizer

- **Perpendicular (P-polarized):** When the electric field is perpendicular to the wires, it interacts weakly, allowing the light to pass through with minimal resistance.
- **Parallel (S-polarized):** When the electric field is parallel to the wires, it interacts strongly, causing the electrons in the wires to oscillate, leading to Reflection and Absorption of light by the wire, absorption of light heating wire slightly.
- **Key takeaway:** The grid acts as a filter, transmitting mostly light with a perpendicular electric field (p-polarized) and blocking or absorbing light with a parallel electric field (s-polarized).



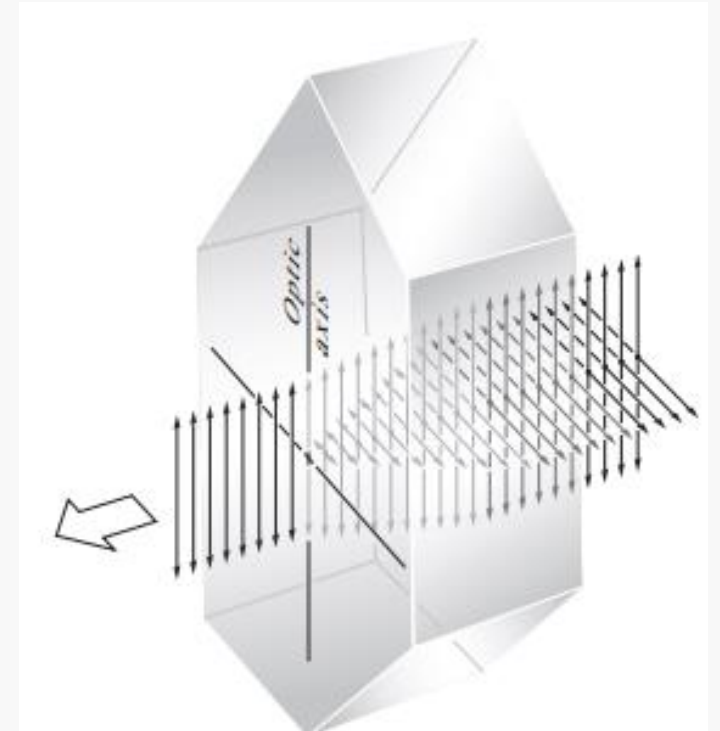
Dichroism: Polarization by Selective Absorption

- Certain materials exhibit dichroism, meaning they absorb light differently depending on the polarization direction of the light waves.
- This selective absorption leads to the transmission of light with a specific polarization and often results in different colors observed through the material depending on the polarization.
- Example: tourmaline, a mineral with a complex crystal structure : $(\text{NaFe}_3\text{B}_3\text{Al}_6\text{Si}_6\text{O}_{27}(\text{OH})_4)$, crystal have a special property: they show different colors depending on the direction of the polarization of light. This is due to their unique crystal structures.
- Light with an electric field parallel to the optic axis passes through easily.



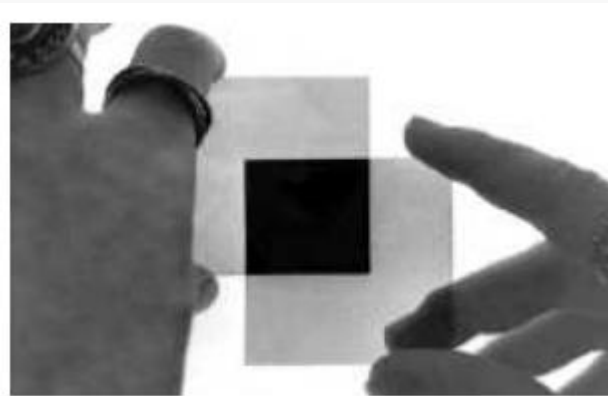
Dichroism: Anisotropic Absorption

- The unique crystal structure of dichroic materials creates an "optical axis."
- Light polarized parallel to the optic axis: Passes through the crystal with minimal absorption (easily transmitted).
- Light polarized perpendicular to the optic axis: Experiences stronger absorption (may be absorbed or transmitted with a different color due to selective wavelength absorption).



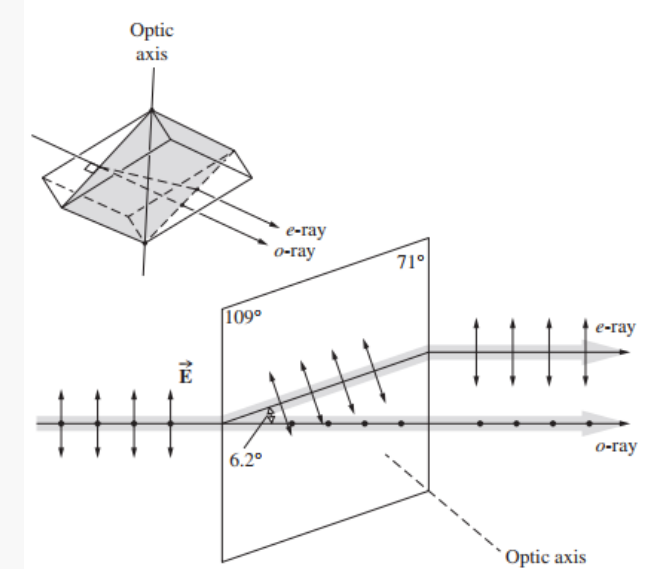
Polaroid

- In 1928, Edwin Herbert Land, a 19-year-old undergraduate at Harvard College, revolutionized polarization technology.
- He invented the first practical dichroic sheet polarizer, commercially known as the Polaroid J-sheet.
- Land's invention provided a relatively inexpensive and convenient way to produce large-area polarizers.
- This breakthrough had a profound impact on various technologies



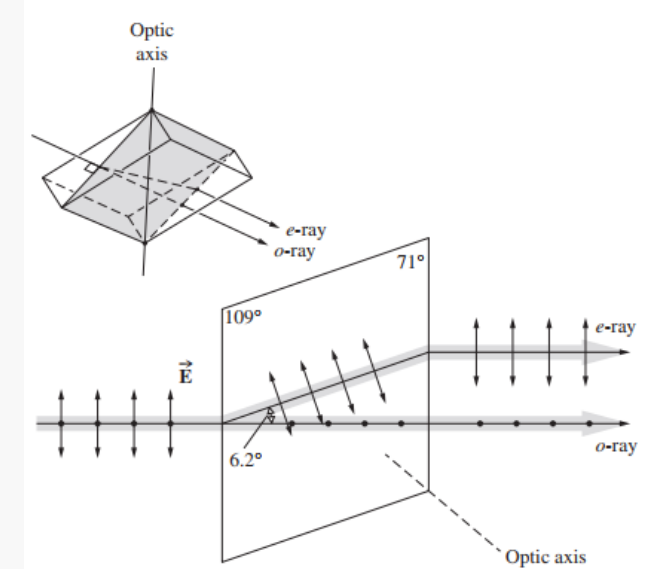
Birefringence: Double Refraction

- Birefringence is a captivating optical property where certain materials cause double refraction: splitting a single beam of unpolarized light into two distinct rays.
- This occurs because these materials exhibit two different refractive indices, depending on the polarization of the light.
- Examples of Birefringent Materials:
 - Calcite (Calcium Carbonate - CaCO_3) Crystal: A classic example that dramatically demonstrates double refraction.
 - Quartz Crystal: Another common birefringent crystal.
 - Stressed Plastics: Internal stresses can induce birefringence in normally isotropic plastics.



Birefringence: Double Refraction

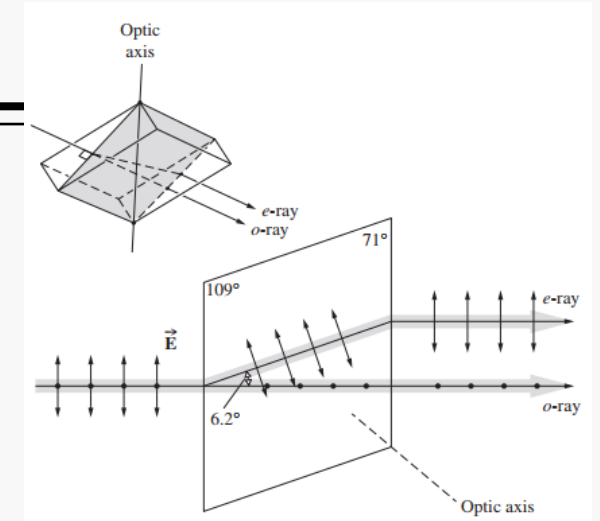
- **Double Refraction:** When light enters a birefringent material, it can be split into two separate rays:
 - Ordinary Ray (o-ray):
 - Polarized perpendicular to the optical axis.
 - Obeys Snell's Law of refraction at the interface.
 - Travels at a constant speed within the material (isotropic behavior in the crystal); straight-line path
 - Extraordinary Ray (e-ray):
 - Polarized in a direction not perpendicular to the optical axis (often varies with direction).
 - Does not strictly obey Snell's Law.
 - Speed varies with direction within the crystal (anisotropic behavior); may follow a curved path within the material due to varying refraction index.



Effects of Birefringence

When light enters a birefringent material, it splits into two rays (ordinary and extraordinary), each traveling at different speeds and directions. This can lead to several observable effects:

- A single object viewed through a birefringent material may appear doubled due to ray separation.
- The material may display interference colors or varying hues depending on the viewing angle and thickness — especially when viewed under polarized light..
- Applications: It has various applications:
 - ✓ Optical mineralogy: Used to identify minerals based on their birefringence properties.
 - ✓ Liquid crystal displays (LCDs): Birefringent materials play a crucial role in how LCDs manipulate light.
 - ✓ Photoelastic Stress analysis: Birefringence can be used to detect stress in materials.



Glan-Foucault Prism

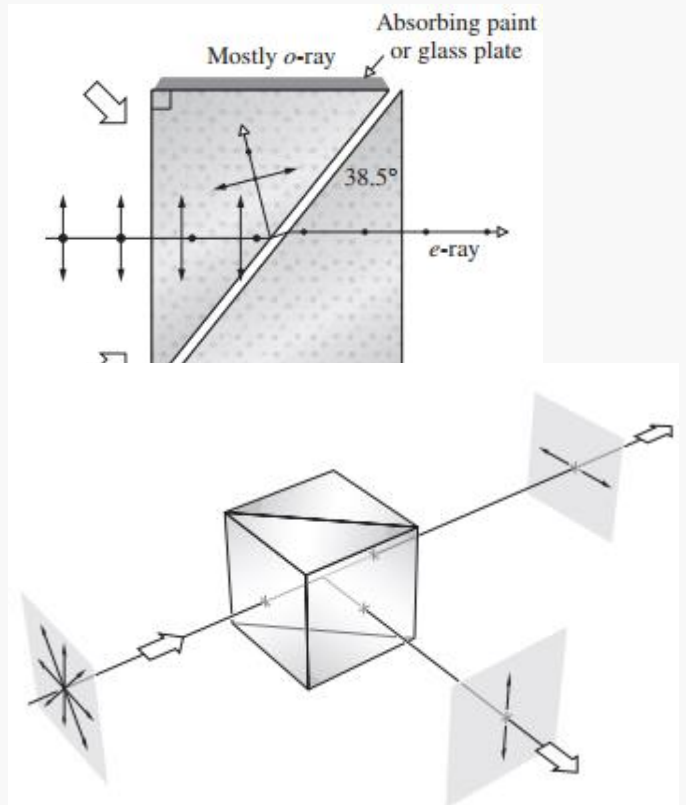
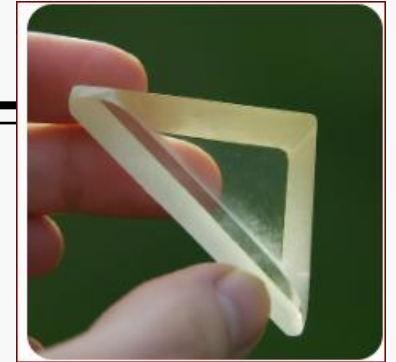
The Glan-Foucault prism is a precision optical device used to filter and control light polarization, commonly applied in laser optics and photonics.

Design:

- Constructed from two right-angled calcite prisms (a birefringent material).
- Separated by a thin air gap — this distinguishes it from the Glan-Thompson prism, which uses optical cement instead of air.)

How it works: When unpolarized light enters the calcite, Birefringence causes it to split into two rays:

- Ordinary Ray (o-ray): Polarized perpendicular to the optical axis
Totally internally reflected at the air gap and absorbed or deflected.
- Extraordinary Ray (e-ray): Polarized parallel to the optical axis.
Transmitted through the prism with minimal deviation — this is the useful output.



Birefringence

Refractive Indices of Some Uniaxial Birefringent Crystals ($\lambda_0 = 589.3 \text{ nm}$)

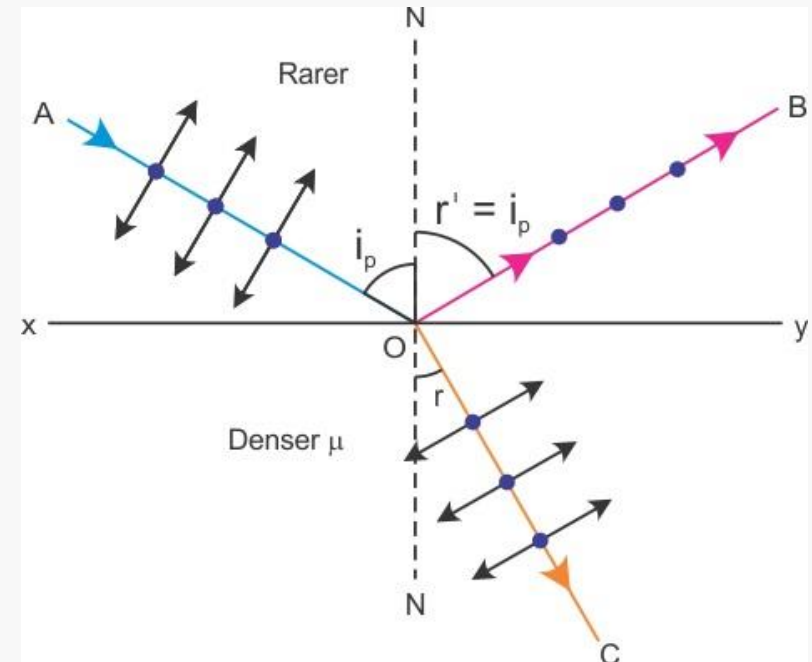
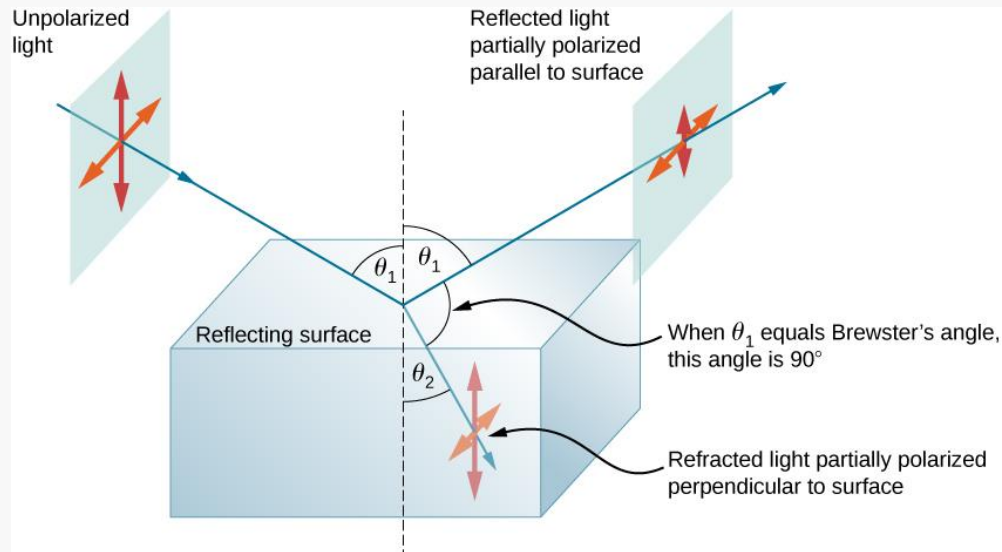
Crystal	n_o	n_e
Calcite	1.6584	1.4864
Ice	1.309	1.313
KDP	1.51	1.47
Lithium niobate	2.30	2.21
Quartz	1.5443	1.5534
Rutile (TiO ₂)	2.616	2.903
Sodium nitrate	1.5854	1.3369
Tourmaline	1.669	1.638

Polarization by Reflection

- Only the component polarized normal to the incident plane (parallel to the surface reflects).
- The particular angle-of-incidence for which this situation occurs is designated by θ_p and referred to as the **polarization angle** or **Brewster's angle**, whereupon $\theta_p + \theta_t = 90^\circ$

$$n_p \sin \theta_p = n_t \sin \theta_t \rightarrow \tan \theta_p = n_t/n_p$$

where $\theta_t = 90^\circ - \theta_p$



Polarization by Reflection



- Photographs of a window with different polarizer filter. The picture on the right is used a polarizer to eliminate the heavily reflected light from window. Here, the polarizer angle is aligned with the polarization angle of reflection from window.

Wave Plates

- The relative optical path length difference between the ordinary and extraordinary rays is given by:

$$\Lambda = d (n_o - n_e)$$

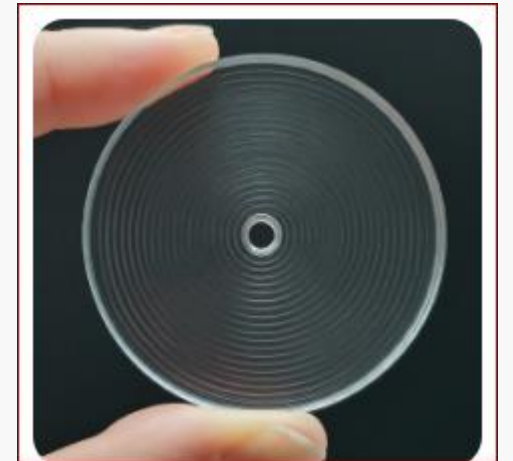
Where:

- d = thickness of the wave plate
- n_o, n_e = refractive indices of the ordinary and extraordinary rays, respectively

- The phase difference $\Delta\phi$, in radians, is:

$$\Delta\phi = k_o \Lambda = \frac{2\pi}{\lambda_o} d (n_o - n_e)$$

Where $\Delta\phi$ is the phase shift introduced between the two polarization components

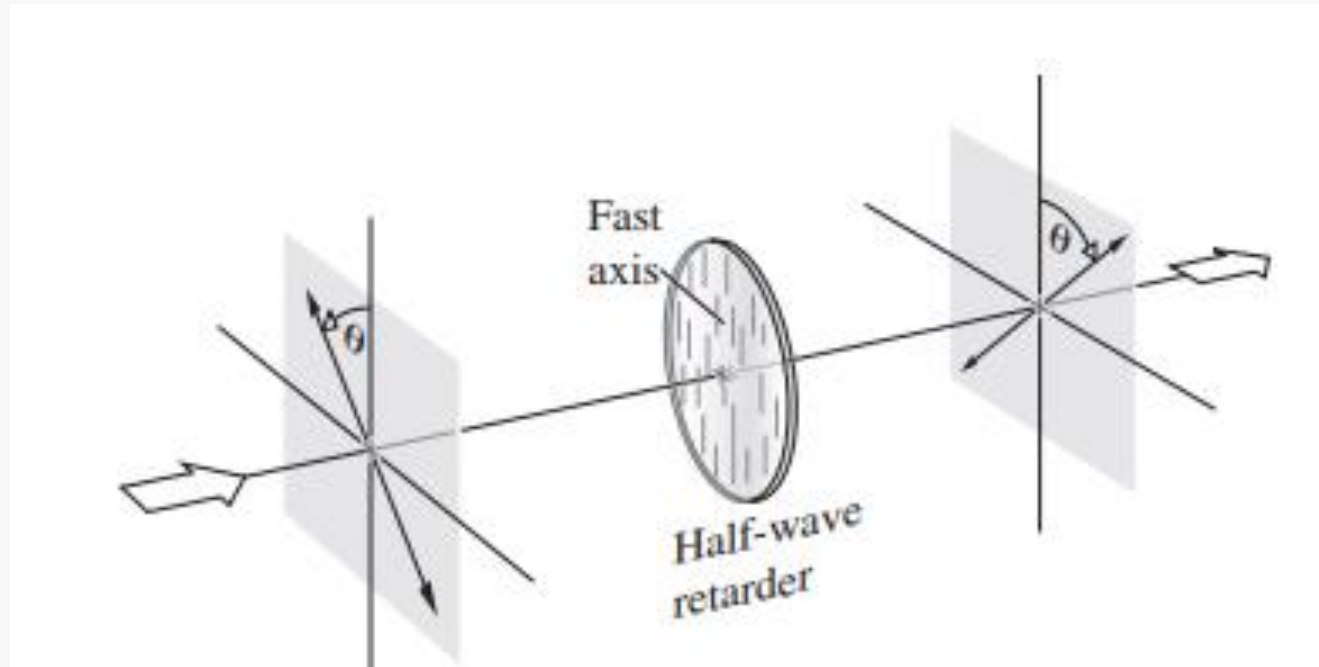


Full Wave Plates

- If $\Delta\phi$ is equal to 2π , the *relative retardation* is one wavelength; the *e*- and *o*-waves are back in-phase, and there is no observable effect on the polarization of the incident monochromatic beam.
- When the *relative retardation* $\Delta\phi$, which is also known as the **retardance**, is 360° the device is called a **full-wave plate** or **full-wave retarder**

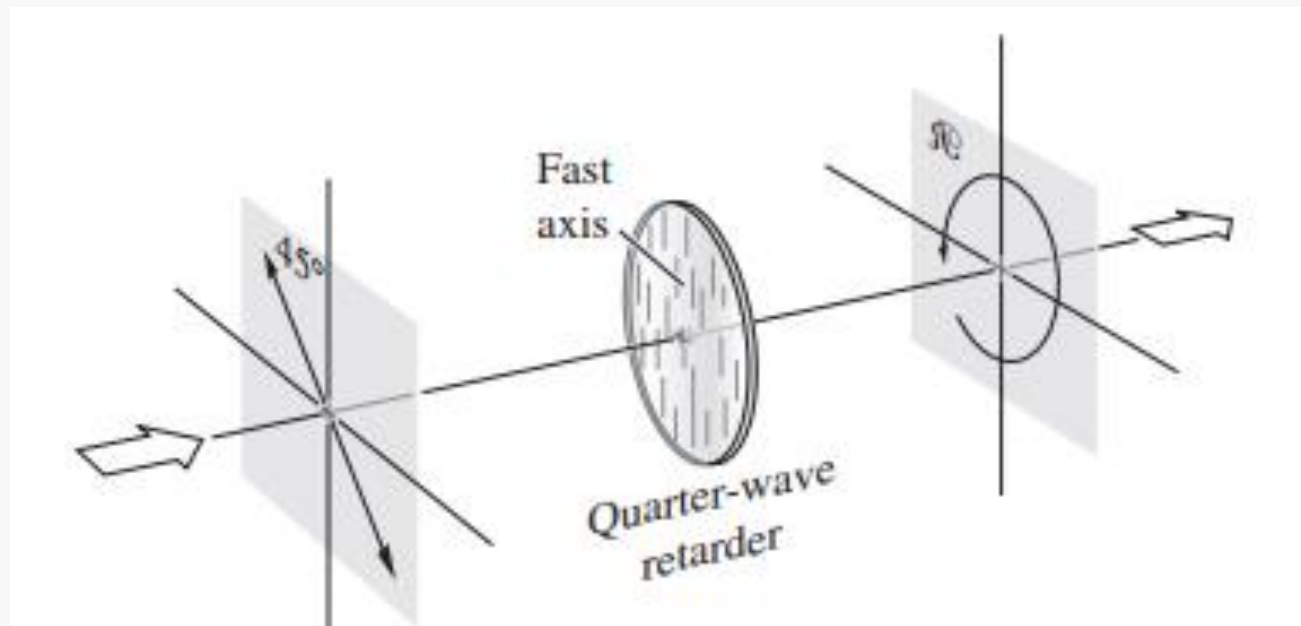
Half Wave Plates

- A retardation plate that introduces a relative phase difference of π radians, or 180° , between the o - and e -waves is known as a **half-wave plate** or **half-wave retarder**.



Quarter-Wave Plates

- The **quarter-wave plate** is an optical element that introduces a relative phase shift of $\Delta\varphi = \pi/2$ between the constituent orthogonal *o*- and *e*-components of a wave.

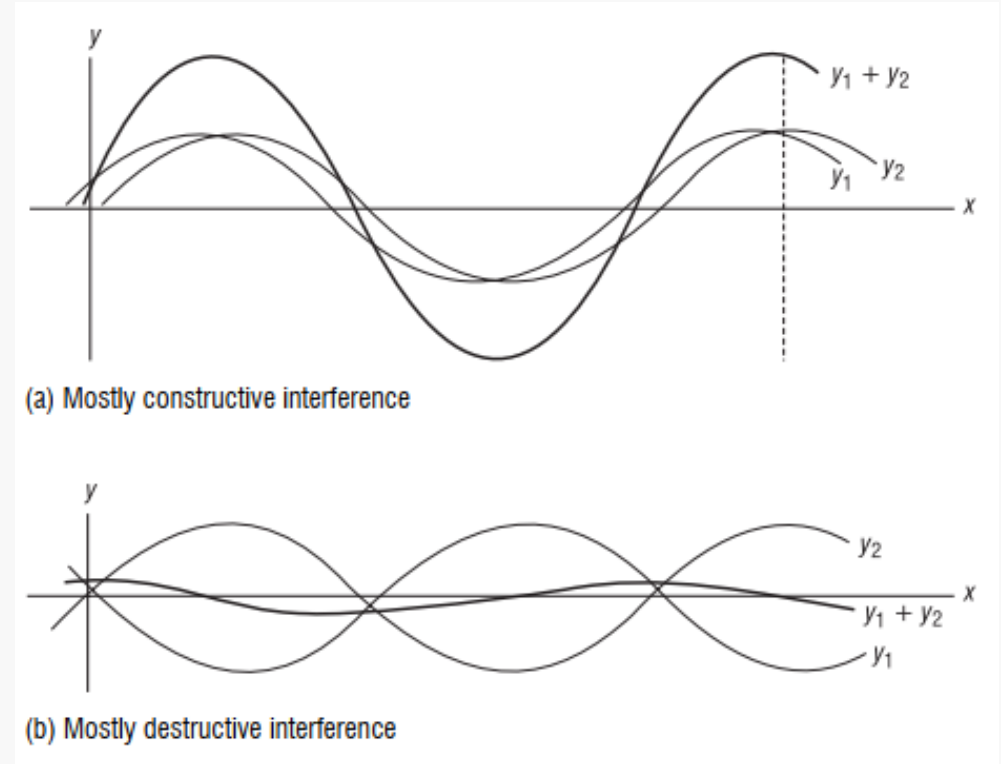


Superposition of Waves

- When two or more waves travel through the same region of space simultaneously, they do so independently of one another.
- At any point in space, the resulting electric field is the vector sum of the electric fields produced by each individual wave.
- This linear addition of fields is known as the superposition principle.

$$\vec{E}_{total} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$

This principle underlies many optical phenomena, including interference and diffraction.

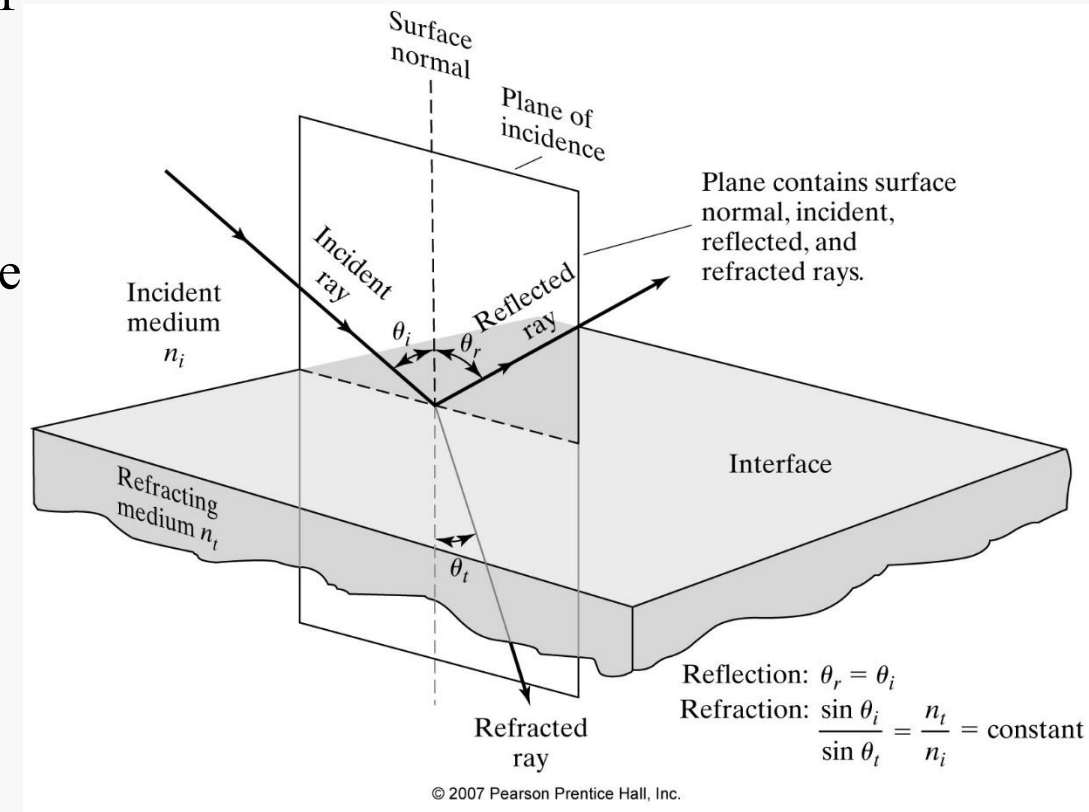


Geometrical Optics

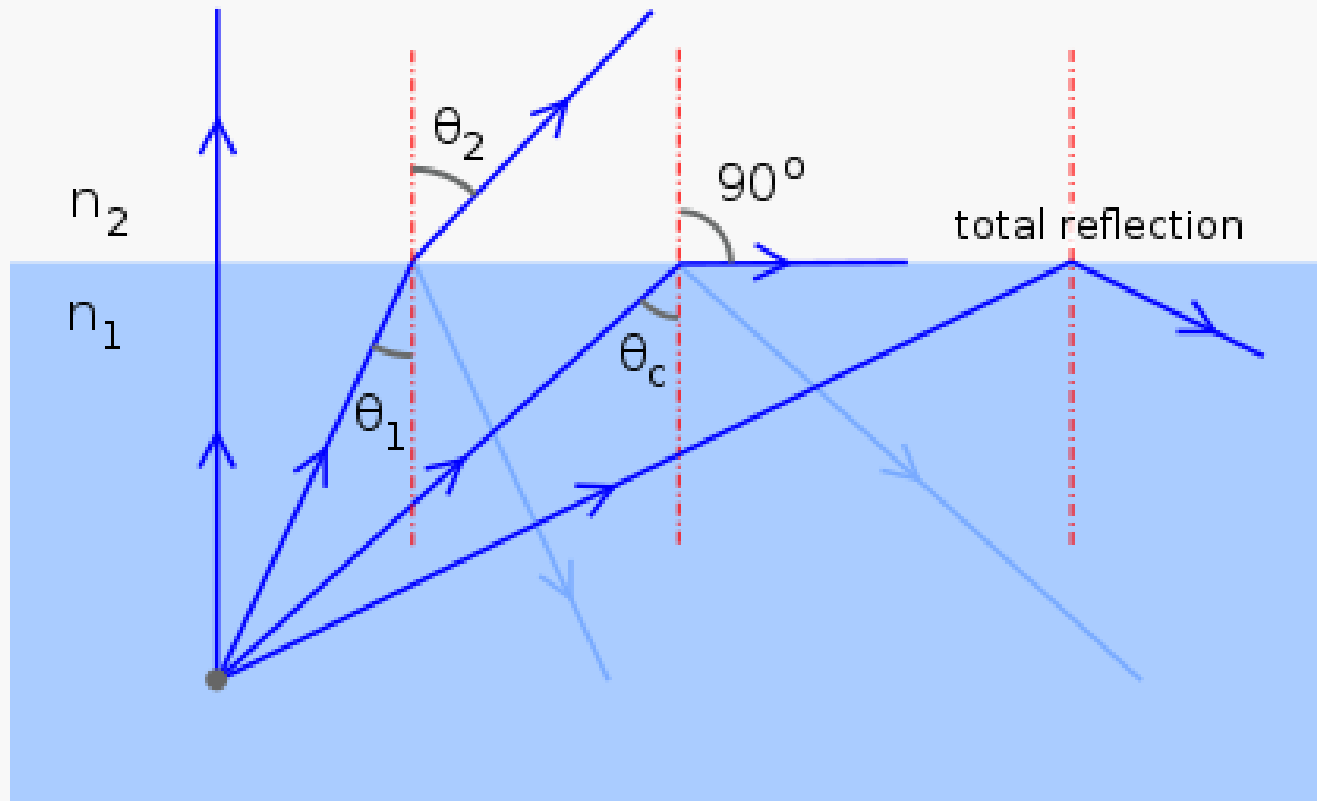
Wavelength of light is negligible compared with dimensions of the relevant components of the optical system.

Law of Reflection: When a ray is reflected at an interface dividing two optical media, the reflected ray remains within the plane of incidence, and the angle of reflection θ_r equals the angle of incidence θ_i .

Law of Refraction (Snell's Law): When a ray of light is refracted at an interface dividing two transparent media, the transmitted ray remains within the plane of incidence and the sine of the angle of refraction θ_t is directly proportional to the sine of the angle of incidence θ_i .



Total Internal Reflection

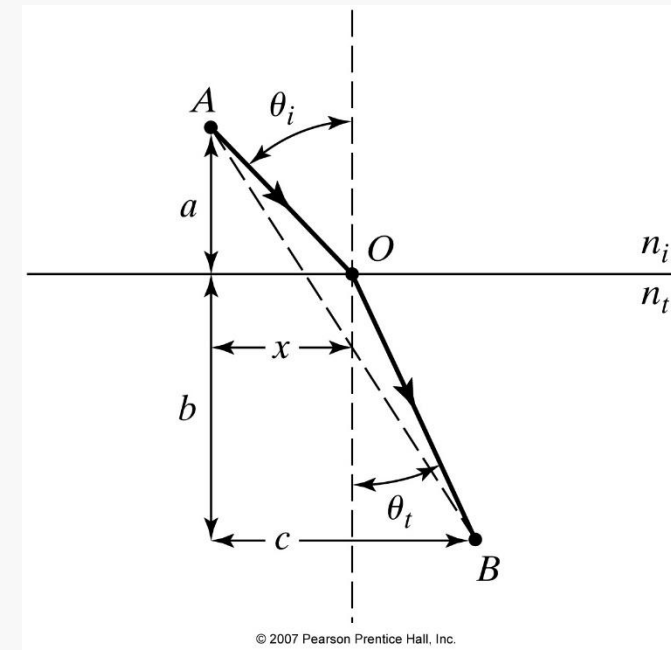
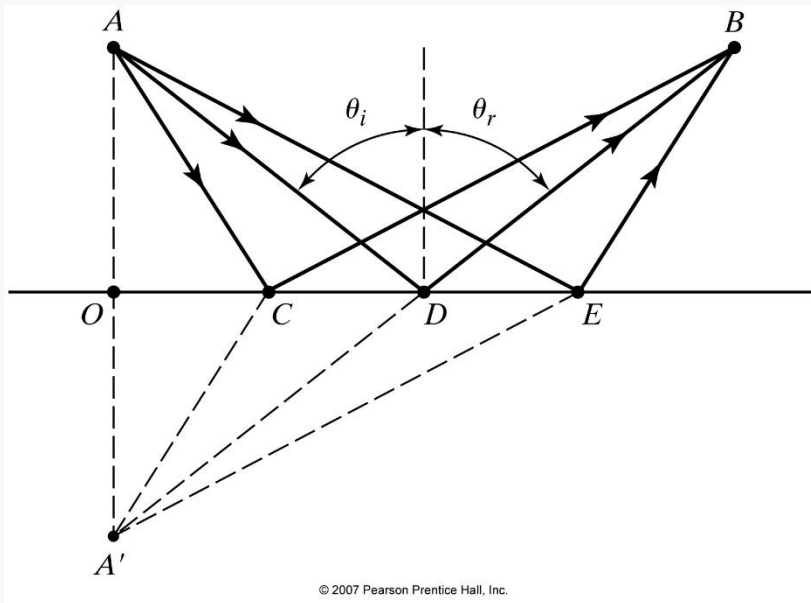


$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \text{ for } n_1 > n_2$$

Fermat's Principle

- A ray of light travels from point A to point B along the path that take the least amount of time.
- **Fermat's principle implies the principle of reversibility:** the path traveled by a light ray from A to B is the same as the path followed (in the opposite direction) from B to A.



Example 3.5

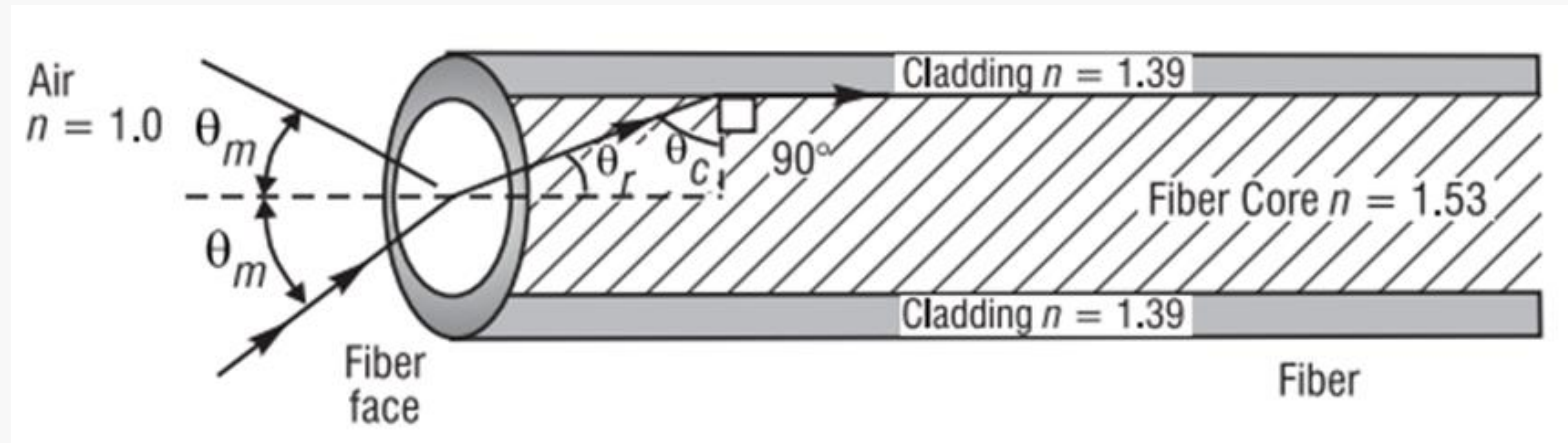
In a handheld optical instrument used under water, light is incident from water onto the plane surface of flint glass at an angle of incidence of 45° . Hints: refractive index of water and flint glass are 1.33 and 1.63.

- (a) What is the angle of reflection of light off the flint glass?
- (b) Does the refracted ray bend toward or away from the normal?
- (c) What is the angle of refraction in the flint glass?

Example 3.5

A step-index fiber 0.0025 inch in diameter has a core index of 1.53 and a cladding index of 1.39. See drawing. Such clad fibers are used frequently in applications involving communication, sensing, and imaging.

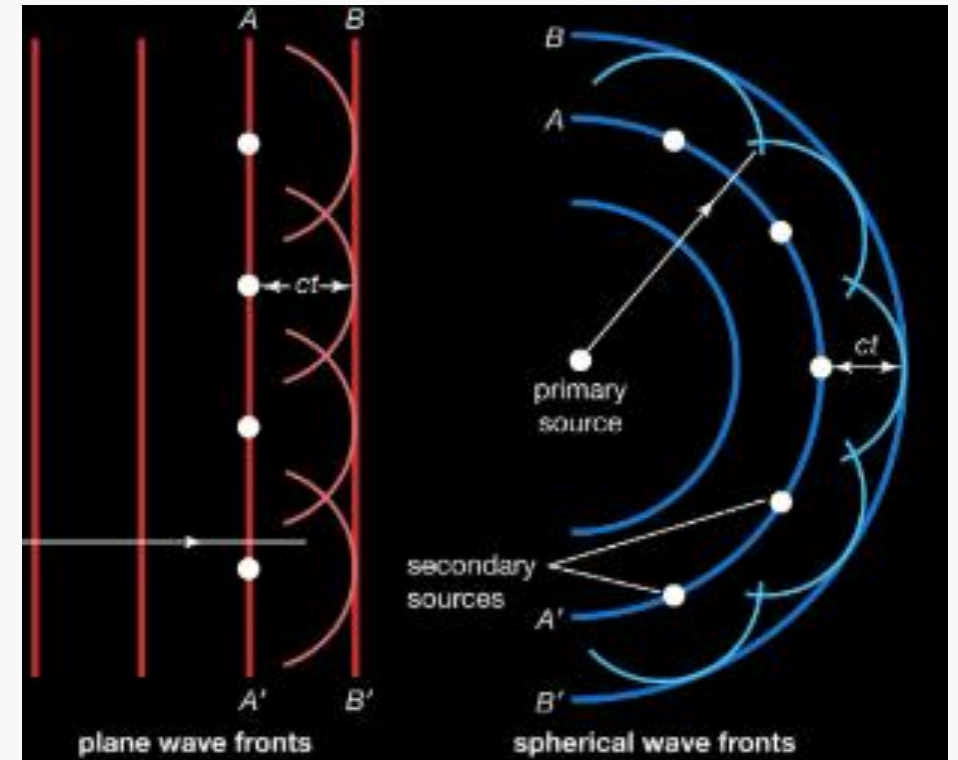
What is the maximum acceptance angle θ_m for a cone of light rays' incident on the fiber face such that the refracted ray in the core of the fiber is incident on the cladding at the critical angle?



Huygens' Principle: Wave Propagation

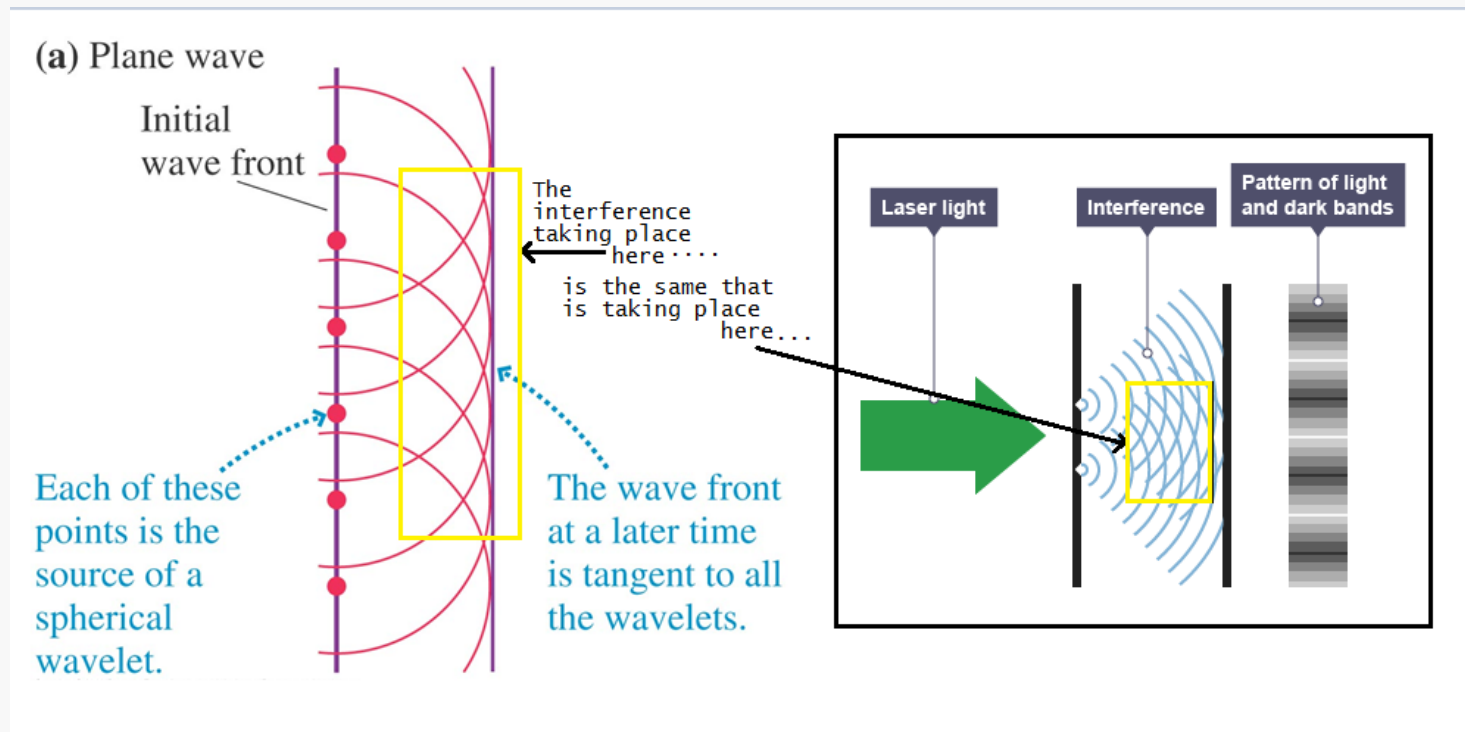
Huygens' Principle, proposed by **Dutch physicist Christiaan Huygens in 1678**, is a foundational concept for understanding how waves propagate through space. It is especially useful in explaining wave phenomena such as diffraction, reflection, and refraction...

- **Core Idea:** Every point on a wavefront can be considered a source of secondary spherical wavelets that spread out in all directions.
- **Wavelets:** These secondary waves, called wavelets, spread out in all directions from their source point.
- **Speed:** The wavelets travel at the same speed as the original wave (e.g., the speed of light for electromagnetic waves).



Huygens' Principle: Wave Propagation

- **New Wavefront:** The new position of the wavefront at a later time (t) is determined by the envelope of all these secondary wavelets.
- **Envelope:** The envelope is the surface (or line in 2D) that is tangent to all the wavelets at that instant in time.



Diffraction: Waves Bending Around Obstacles

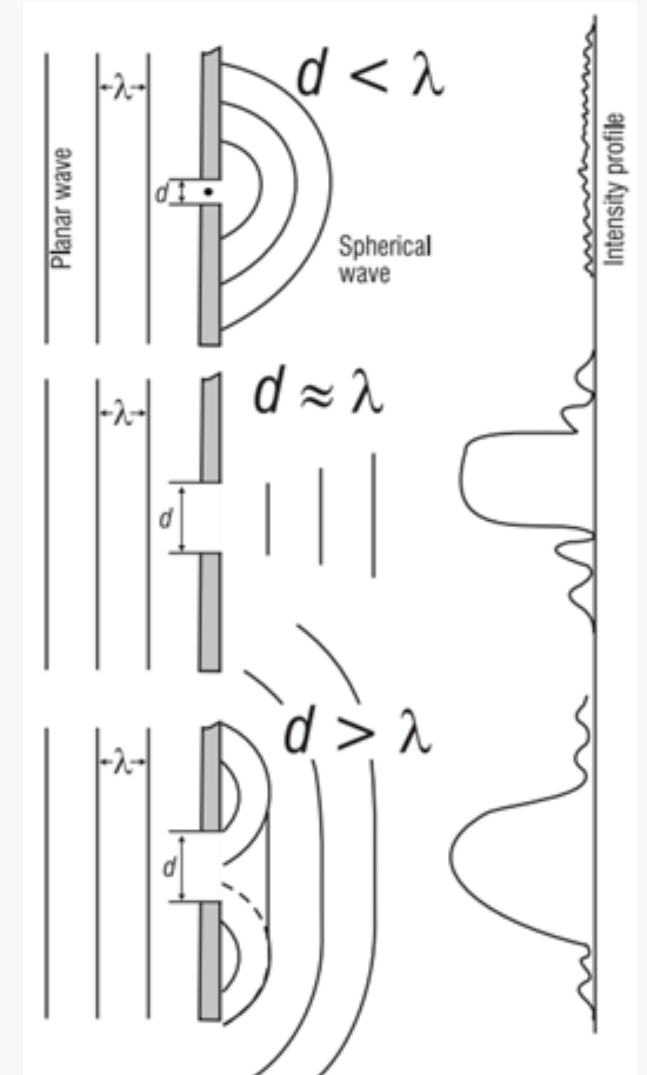
Diffraction is the bending and spreading of waves when they encounter an obstacle or pass through a narrow opening.

It occurs with all types of waves: light (EM), sound, water, and even quantum waves (matter waves such as electrons, neutrons).

Key Factors Influencing Diffraction:

Wavelength vs. obstacle size: Greater diffraction occurs when the wavelength is comparable to or larger than the size of the opening or obstacle.

Wave type: Longer wavelengths (e.g., sound waves) diffract more easily than shorter wavelengths (e.g., visible light).



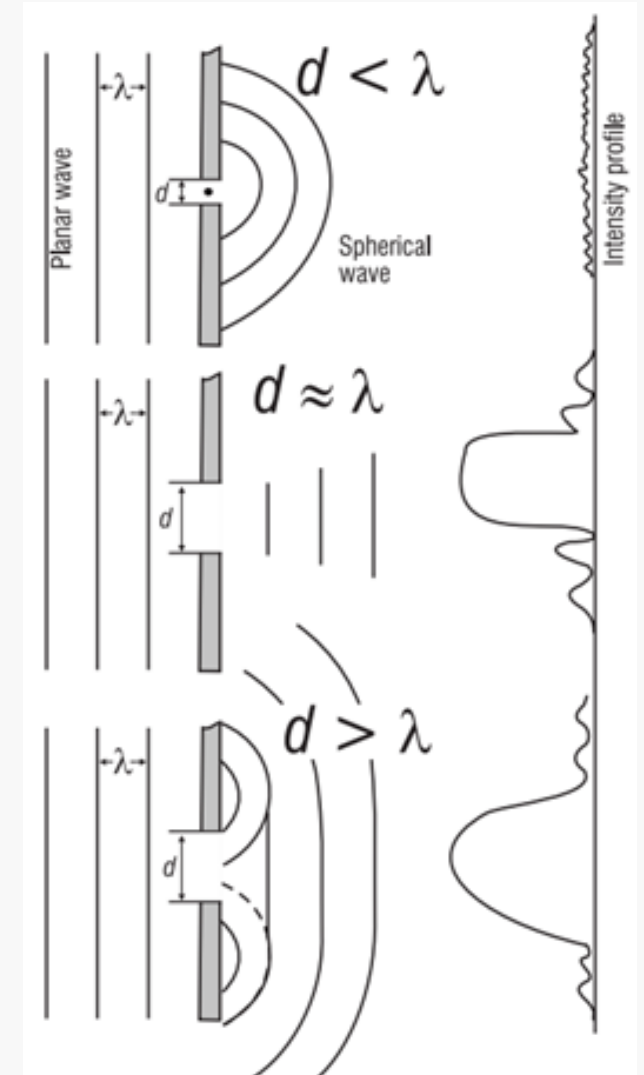
Types of Diffraction (Based on Observation Distance)

Fresnel Diffraction (Near-Field):

- Occurs when the observation screen is close to the diffracting object.
- Characterized by complex, non-uniform diffraction patterns.
- The wavefronts arriving at the screen are still significantly curved.

Fraunhofer Diffraction (Far-Field):

- Occurs when the observation screen is effectively at an infinite distance from the diffracting object.
- Achieved practically by using lenses to focus parallel rays.
- Characterized by simpler, well-defined diffraction patterns with parallel wavefronts incident on the screen.

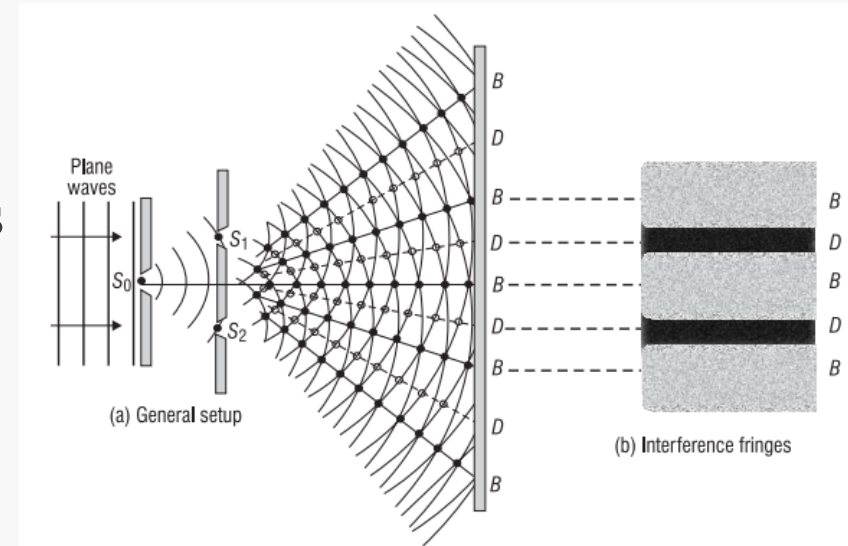


Wave Interference: Constructive and Destructive Effects

Interference occurs when two or more waves overlap (superimpose), resulting in a new wave pattern. The resulting amplitude depends on how the waves align.

Phase Difference and Interference:

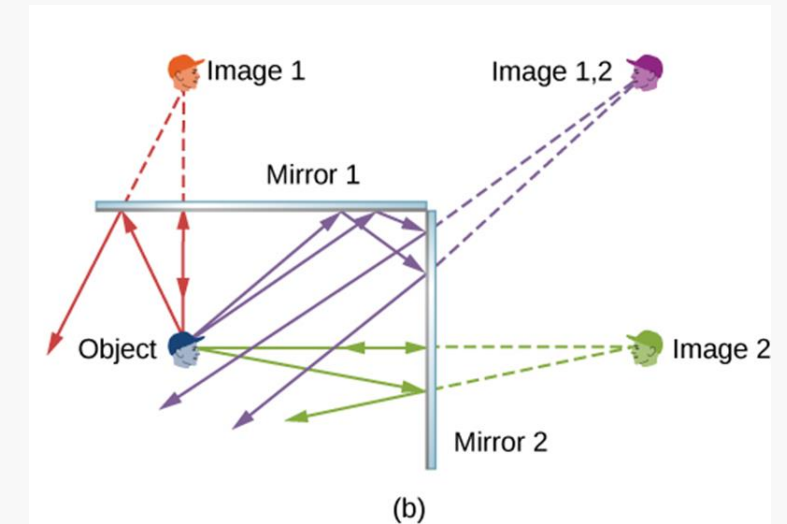
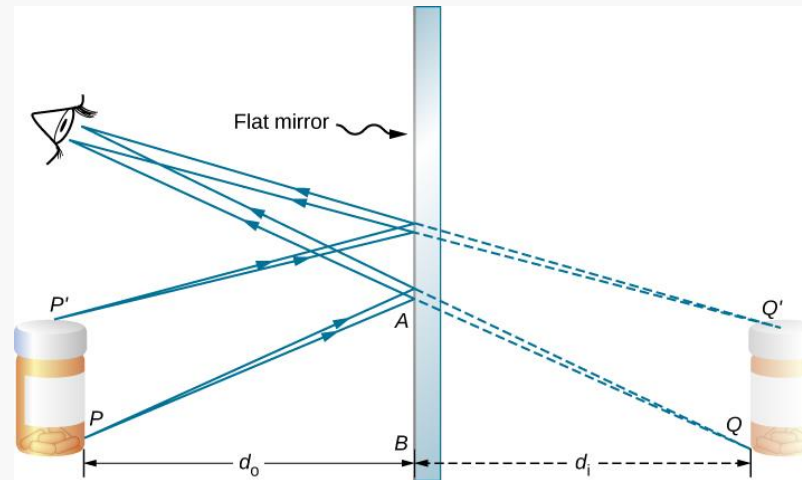
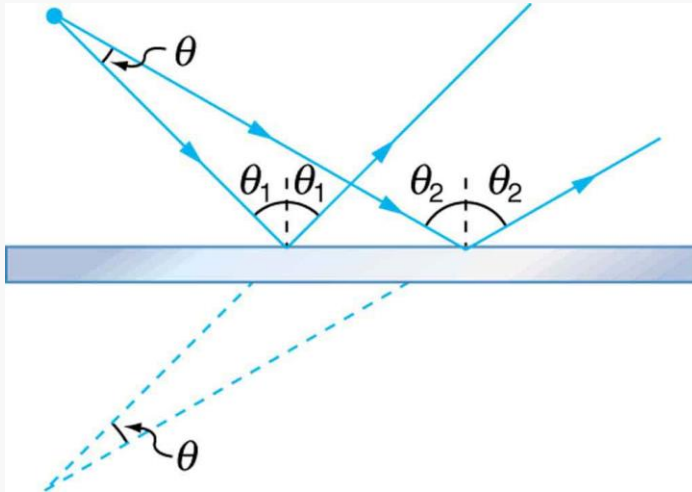
- Phase Difference: The relative shift between the wave cycles of two interacting waves.
 - 0° Phase Difference (In-Phase)
 - Constructive Interference
 - Amplitudes add, producing a larger wave.
 - 180° Phase Difference (Out-of-Phase)
 - Destructive Interference
 - Amplitudes cancel, producing a reduced or zero-amplitude wave.



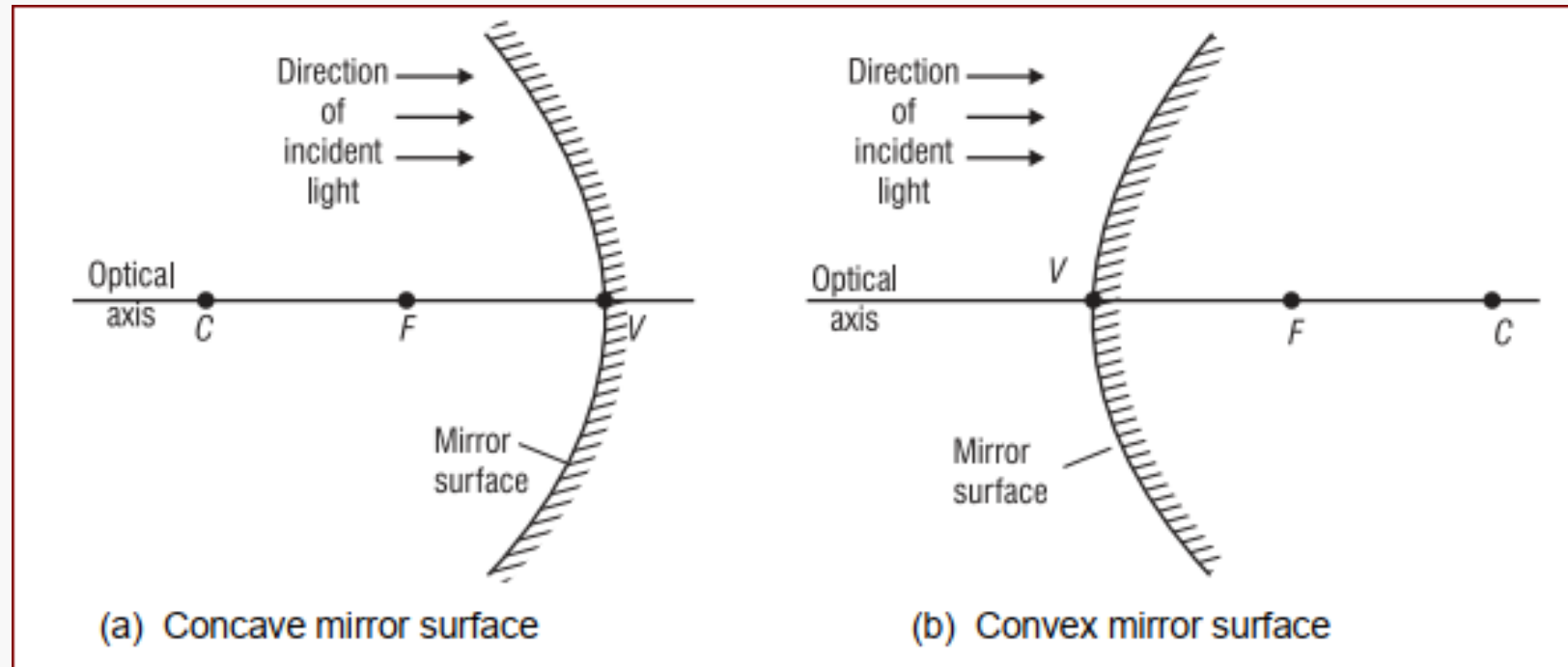
Demonstrated by **Thomas Young** in the famous **Double-Slit Experiment (1801)**.

Images Formed by Plane Mirrors

Images are formed when many nonparallel rays from a given point on a source are reflected from the mirror surface and form a corresponding image point.



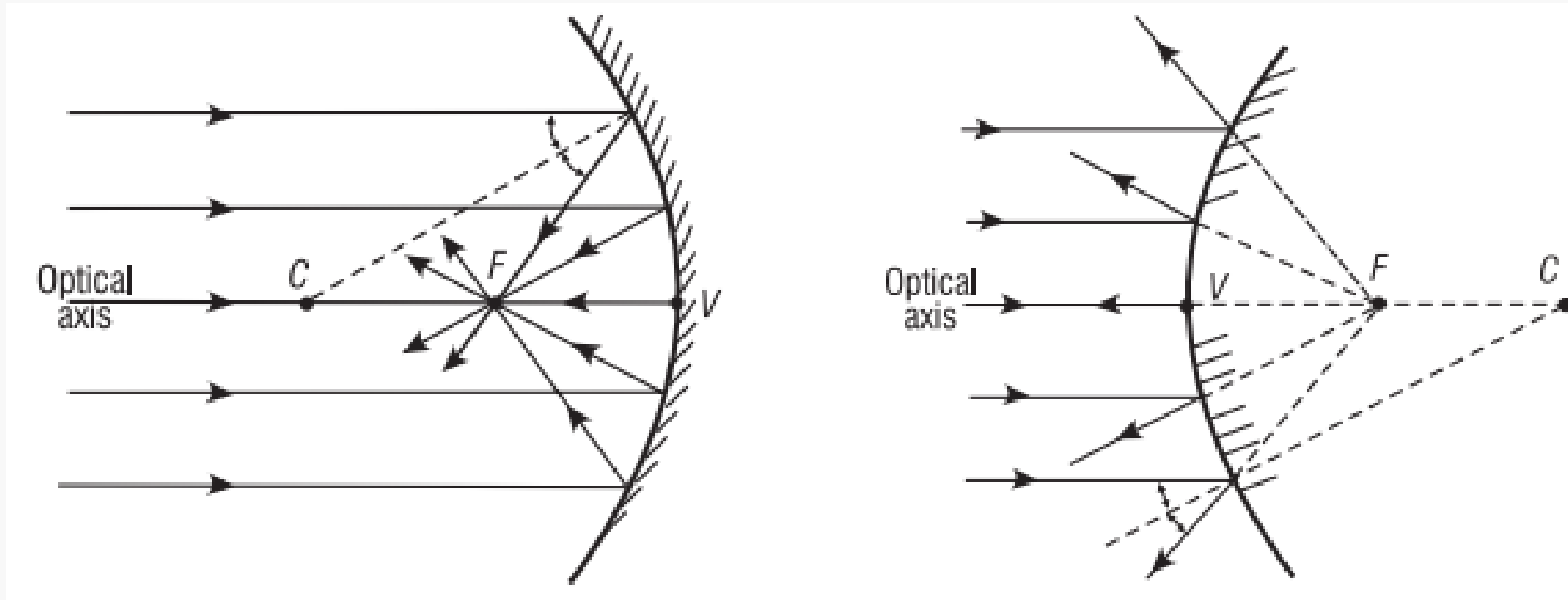
Spherical Mirrors



Defining Point concave and convex mirrors.

Spherical Mirrors

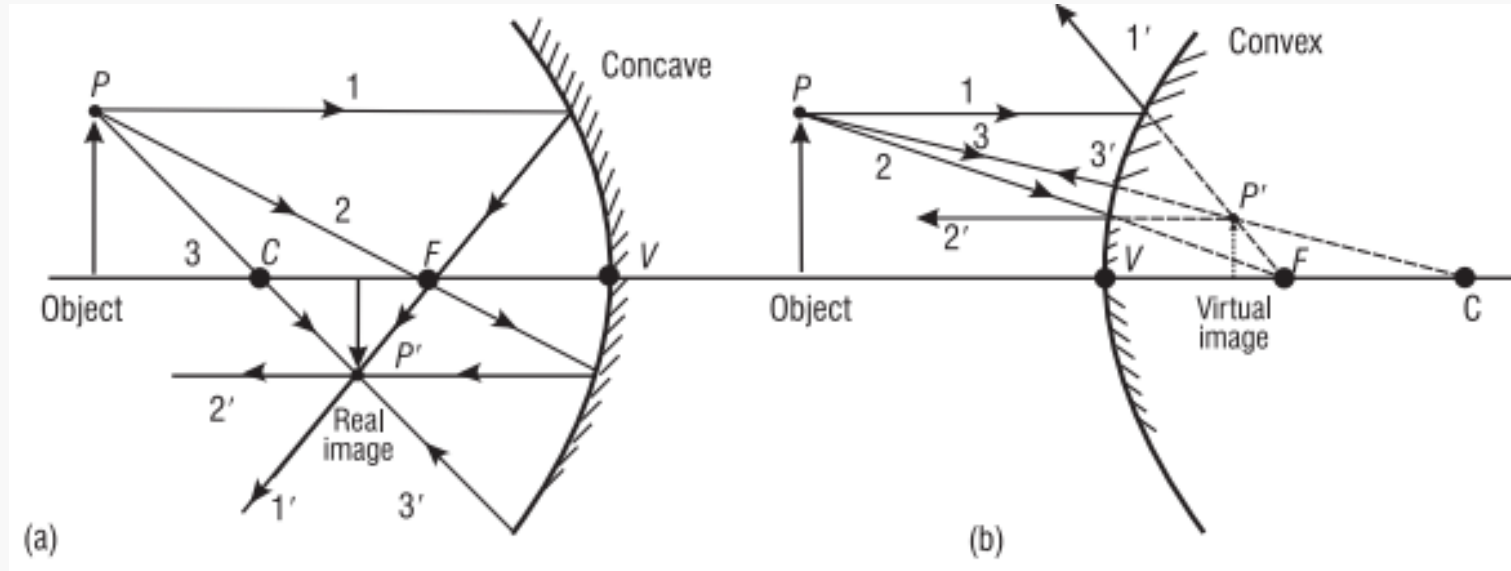
Concave and convex mirrors: Parallel rays and focal points.



Where C is center of curvature and F is focal point.

Spherical Mirrors

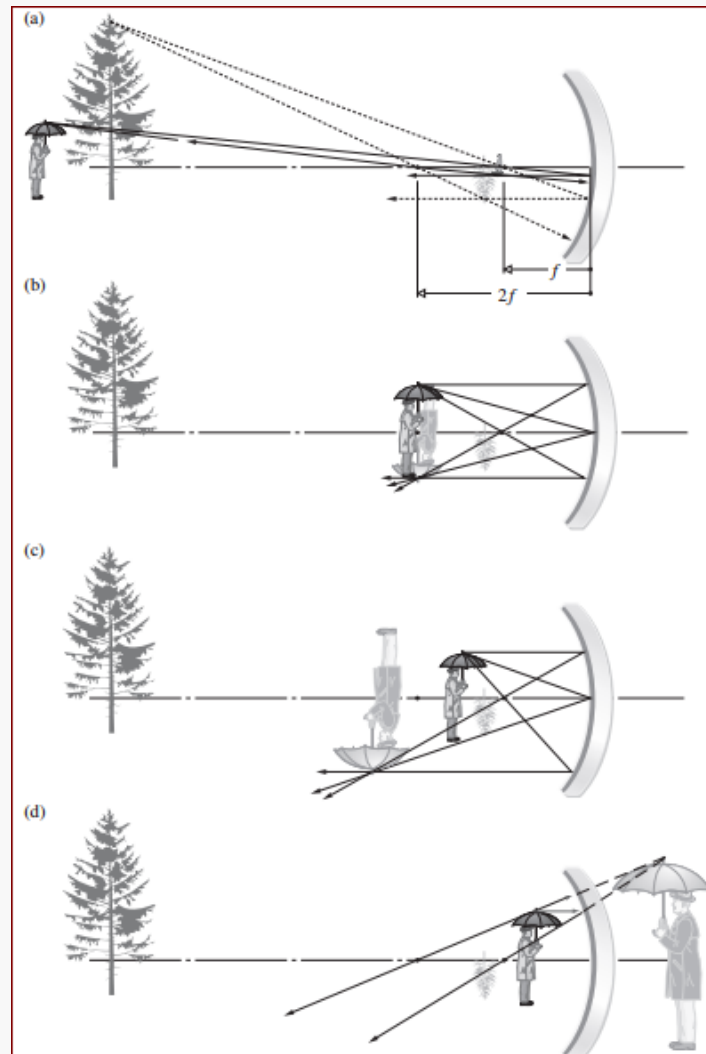
Concave and convex mirrors: Parallel rays and focal points.



Where C is center of curvature and F is focal point.

The mirror equation is $\frac{1}{d_o} + \frac{1}{d_i} = \frac{2}{r} = \frac{1}{f}$ and Magnification m to be $m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$; $m > 0$: image is upright and $m < 0$: image is inverted.

Images of Real Objects Formed by Spherical Mirrors



Concave				
Object		Image		
Location	Type	Location	Orientation	Relative Size
$\infty > s_o > 2f$	Real	$f < s_i < 2f$	Inverted	Minified
$s_o = 2f$	Real	$s_i = 2f$	Inverted	Same size
$f < s_o < 2f$	Real	$\infty > s_i > 2f$	Inverted	Magnified
$s_o = f$		$\pm \infty$		
$s_o < f$	Virtual	$ s_i > s_o$	Erect	Magnified
Convex				
Object		Image		
Location	Type	Location	Orientation	Relative Size
Anywhere	Virtual	$ s_i < f $, $s_o > s_i $	Erect	Minified

Refraction at Spherical Surface

From Fermat's Principle that Optical Path Length (OPL) will be stationary.

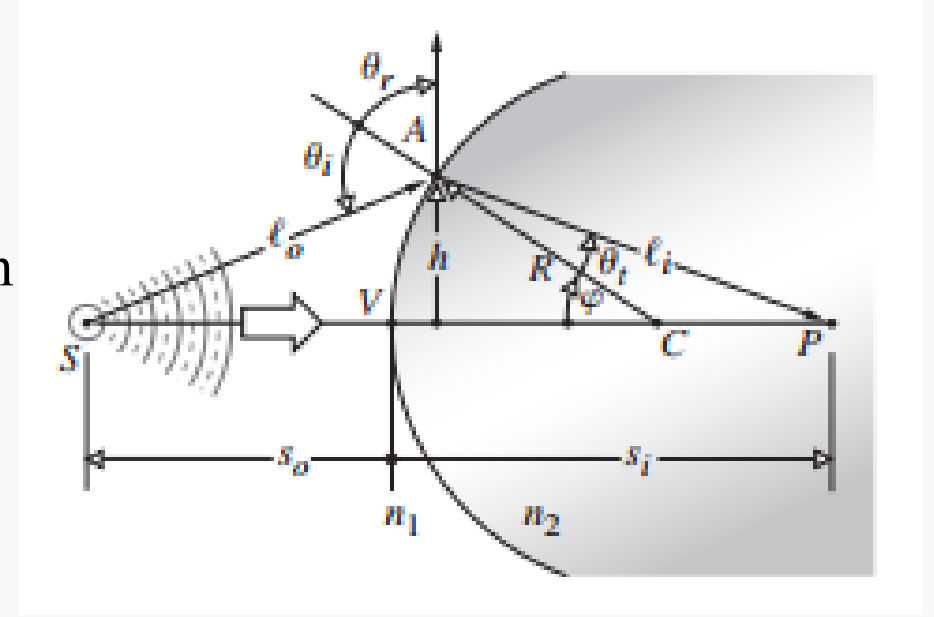
$$OPL = n_1 \ell_o + n_2 \ell_i$$

Let's apply law of cosines in triangles SAC and ACP with that $\cos \varphi = -\cos(180^\circ - \varphi)$, we find that

$$\ell_o = [R^2 + (s_o + R)^2 - 2R(s_o + R) \cos \varphi]^{1/2}$$

Similarly

$$\ell_i = [R^2 + (s_i - R)^2 + 2R(s_i - R) \cos \varphi]^{1/2}$$



$$OPL = n_1 [R^2 + (s_o + R)^2 - 2R(s_o + R) \cos \varphi]^{1/2} + n_2 [R^2 + (s_i - R)^2 + 2R(s_i - R) \cos \varphi]^{1/2}$$

Because R is constant and φ is positive variable, we can set $\frac{d(OPL)}{d\varphi} = 0$

Refraction on Spherical Surface

$$\frac{n_1 R(s_o + R) \sin \varphi}{2 \ell_o} - \frac{n_i R(s_o - R) \sin \varphi}{2 \ell_i} = 0$$

We get,

$$\frac{n_1}{\ell_o} + \frac{n_2}{\ell_i} = \frac{1}{R} \left(\frac{n_2 s_i}{\ell_i} - \frac{n_1 s_o}{\ell_o} \right)$$

Where $\cos \varphi \sim 1$ and $\sin \varphi \sim \varphi$.

Now ℓ_o and ℓ_i yields: $\ell_o \approx s_o$ and $\ell_i \approx s_i$

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{(n_2 - n_1)}{R}$$

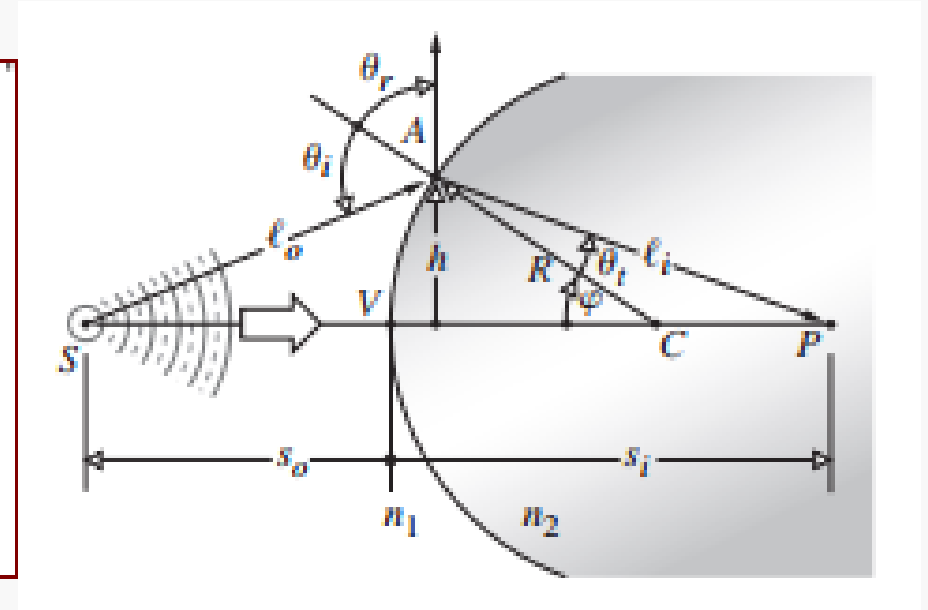
Refraction on Spherical Surface

- The approximation ($\cos \varphi \approx 1$ and $\sin \varphi \approx \varphi$) is called the *first-order theory*, there is *third-order theory* $\sin \varphi \approx \varphi - \frac{\varphi^3}{3!}$, but we will look into this if we get sufficient time.
- h are appropriately small are known as **paraxial rays** ($\sin \varphi \approx \varphi$). The emerging wave front segment corresponding to these paraxial rays is essentially spherical and will form a “perfect” image at its center P located at s_i .
- Gauss, in 1841, was the first to give a systematic exposition of the formation of images under the above approximation, and the result is variously known as *first-order*, *paraxial*, or **Gaussian Optics**.

Refraction on Spherical Surface

Sign Convention for Spherical Refracting Surfaces and Thin Lenses* (Light Entering from the Left)

s_o, f_o	+ left of V
x_o	+ left of F_o
s_i, f_i	+ right of V
x_i	+ right of F_i
R	+ if C is right of V
y_o, y_i	+ above optical axis



Refraction on Spherical Surface

Let's consider, imaged at infinity ($s_i = \infty$)

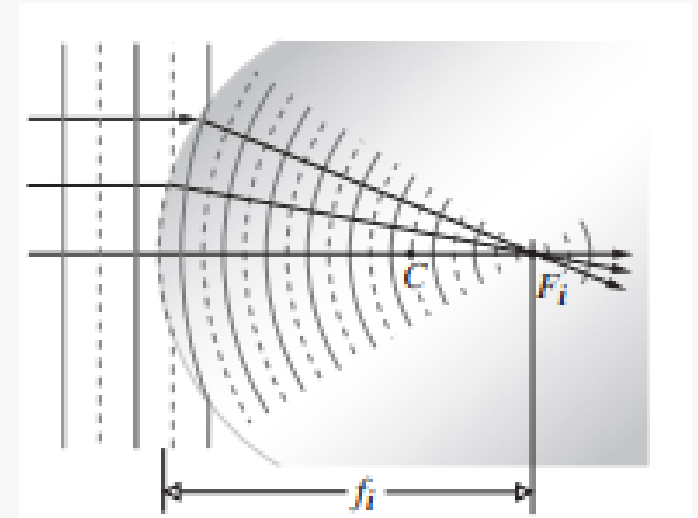
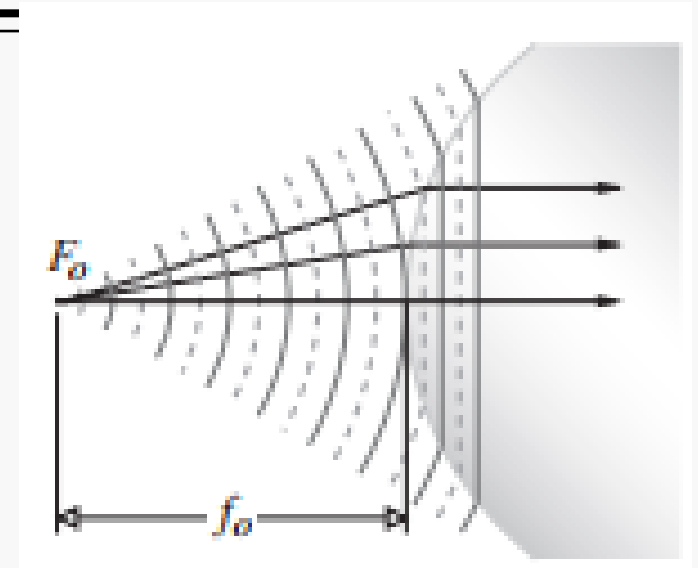
$$\frac{n_1}{s_o} + \frac{n_2}{\infty} = \frac{(n_2 - n_1)}{R}$$

Then $s_o = f_o$ (**1st focal length** or **object focal length**),
and so that

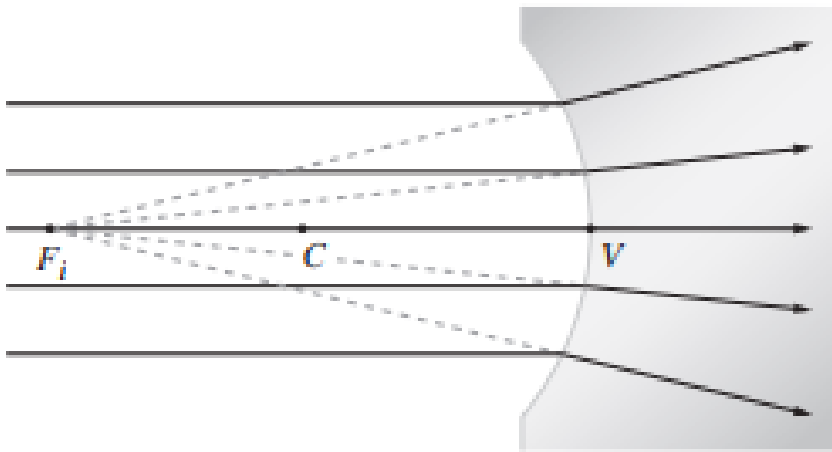
$$f_o = \frac{n_1}{n_2 - n_1} R$$

Let's consider the **image focus** or **2nd focus**, ($s_o = \infty$)

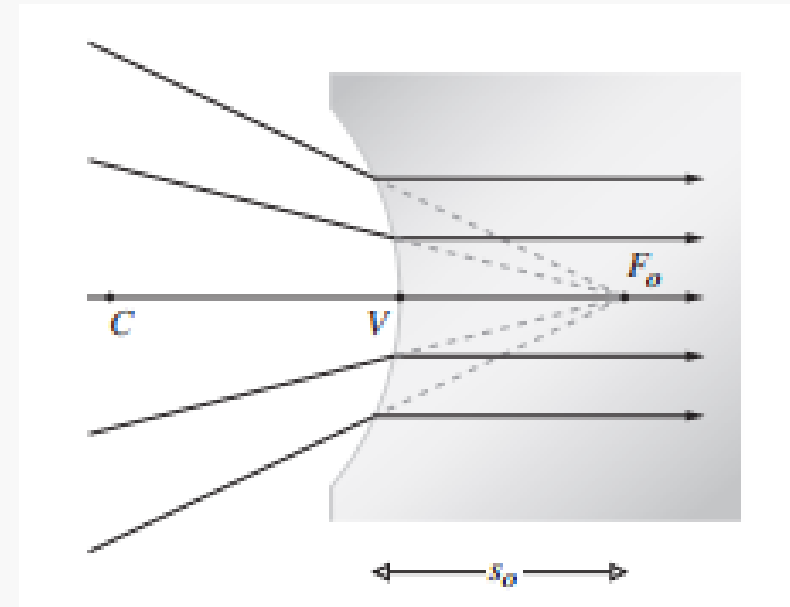
$$f_i = \frac{n_2}{n_2 - n_1} R$$



Refraction on Spherical Surface



A virtual image point



A virtual object point

Example 3.6

A long horizontal flint-glass ($n_g = 1.800$) cylinder is 20.0 cm in diameter and has a convex hemispherical left end ground and polished onto it. The device is immersed in ethyl alcohol ($n_a = 1.361$) and a tiny LED is located on the central axis in the liquid 80.0 cm to the left of the vertex of the hemisphere. Locate the image of the LED. What would happen if the alcohol was replaced by air?

Thin Lens

1st surface

$$\frac{n_m}{s_{o1}} + \frac{n_l}{s_{i1}} = \frac{(n_l - n_m)}{R_1}$$

And 2nd surface

$$\frac{n_l}{(-s_{il} + d)} + \frac{n_m}{s_{i2}} = \frac{(n_m - n_l)}{R_2}$$

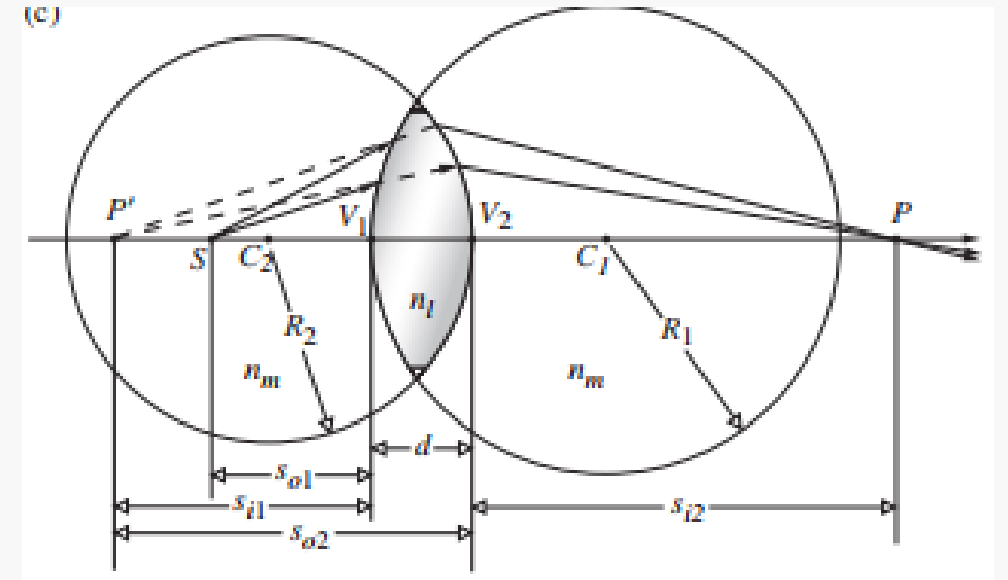
Here $n_l > n_m$ and R_2 is negative

$$\frac{n_m}{s_{o1}} + \frac{n_m}{s_{i2}} = (n_l - n_m) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) + \frac{n_l d}{(s_{il} - d)s_{il}}$$

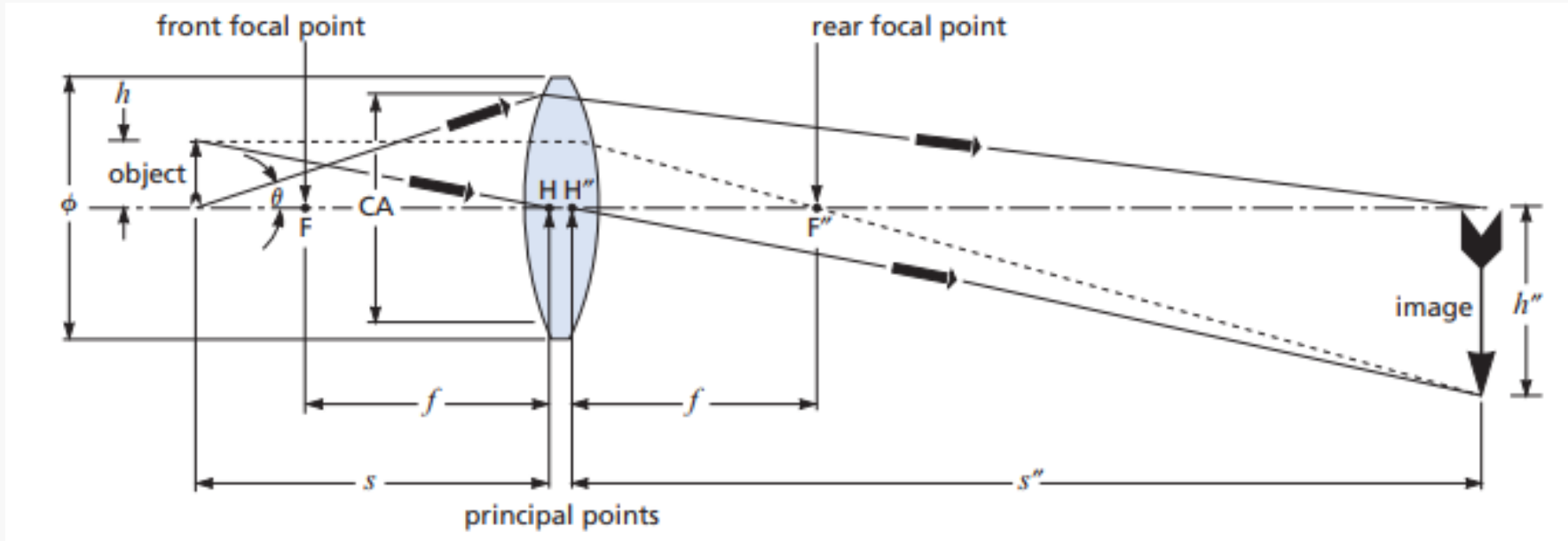
If the lens is thin enough ($d \rightarrow 0$)

we have the very useful **Thin-Lens Equation**, often referred to as the **Lensmaker's Formula**:

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f} = \frac{n_l - n_m}{n_m} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$



Thin Lens



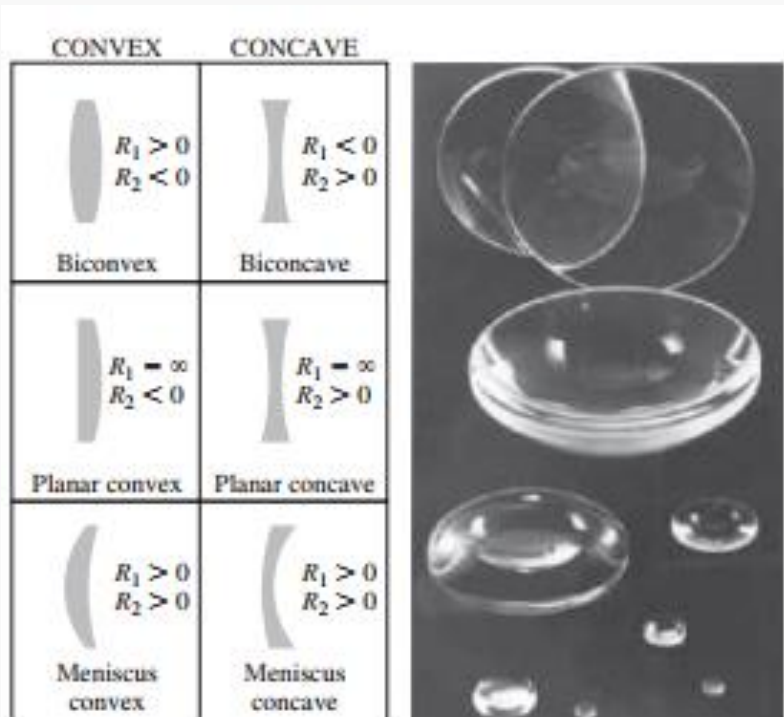
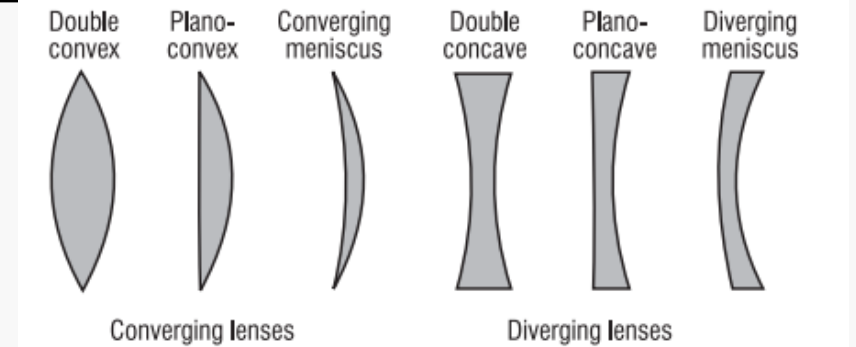
Equations for this lens:

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f} \quad \text{and} \quad \frac{1}{f} = \frac{n_g - n_m}{n_m} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

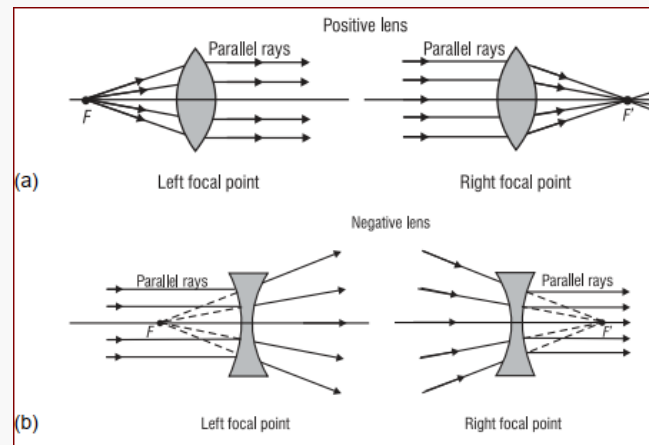
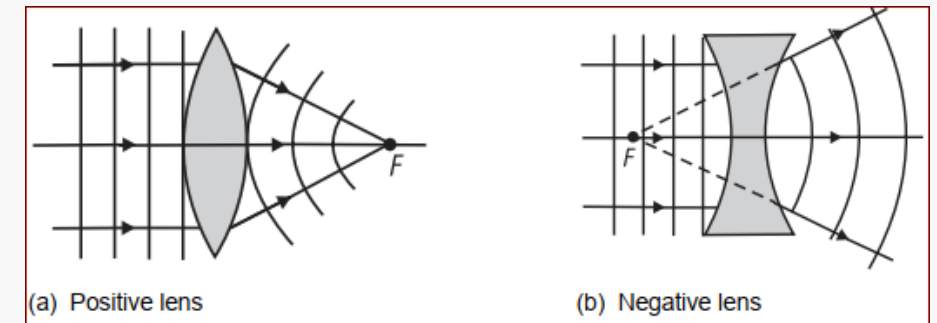
Magnification m produced by a thin lens is $m = \frac{h''}{h} = -\frac{s''}{s}$

Thin Lens

- Two types of lenses: Converging (or positive) and Diverging (or negative),
- Converging lenses are thicker in the middle than at the edges,
- Diverging lenses are thinner in the middle than at the edges.

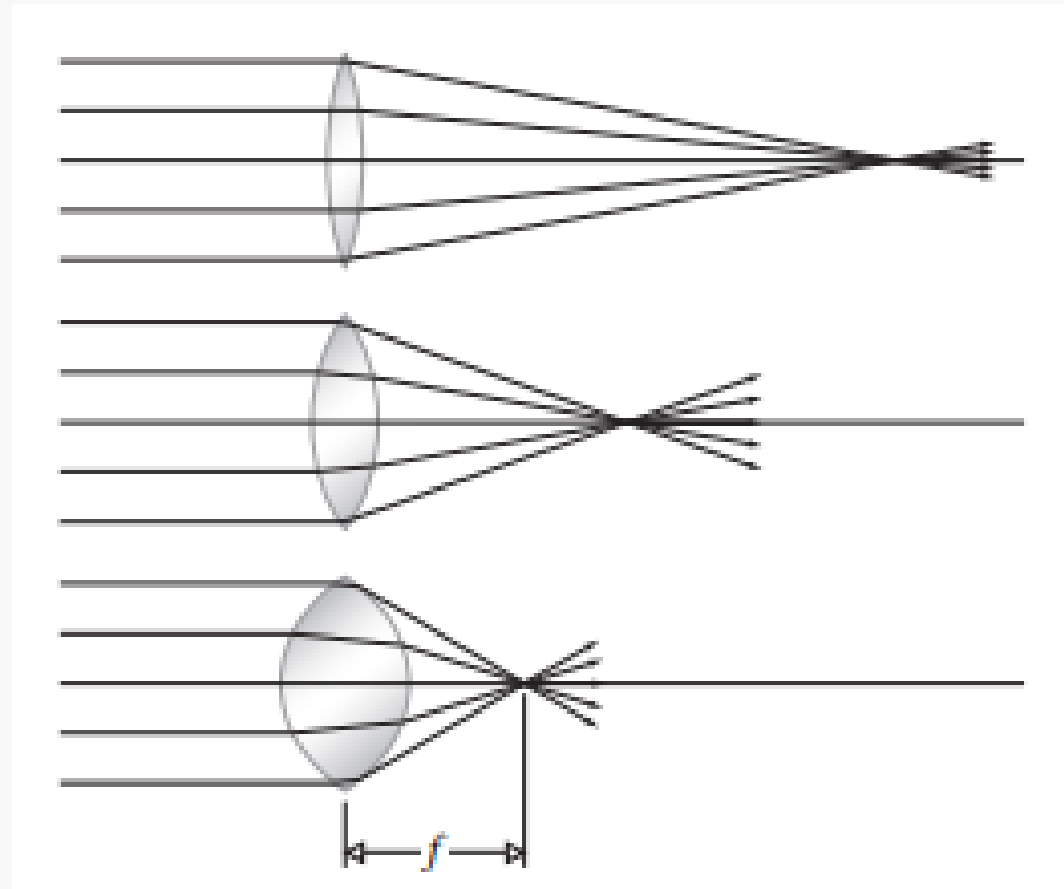


3/2/2020



Function of a Lens

The greater the curvature ($1/R$), the shorter the focal length.



Newtonian Form

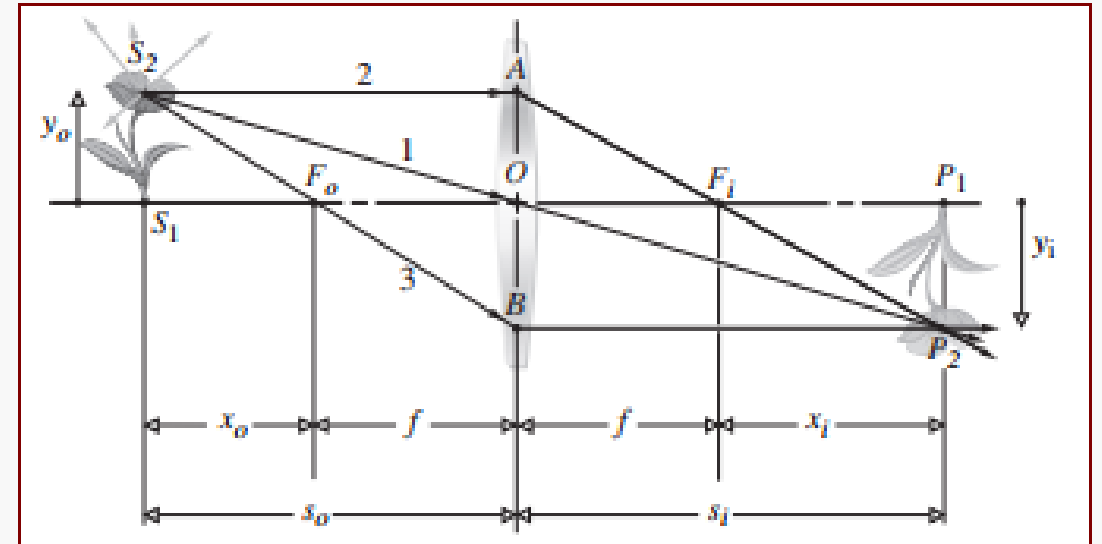
From AOF_i and $P_2P_1F_i$

$$\frac{y_0}{|y_i|} = \frac{f}{s_i - f} = \frac{f}{x_i}$$

Similarly

$$\frac{|y_i|}{y_0} = \frac{f}{s_0 - f} = \frac{f}{x_0}$$

$$x_0 x_i = f^2$$

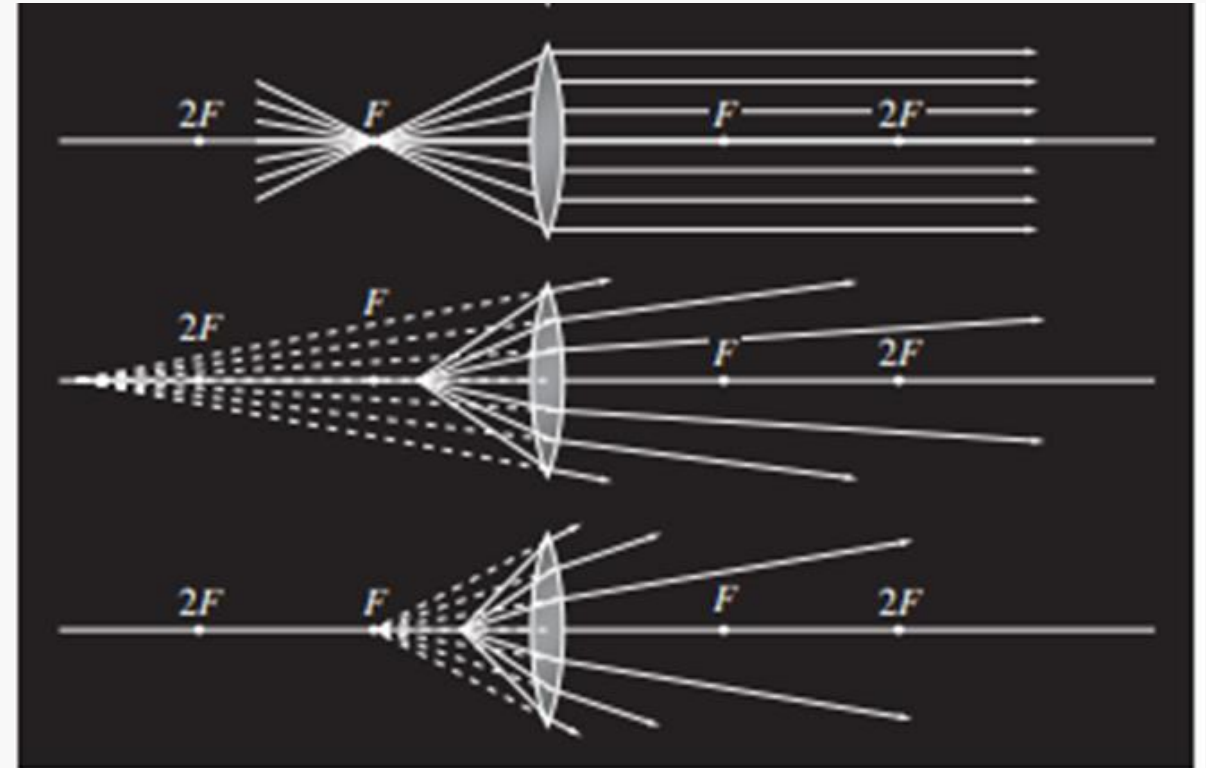
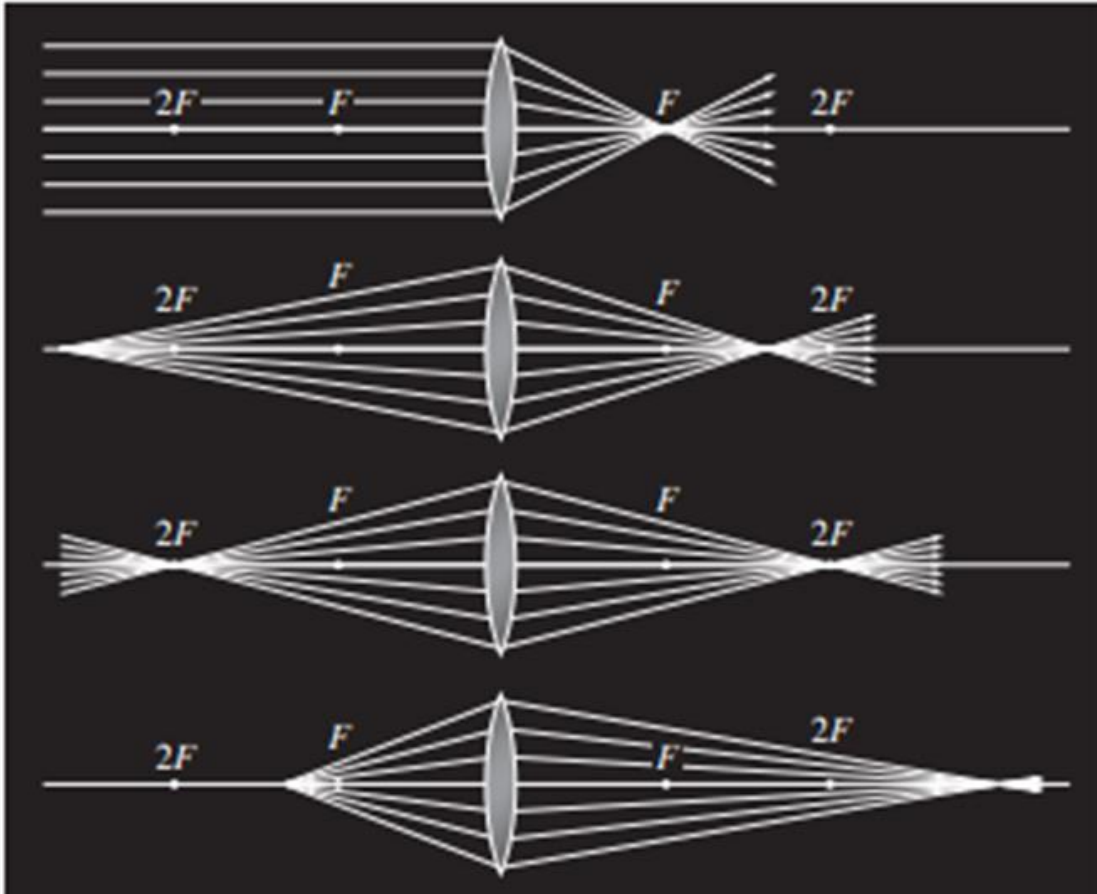


Newtonian Form

Meanings Associated with the Signs of Various Thin Lens and Spherical Interface Parameters

Quantity	Sign	
	+	-
s_o	Real object	Virtual object
s_i	Real image	Virtual image
f	Converging lens	Diverging lens
y_o	Erect object	Inverted object
y_i	Erect image	Inverted image
M_T	Erect image	Inverted image

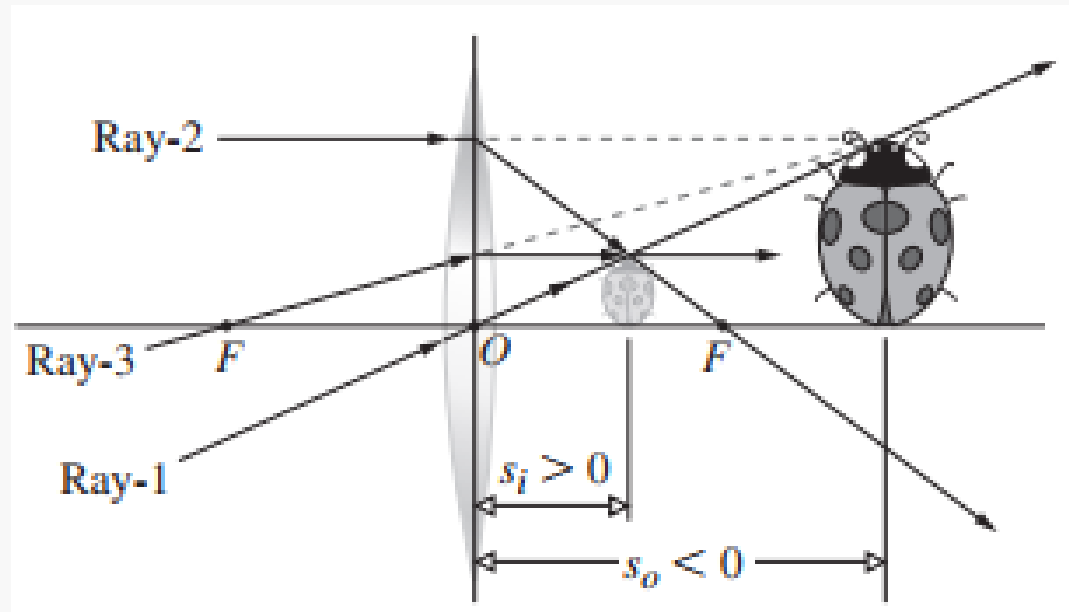
Function of a Lens



Example 3.6

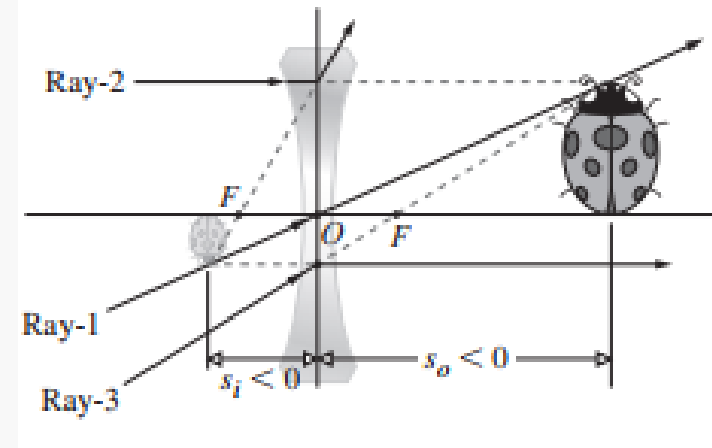
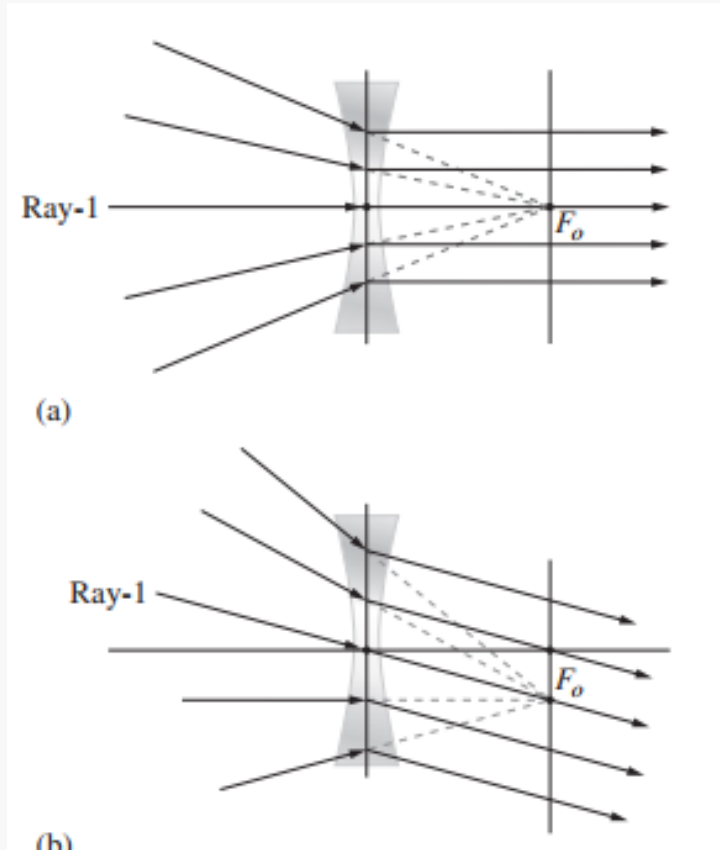
Compute the focal length in air of a thin planar-convex lens having a radius of curvature of 50 mm and an index of 1.5

Virtual Objects

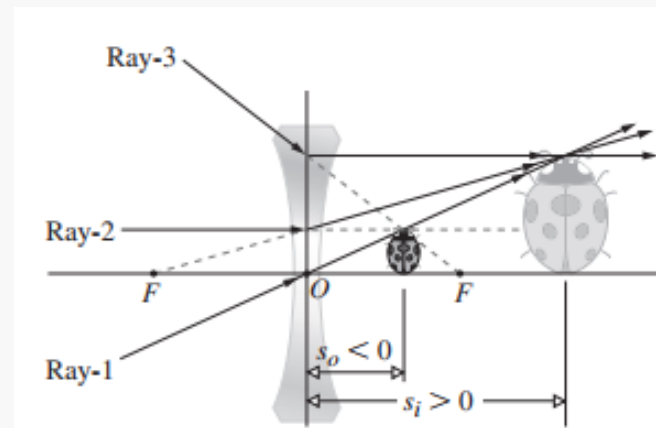


A virtual object (far right) and its real image (upright image). It is possible with a multi-lens system.

Virtual Objects

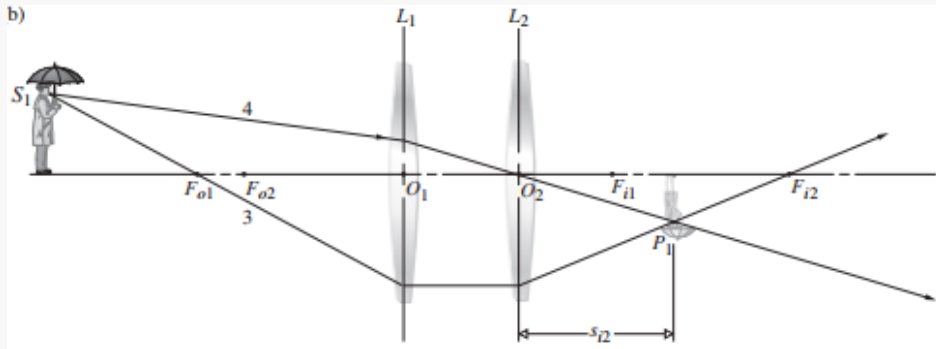


A virtual object (far right) and its virtual inverted image. It is possible with a multi-lens system.

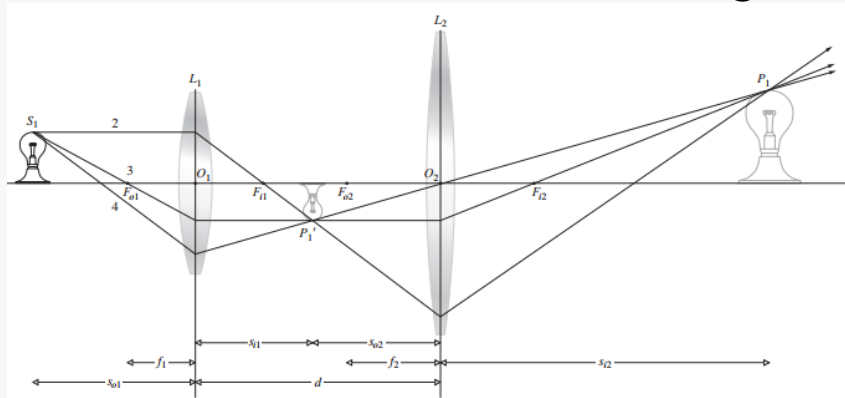


A virtual object and its real enlarge, upright image. It is possible with a multi-lens system that causes the rays to initially converge.

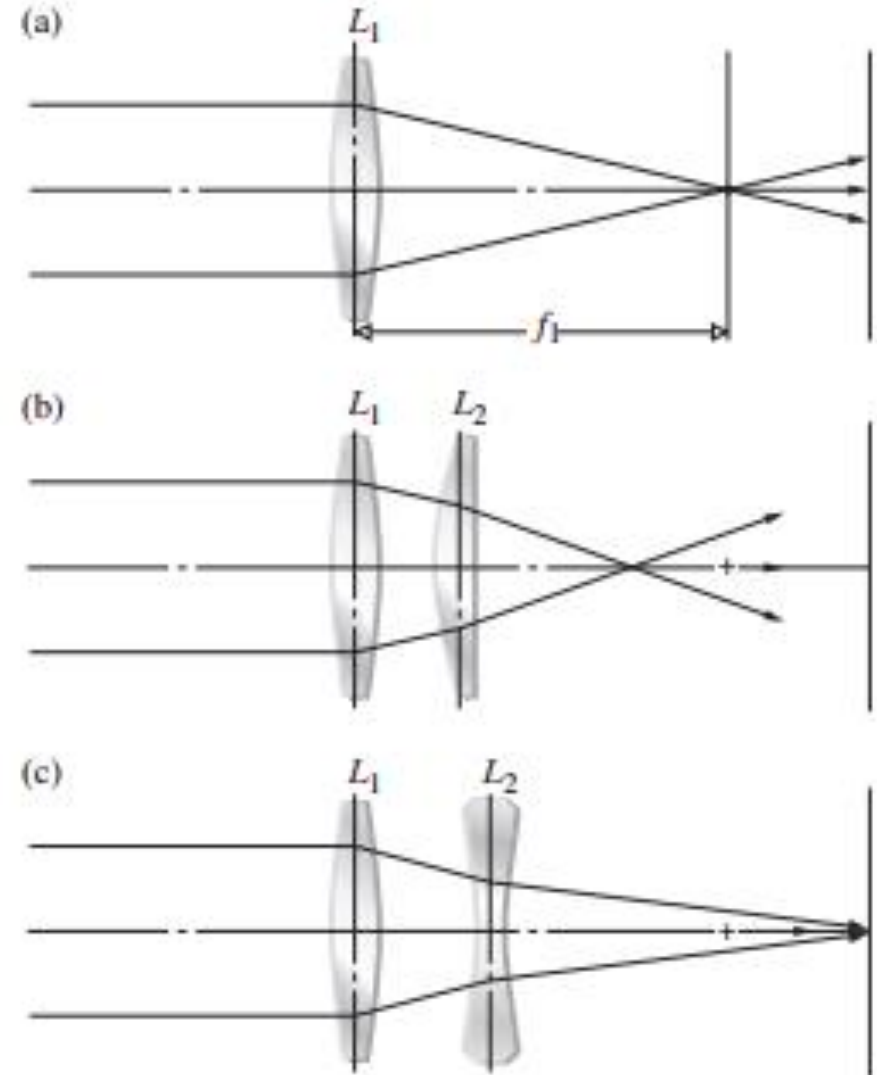
Effective Focal Length



Let's consider two thin lenses are separated by a distance smaller than either focal length.

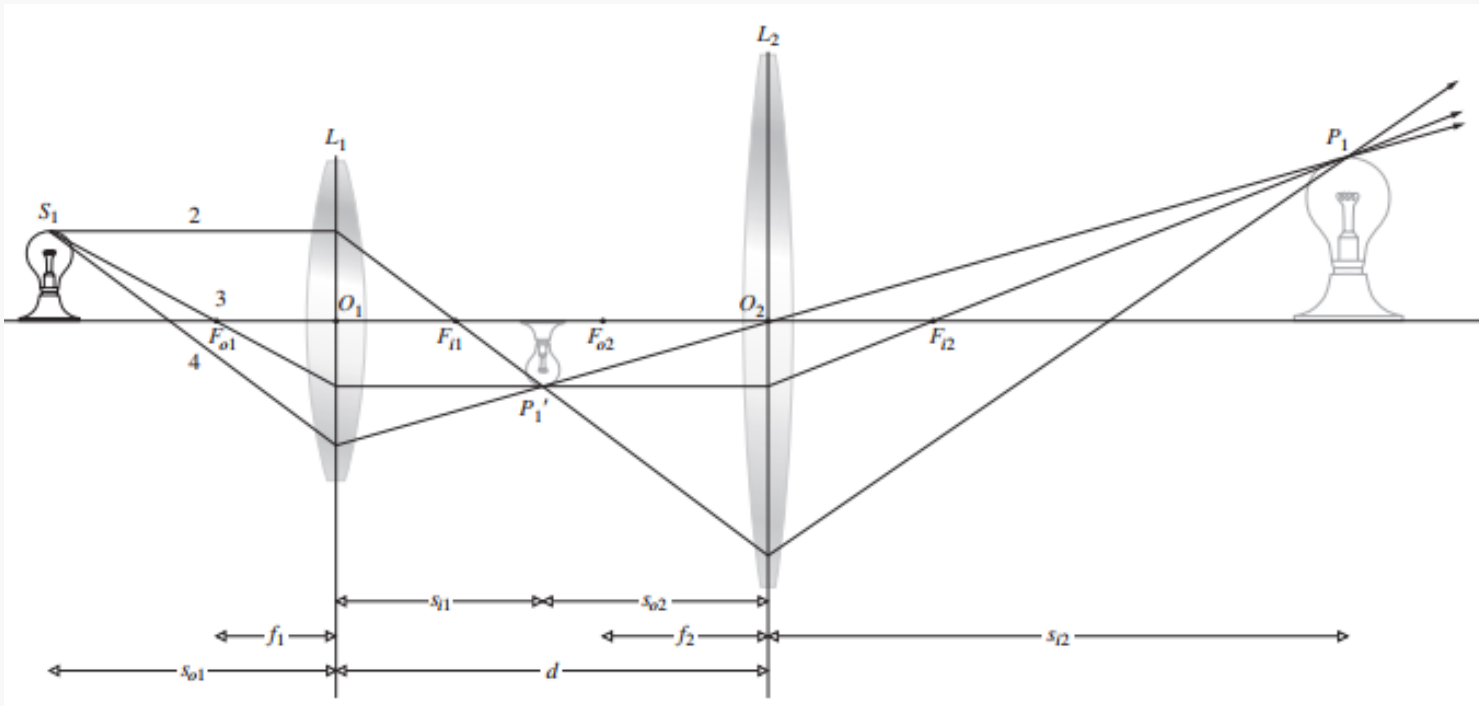


Two thin lenses separated by a distance greater than the sum of their focal lengths.



Effective Focal Length

Two thin lenses separated by a distance greater than the sum of their focal lengths.



Effective Focal Length

For L_1 lens:

$$\frac{1}{s_{il}} = \frac{1}{f_l} - \frac{1}{s_{ol}} \rightarrow s_{il} = \frac{s_{ol}f_l}{s_{ol} - f_l}$$

And L_2 lens:

$$s_{i2} = \frac{(d - s_{il})f_2}{d - s_{il} - f_2}$$

$$s_{i2} = \frac{f_2d - f_2s_{ol}f_l/(s_{ol} - f_l)}{d - f_2 - s_{ol}f_l/(s_{ol} - f_l)}$$

Transverse Magnification of Compound lens

Total transverse magnification of the compound lens is the product of the individual magnifications.

$$M_T = M_{T1}M_{T2}$$

- It can be shown that.

$$M_T = \frac{f_1 s_{i2}}{d(s_{o1} - f_1) - s_{o1}f_1}$$

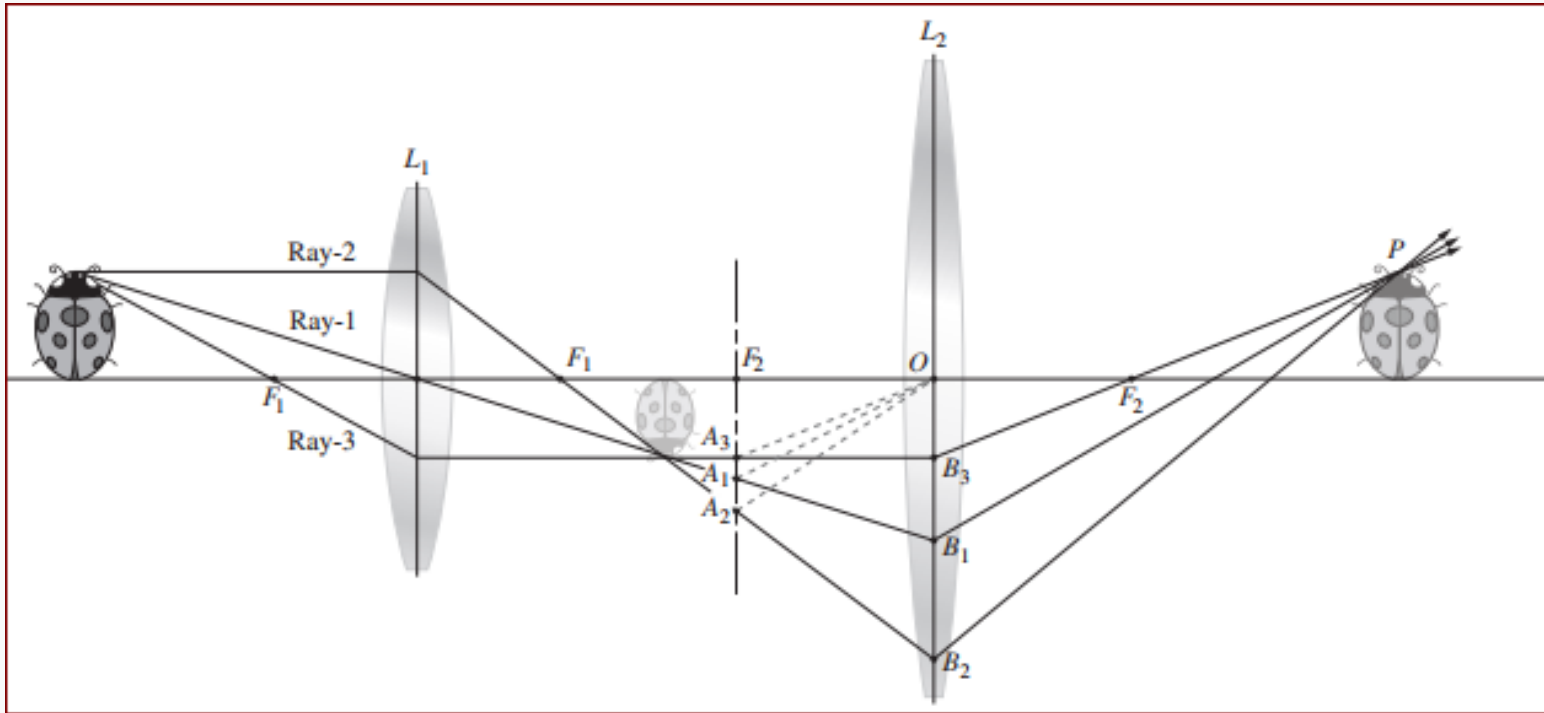
Transverse Magnification of Compound lens

A thin biconvex lens having a focal length of $+40.0$ cm is located 30.0 cm in front (i.e., to the left) of a thin biconcave lens of focal length -40.0 cm. If a small object is situated 120 cm to the left of the positive lens

- a. Determine the location of its image by calculating the effect of each lens.
- b. Compute the magnification.
- c. Describe the image.

Transverse Magnification of Compound lens

Two positive lenses, L_1 and L_2 and are separated by a distance greater than the sum of both focal lengths



Back and Front Focal Length

Front Focal length (f.f.l) when the $s_{i2} \rightarrow \infty$

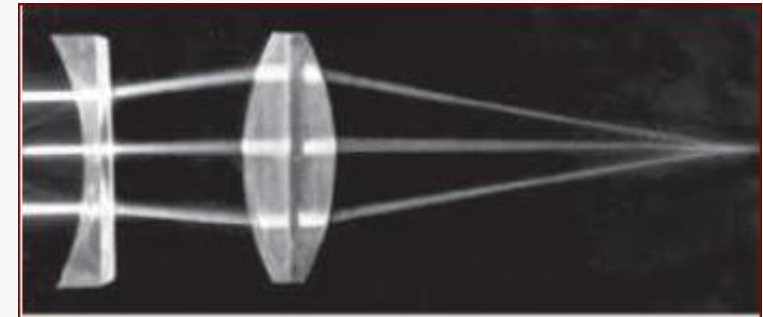
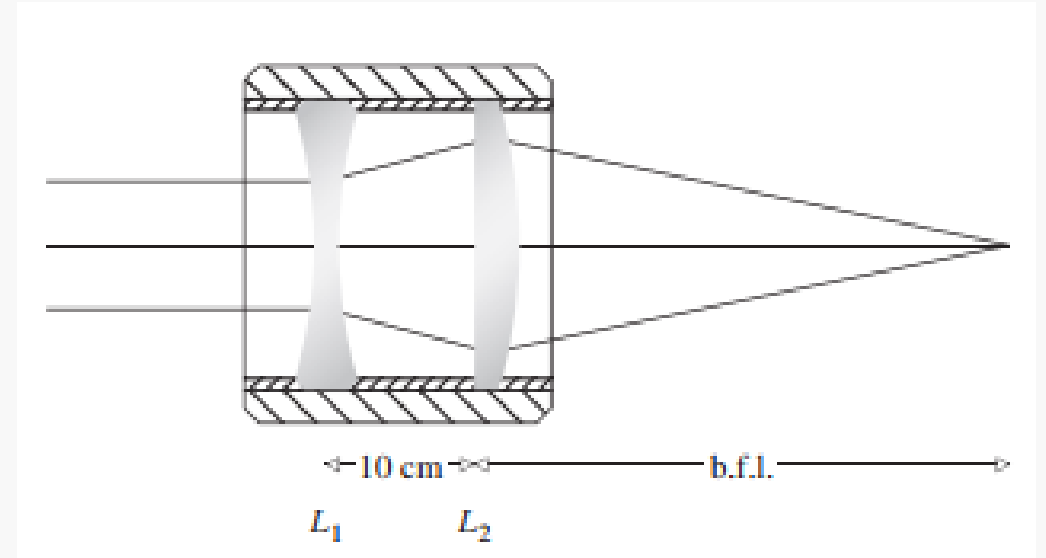
$$b.f.l = \frac{f_2(d-f_1)}{d-(f_1+f_2)}$$

Similarly, letting $s_{o1} \rightarrow \infty$

$$f.f.l = \frac{f_1(d-f_2)}{d-(f_1+f_2)}$$

Back focal length for lens combination of the figure:

$$b.f.l = \frac{20(10 - (-30))}{10 - (-30 + 20)} = 40cm$$



Effective Focal Length

If two thin lenses in contact, the resultant this lens has an *effective focal length* , f , such that

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

And N such lenses

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \dots + \frac{1}{f_N}$$

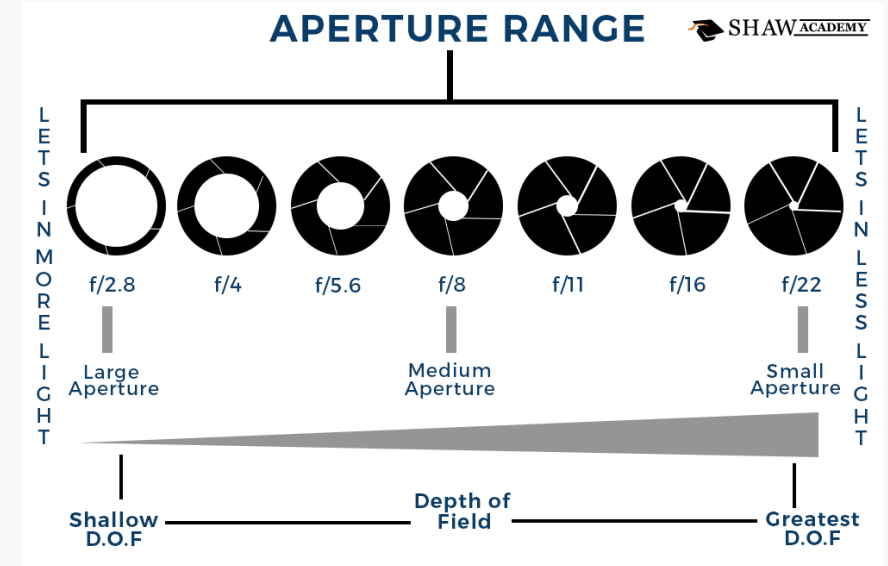
f-number and Numerical Aperture

The size of a lens directly affects its light gathering power, which determines the brightness of the final image.

Two commonly used indicators are: f-number and numerical aperture

F-number is a dimensionless parameter that indicates the light-gathering ability of an optical system, such as camera lens. It is defined simply as the ratio of the focal length *f* divided by diameter of the entrance pupil (the effective aperture)):

$$f_{\text{number}} = \frac{\text{Focal length}}{\text{Aperture diameter}} = \frac{f}{D}$$



f/32

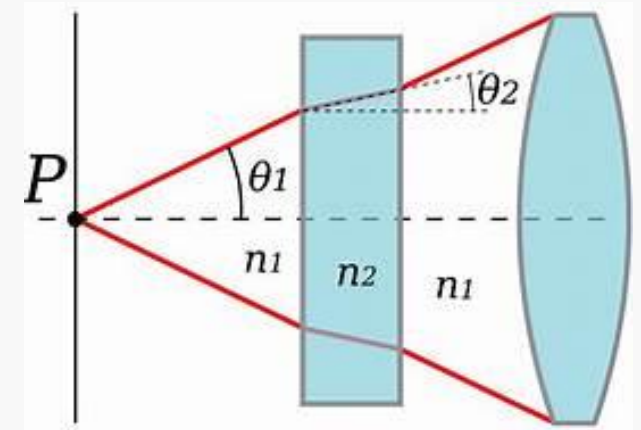


f/5

f-number and Numerical Aperture

Another commonly used important designing parameters for a lens is numerical aperture (NA), which directly relate how much light the lens gathers. It goes a step further than f-number by considering the refractive index of the medium in which lens in it.

- It's a dimensionless number between 0 and 1 that characterizes the range of angles over which the system can accept light.
- It considers both the aperture size and the refractive index of the medium the light travels through (typically air or a specific immersion medium in microscopy).
- NA is particularly important in microscopy and fiber optics, as it affects the resolving power and the brightness of the image.



f-number and Numerical Aperture

The NA of a lens is exhibited the difference in solid angles (cone) of light accepted. So, lens in the air:

$$NA = \frac{CA}{2f} \rightarrow NA = \frac{1}{2(f_{number})}$$

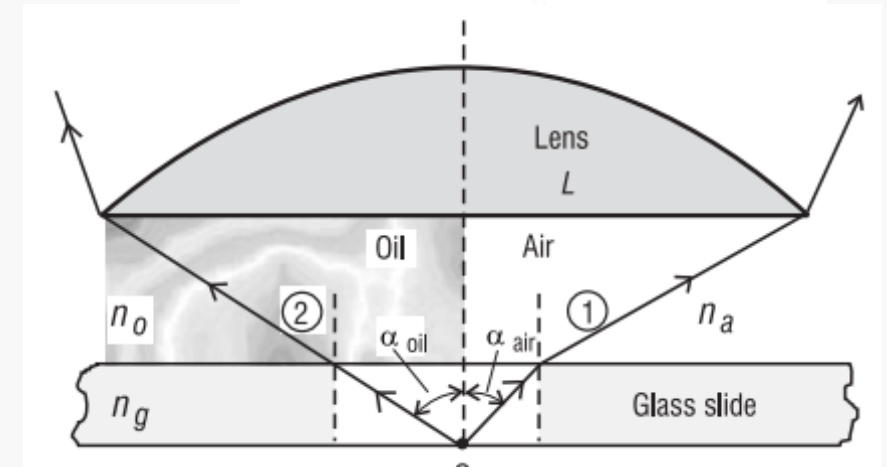
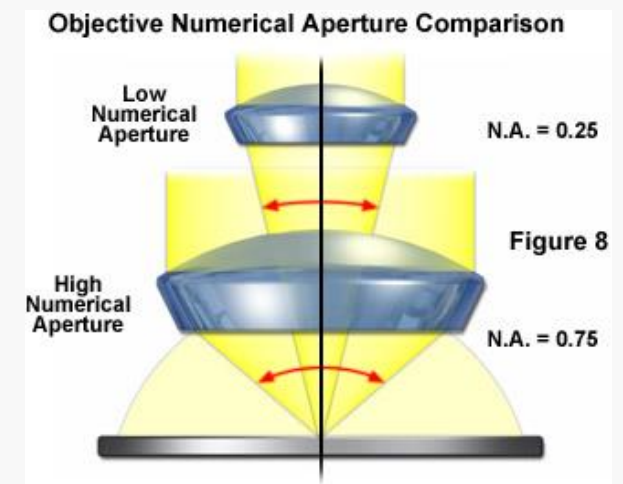
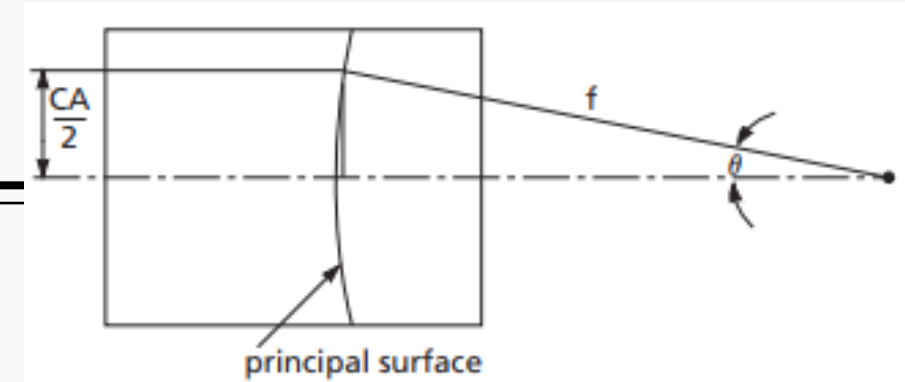
$$image\ brightness \propto \frac{1}{(f_{number})^2} = (NA)^2$$

Lens in a medium

$$NA = n \sin \alpha$$

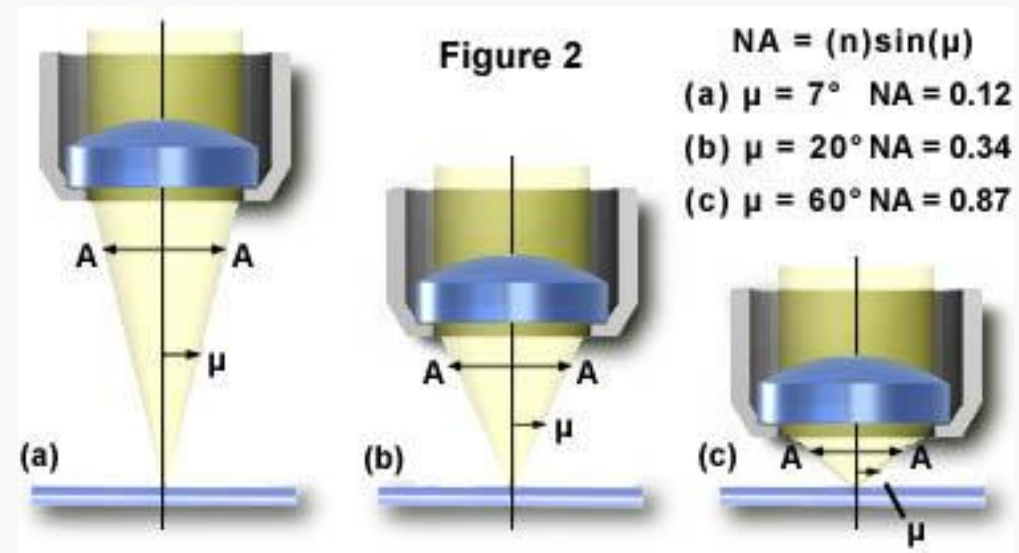
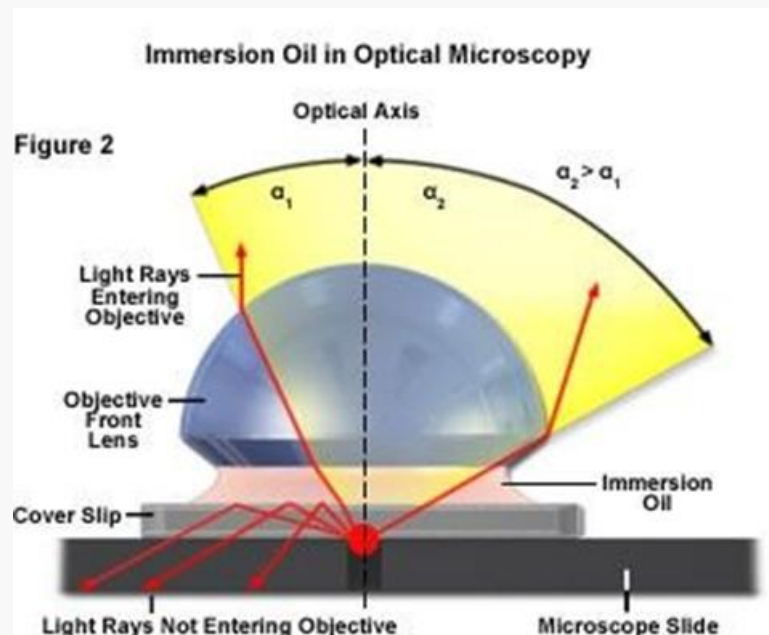
Here, n is the index of refraction of the intervening medium b-t object and lens and α is half-angle.

Example of increasing **NA** via introducing medium between lens and object. In the figure, ray-1 suffers refraction at the glass-air interface.



f-number and Numerical Aperture

$$NA = n \sin \alpha$$

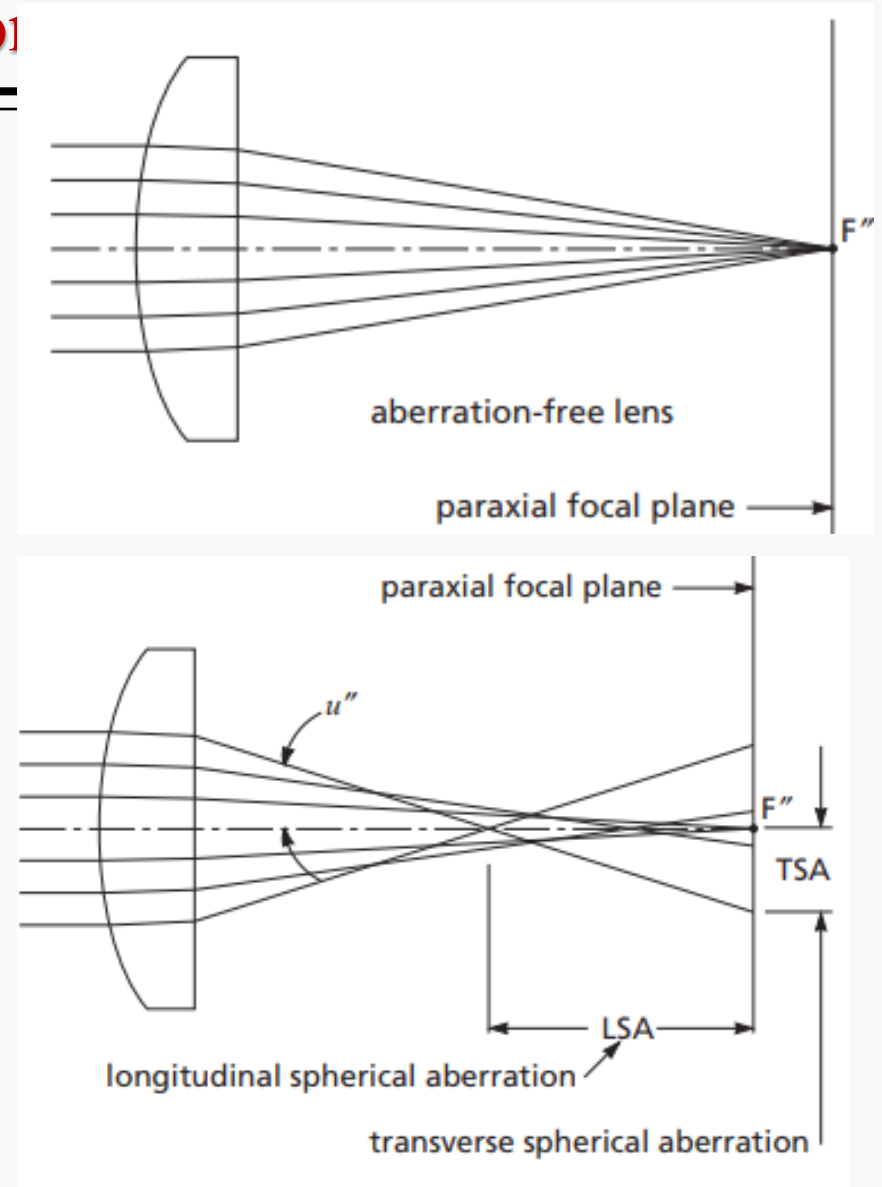


Aberration

- Modern imaging system have largely overcome lens aberrations thanks to advancements in lens technology. It is possible with the present technology to construct high-quality spherical lens systems,
- **Aberrations** are mainly due to diffraction. Diffraction is a fundamental limitation on any optical system due to the wave nature of light.

Ideal Lens Behavior vs. Spherical Aberration

- An ideal lens would focus all incoming parallel rays (collimated light) to a single, perfect point.
- In reality, spherical aberration occurs: rays striking different parts of the lens focus at different points along the optical axis.
- This aberration is influenced by:
 - Lens shape and curvature
 - Lens orientation
 - Object and image distance (conjugate ratio)
- The variation of focus along the optical axis is called longitudinal spherical aberration.



Spherical Aberration

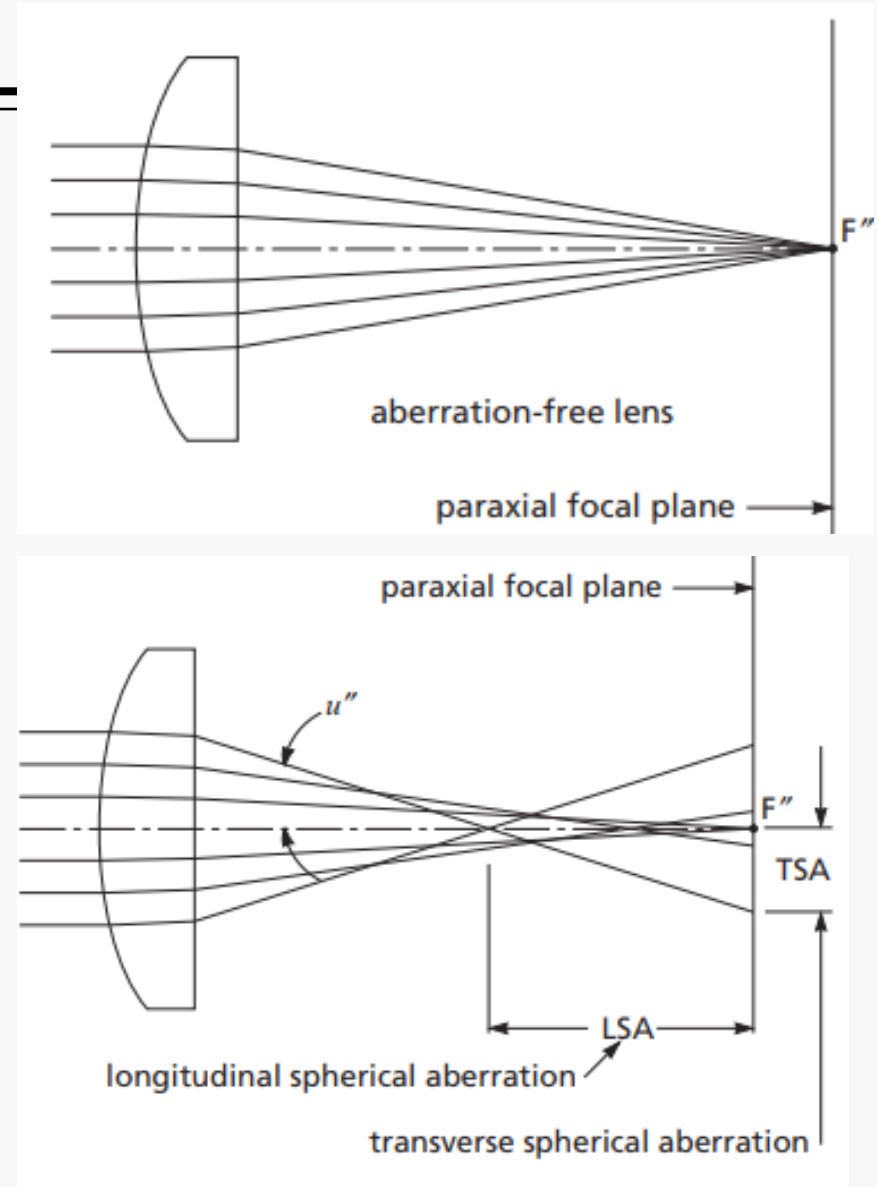
- Longitudinal Spherical Aberration (LSA): Refers to the difference in focus between rays hitting the center and the edge of the lens (along the optical axis).
- Transverse Spherical Aberration (TSA): Represents the distance between the paraxial focal plane (ideal focus for central rays) and where the edge rays focus (perpendicular to the optical axis).

Longitudinal spherical aberration (LSA) and transverse spherical aberration (TSA) are related by:

$$TSA = LSA \times \tan u''$$

Spherical aberration due to higher order terms of sine function in Snell's law:

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots$$



Spherical Aberration

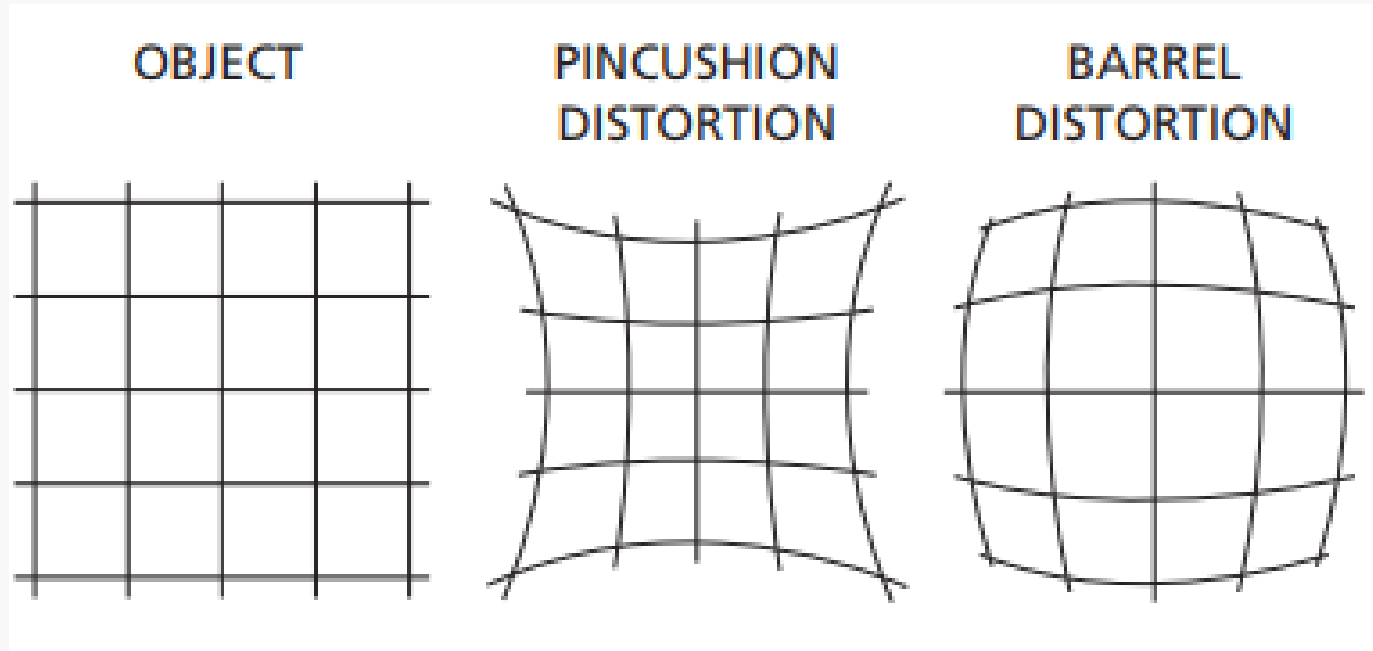
- In 1857, Ludwig von Seidel made a significant contribution to optics by analyzing the effects of a specific mathematical term ($\theta^3/3!$) on lens aberrations. These aberrations are now known as Seidel aberrations.
- Seidel's work categorized the various imperfections that can occur in an optical system, particularly for monochromatic light (single color). These categories include:
 - Spherical Aberration
 - Astigmatism
 - Field Curvature
 - Coma
 - Distortion

Spherical Aberration

- Ludwig von Seidel also identified chromatic aberration and lateral color as issues arising in systems using polychromatic light (multiple colors).
- Notably, Seidel developed a method to approximate these aberrations using sine functions, eliminating the need for tracing every light ray. This approach is still relevant today, with powerful ray-tracing software even utilizing Seidel's formula for spherical aberration calculations.

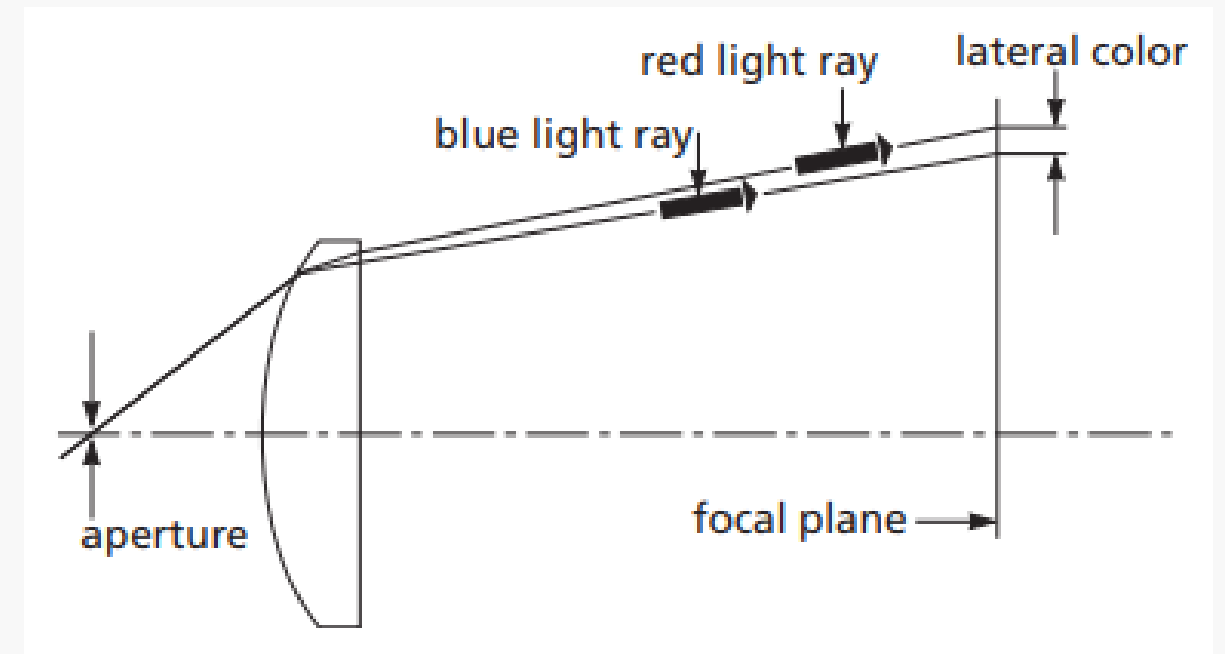
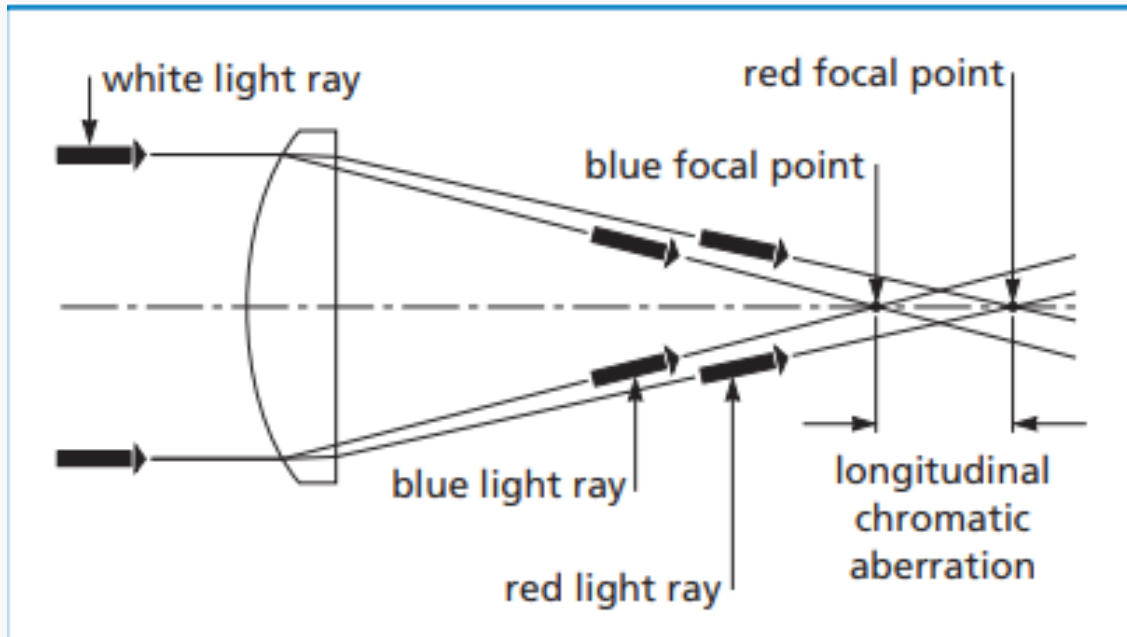
Distortion

The image field can exhibit two imperfections: curvature and distortion. These aberrations are caused solely by the shape of the lens elements and are independent of the wavelength of light (meaning they occur even with monochromatic light).



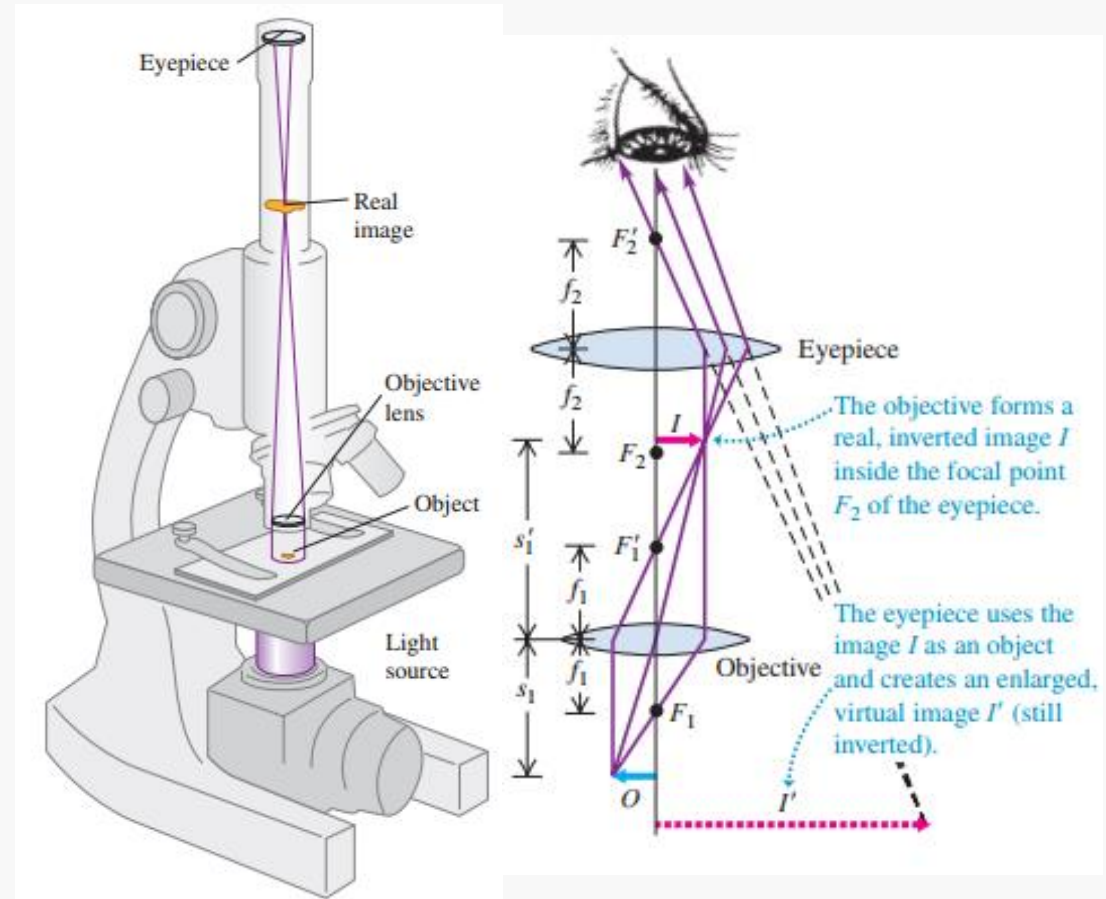
Chromatic Aberration

Chromatic aberration arises specifically when using optics with light containing multiple wavelengths (polychromatic light). The figure below illustrates longitudinal chromatic aberration.



Basic Concepts in Optical Microscopy

For over 150 years, the optical microscope has been a cornerstone of scientific discovery in life sciences and material science. This remarkable tool allows us to visualize objects as small as 200 nanometers ($2 \times 10^{-7} m$), enabling us to study single-celled organisms like bacteria, which are typically around 0.2 millimeters ($2 \times 10^{-4} m$) in size.

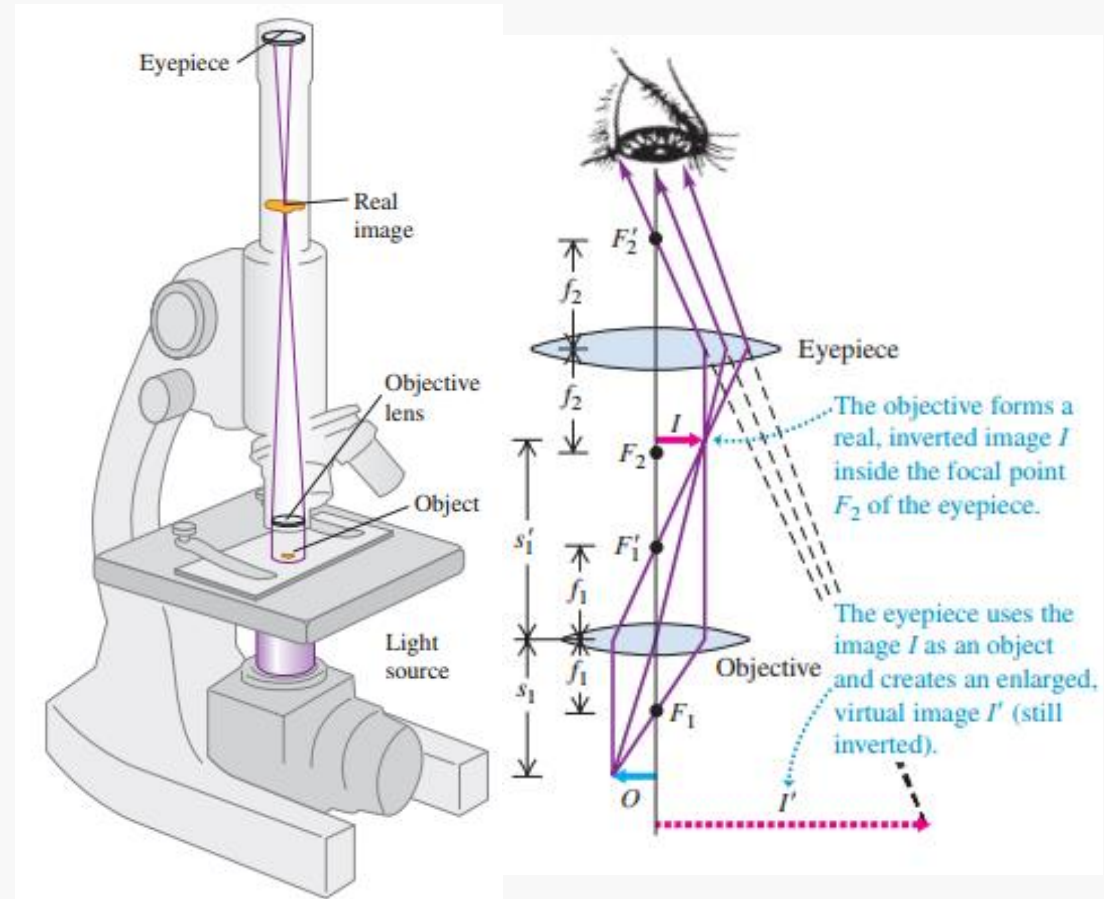


Basic Concepts in Optical Microscopy

The Essential Components of an Optical Microscope:

At its core, an optical microscope utilizes a minimum of two lenses to achieve its function:

- Illumination: A light source illuminates the specimen being studied.
- Magnification: A series of lenses magnify the image of the illuminated specimen, allowing us to view it in much greater detail



Basic Concepts in Optical Microscopy

Microscope magnification results from the combined effect of two factors

- Objective Lens Magnification (M_1): This determines the linear size (actual dimensions) of the magnified image formed inside the microscope.
- Eyepiece Magnification (M_2): This relates to the angular size (how big something appears) of the final image seen through the eyepiece. It essentially determines how much larger the image appears compared to its size when viewed without magnification.

The overall magnification of a microscope is calculated by multiplying these two individual magnifications. Angular magnification of a microscope (M) is:

$$M = M_1 M_2$$

Magnifier

Our perception of an object's size depends on the angle it creates (subtended angle) when viewed by our eye. Bringing an object closer increases this angle, making the image appear larger on our retina. However, our eyes have a limit to how close they can focus clearly - typically around 25 cm (near point).

Converging lens can be used to form a virtual image that is larger and farther from the eye than the object itself.

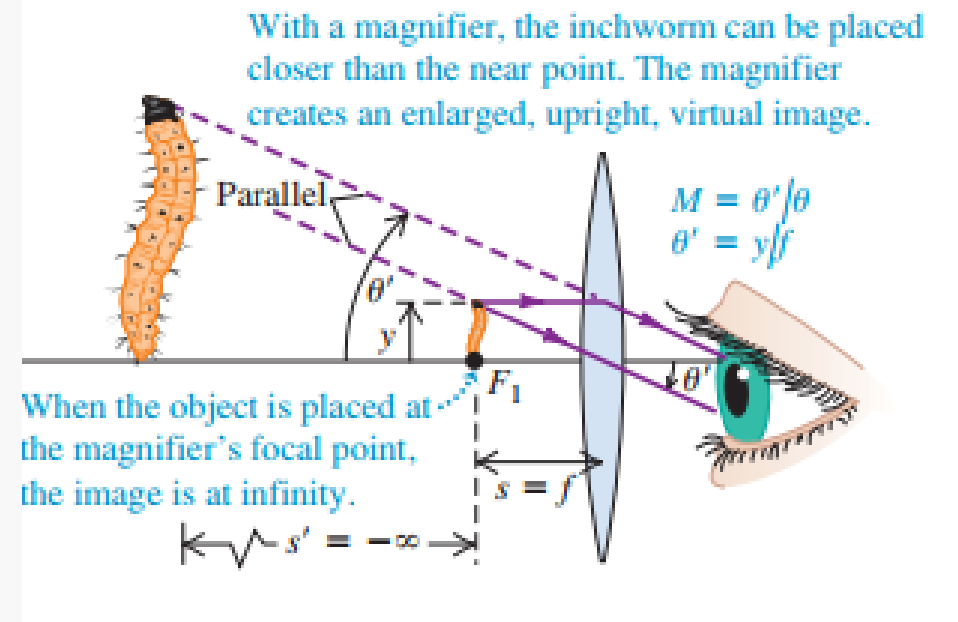
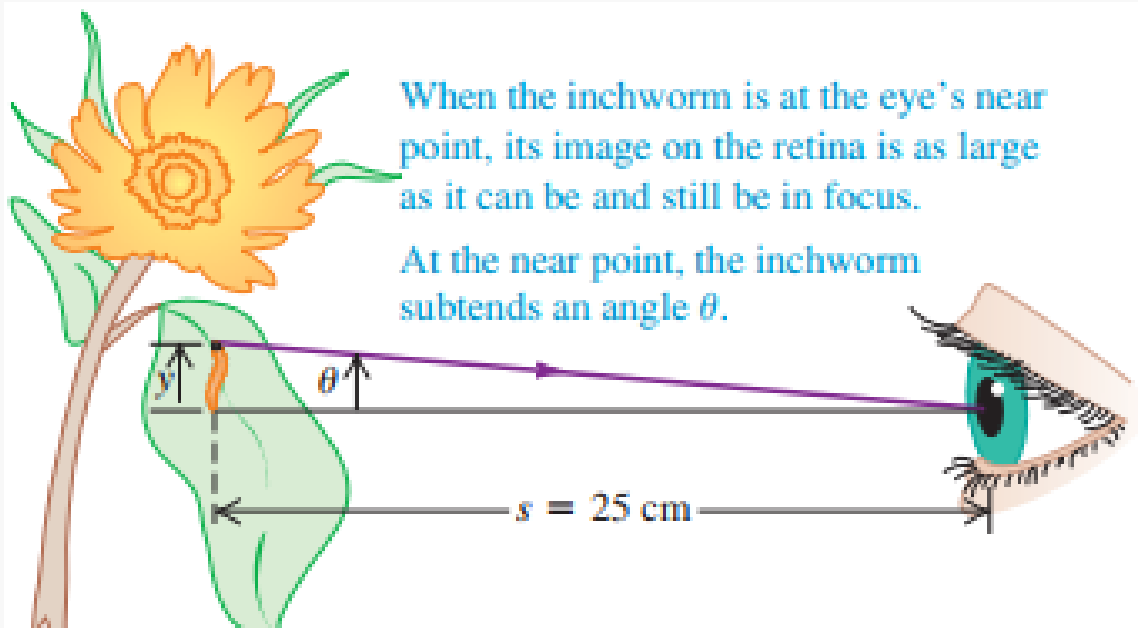
- **Angular Magnification (M):** This refers to the increase in the apparent size of an object when viewed through a lens. It's calculated as the ratio of the image's angular size (θ') to the object's angular size (θ).

$$M = \frac{\theta'}{\theta} (\text{Angular magnification})$$

- **Lateral magnification (M_L):** This refers to the actual change in size of the image compared to the object..

$$M = -s_i/s_o$$

Magnifier



$$\theta = \frac{y}{25 \text{ cm}} \quad \text{and} \quad \theta' = \frac{y}{f}$$

Angular magnification for a simple magnifier is:

$$M = \frac{\theta'}{\theta} = \frac{25 \text{ cm}}{f}$$

Magnifier of Basic Optical Microscope

the overall magnification formula becomes:

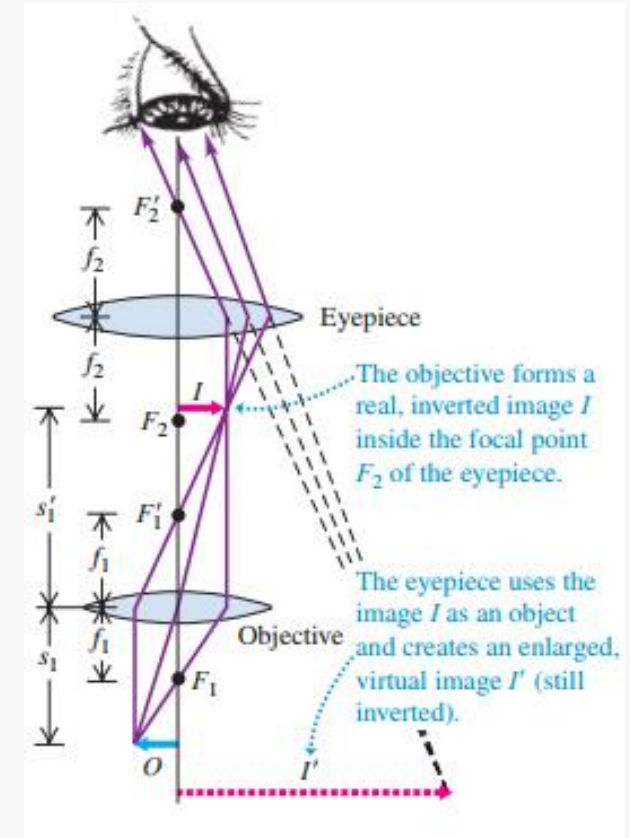
$$M = M_1 M_2 = \left(-\frac{s_i}{s_o} \right) \left(\frac{25 \text{ cm}}{f} \right)$$

The total magnification (M) of a microscope is the product of two individual magnifications:

Objective Lens Magnification (M_1): This is typically denoted by a negative value representing the image size (s_i) relative to the object size (s_o).

In most cases, the objective lens positions the object very close to its focal point ($s_o \approx f_1$).

Eyepiece Magnification (M_2): This represents the angular magnification of the eyepiece, typically denoted by the ratio of a standard viewing distance (often 25 cm) to the eyepiece focal length (f_2).



Magnifier of Basic Optical Microscope

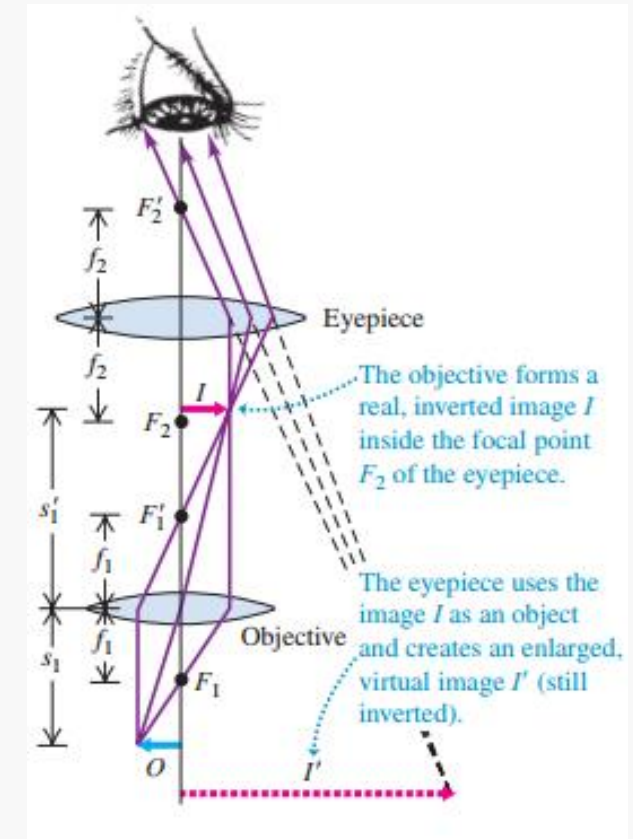
Typically, objective is very close to the focal point ($s_o \approx f_1$), so image distant s_i is very great in comparison to the focal length f_1

$$M = \frac{(25 \text{ cm}) s_i}{f_1 f_2}$$

One way to increase magnification is by using an objective lens with a shorter focal length (f_1). Most microscopes have a rotating turret containing several objectives with different focal lengths, allowing for varying magnification levels.

A significant challenge in microscopy is achieving good contrast. This can be difficult when observing very thin specimens or highly reflective surfaces, as light might simply pass through or bounce off without revealing details.

To address this issue, scientists have developed various optical techniques that enhance contrast and even introduce color variations in the specimen. These techniques, often referred to as "staining" or "contrast enhancement," enable researchers to visualize critical features within the sample



Thank you very much for your attention